Recent Advances on Non-Line-of-Sight Imaging: Conventional Physical Models, Deep Learning, and New Scenes (Updated 2022–2026)

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***Abstract*—As an emerging technology that has attracted huge attention, non-line-of-sight (NLOS) imaging can reconstruct hidden objects by analyzing the diffuse reflection on a relay surface, with broad application prospects in the fields of autonomous driving, medical imaging, and defense. Despite the challenges of low signal-to-noise ratio (SNR) and high ill-posedness, NLOS imaging has been developed rapidly in recent years. Most current NLOS imaging technologies use conventional physical models, constructing imaging models through active or passive illumination and using reconstruction algorithms to restore hidden scenes. Moreover, deep learning algorithms for NLOS imaging have also received much attention recently. This paper presents a comprehensive overview of both conventional and deep learning-based NLOS imaging techniques. It is an updated version of our peer-reviewed journal paper published in *APSIPA Transactions on Signal and Information Processing* (2022, vol. 11, pp. 1–48, DOI: 10.1561/116.00000019), extended to cover significant advances from 2022 to 2026; the content added in this update has not undergone peer review. In particular, we highlight the emergence of transformer-based and diffusion-model-based reconstruction networks, the achievement of real-time dynamic NLOS video, the extension of NLOS imaging to non-optical modalities such as radar and ultrasound, and the exploration of new application scenarios including human pose estimation and super-field-of-view imaging. Besides, we survey new proposed NLOS scenes, and discuss the challenges and prospects of existing technologies. Such a survey can help readers have an overview of different types of NLOS imaging, thus expediting the development of seeing around corners. A curated list of 210+ NLOS papers and an interactive project homepage are available at https://github.com/ruixv/NLOS_Overview and https://ruixv.github.io/NLOS_Overview/.**

***Index Terms*—Non-line-of-sight (NLOS), deep learning, active NLOS imaging, passive NLOS imaging, transformer, dynamic NLOS, radar NLOS, acoustic NLOS, neural implicit surface, diffusion model, real-time reconstruction.**

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I. Introduction

With the rapid development of photon-sensitive sensors and imaging algorithms, optical imaging capabilities have been greatly improved in recent years. Within the line of sight, high-quality imaging can be achieved at a relatively long distance. However, due to the inherent physical constraint of visible light, traditional optical imaging is difficult to see objects outside the line of sight. To break that restriction, non-line-of-sight(NLOS) imaging analyzes the diffuse reflection from a relay wall to image hidden objects, which has broad applications in many fields, such as medical imaging, autonomous driving, and robotic vision [1], [2].

According to whether a controllable light source is used, NLOS imaging can be divided into active imaging and passive imaging. Active NLOS imaging often uses expensive external light sources with high temporal resolution(e.g., ultrafast laser) to illuminate the diffuse relay surface. Simultaneously, a sensitive time-resolved detector is used to detect the light reflected on the relay surface, hidden objects, and relay surface in sequence. The collected effective light is often referred as the three-bounce light since it is reflected by three surfaces (relay surface, hidden objects, and relay surface) in succession. Then, the collected three-bounce light is analyzed by different algorithms(e.g., backprojection [3], [4], inverse methods [5] and wave-based methods [6], [7]) to reconstruct the hidden scene. Since the active methods can collect different kinds of information, including intensity, time, and coherence, the active methods can perform a high-resolution 3D reconstruction. On the other hand, passive methods do not use a controllable external light source but use ambient light or light emitted by hidden objects to complete NLOS imaging. Despite the low cost, passive methods usually can only collect intensity [8] or limited coherence information [9], [10] and usually complete low-quality 2D reconstruction or localization, while a few recent works can estimate hidden scene position or depth with partial occluders [8], [11].

Four typical NLOS imaging setups are shown in Figure 1, and there are many challenges when realizing the NLOS imaging. First, due to the high-order loss with distance and environmental noise during the light transport process, NLOS imaging is an ill-posed problem with low SNR, making high-quality reconstruction extremely difficult [12]. Different hidden scenes may produce the same measurement, which deepens the ill-posedness of the problem [13]. Second, the spatial resolution of the reconstruction result is limited by

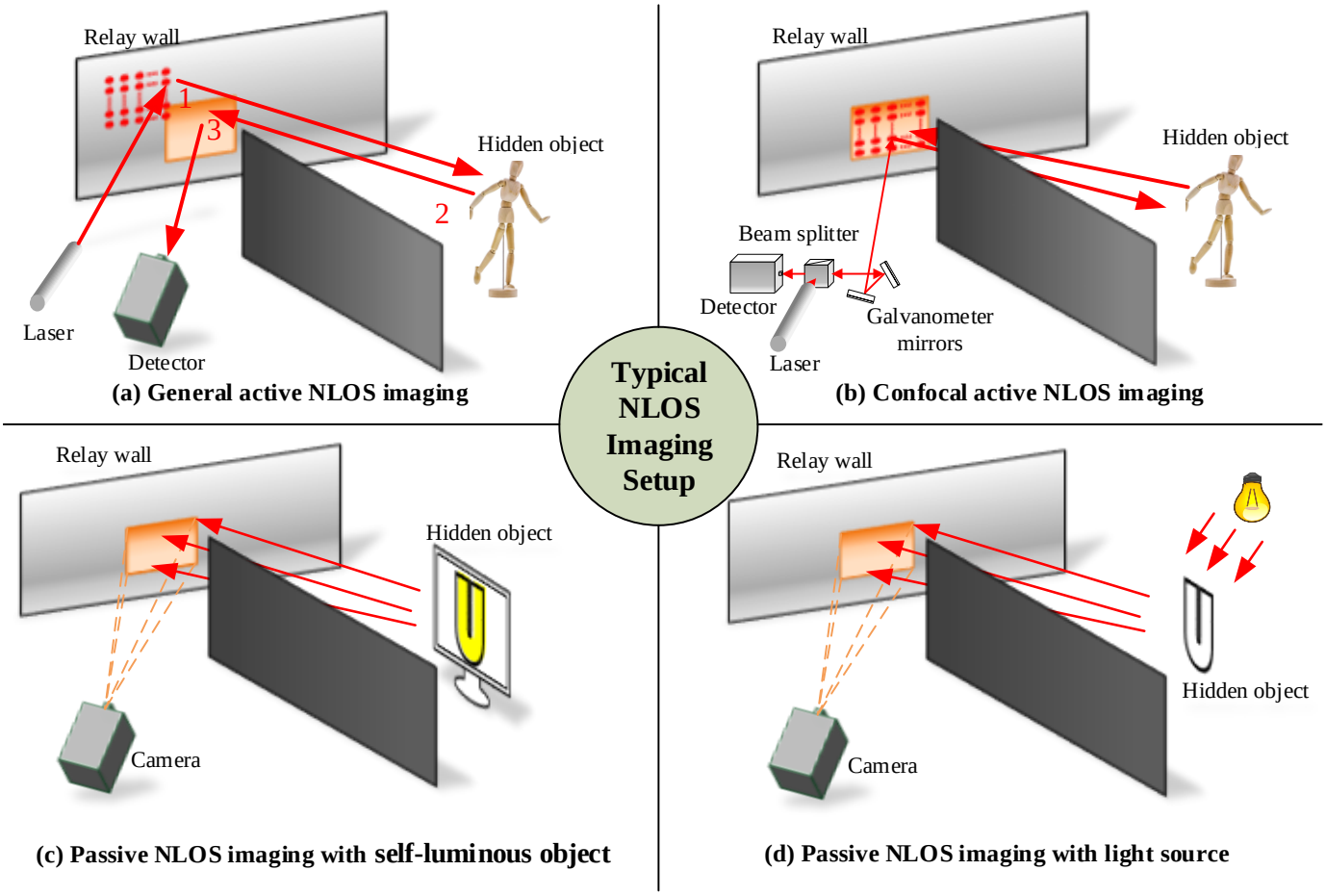


Fig. 1. Four typical NLOS imaging setups

the size of the scanning area (aperture size) and the system's temporal resolution. When the size of the hidden scene is large or complex, the spatial resolution will be limited by the computational complexity [1]. Third, the data collection time is too long [14]. Typical time-resolved NLOS imaging needs a scanning process to obtain data, making it difficult to reconstruct in real-time with high quality. Although array detectors (e.g., SPAD array) are promising to eliminate the scanning process [15], they have not yet been fully explored.

Despite the difficulties, many emerging innovative methods have achieved NLOS imaging under certain scenarios in recent years. Figure 2 summarizes recent NLOS imaging research. It can be seen that the conventional methods based on physical imaging models, are still the mainstream of research in recent years. Conventional methods rely on imaging setup and illumination (e.g., active or passive), which are committed to developing three factors in NLOS imaging: advanced hardware systems, accurate forward imaging models and effective reconstruction algorithms. For example, [16] deployed confocal settings and dual-telescope to complete an amazing 1.43km NLOS imaging. Liu et al. exploited the phasor field to convert NLOS imaging model to LOS imaging model [2], [6], thus achieving high-quality reconstruction of complex scenes. [1] subtly converted NLOS reconstruction into a three-dimensional deconvolution problem. All these works have greatly promoted the development of NLOS imaging.

Besides conventional methods, the application of deep learning in NLOS imaging has also been rapidly developed. According to network design principles, the deep learning methods used in NLOS imaging are divided into end-to-end networks [17] and physics-based networks [18], [19]. Compared with conventional algorithms, deep learning algorithms can thoroughly learn the scene prior, automatically extract features, and complete the reconstruction of hidden objects. Although NLOS imaging based on deep learning is still at the early stage, it has broad research prospects for practical applications. Moreover, some new types of NLOS scenes, such as "imaging behind occluders" [20] that used two reflections behind obstacles and keyhole imaging [21], have also been proposed, enriching the applications of NLOS imaging. This article aims to make a detailed review of the various types of NLOS imaging research mentioned above, including conventional active and passive methods, deep learning-based NLOS imaging, and new NLOS scenarios. The corresponding challenges and opportunities will also be discussed. We hope this article can help readers have a systematic understanding of existing NLOS imaging research.

Compared to the existing surveys [22], [23], this article is much more comprehensive with the coverage of different NLOS imaging scenes, deep learning algorithms, and new

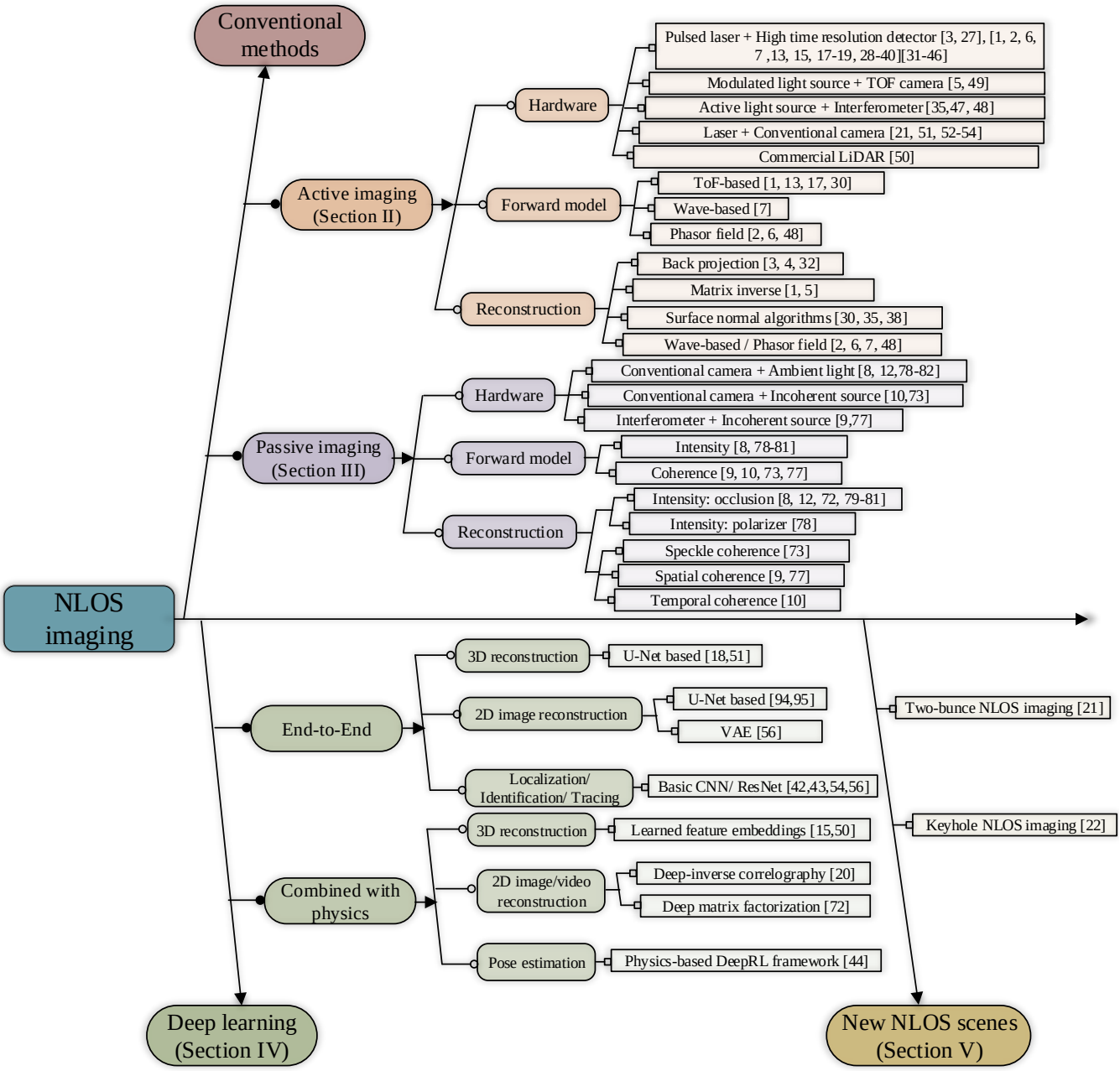


Fig. 2. An overview of NLOS imaging.

NLOS imaging scenarios. As an emerging technology that has developed rapidly in recent years, the algorithms and applicable scenarios of NLOS imaging are constantly growing. Survey [22] summarized the research of NLOS imaging in detail by classifying existing methods based on ToF information, coherent information, and intensity information. However, it completes the review from the perspective of the information used, and has less introduction to imaging models, reconstruction algorithms, and recent deep learning methods. Besides, in [22], the passive imaging problem was only a subset of different exploited information and thus not described in detail. Another review [23] provided and summarized the key technologies of laser-based active NLOS imaging, which however lacked the description of passive NLOS imaging, deep learning methods and new NLOS scenes. On the contrary, this article summarizes the latest active and passive NLOS imaging research based on physical methods and provides a detailed summary and analysis of the latest deep learning algorithms, including the advantages, types, challenges, and prospects. Besides, this article also summarizes several new types of NLOS imaging scenarios. Notice that such summarization can help readers have an overall understanding of NLOS imaging, which however cannot be found in the existing surveys [22], [23].

The remainder of this article is organized as follows. In Section II, we first introduce the existing work of active NLOS imaging from three aspects: the hardware, forward propagation model, and reconstruction algorithms. We also discuss the challenges and prospects of active NLOS imaging. Then, in Section III, the related work, including the hardware, forward model, and reconstruction algorithms, as well as the challenges and prospects of passive NLOS methods, are summarized respectively. In Section IV, we review the deep learning algorithms that have emerged in recent years from their motivations, network structures, loss functions, corresponding challenges, and prospects. The new NLOS imaging scenarios are discussed in Section V and conclusions are drawn finally in Section VI. Figure 2 clearly illustrates the relationship between the structure of this article and the current development of NLOS imaging technologies. It should be noted that imaging through a scattering medium [24], [25] does not belong to the NLOS imaging scope in this article.

Update (2022–2026). Since the original version of this survey (published in 2021), NLOS imaging has undergone rapid and dramatic advances across all fronts. On the hardware side, novel detector designs such as gradient-gated SPAD arrays [26] and superconducting nanowire single-photon detectors (SNSPDs) operating at infrared wavelengths [27] have extended the capability of time-resolved systems. On the algorithmic side, the introduction of transformer-based architectures [28], [29], state-space models [30], graph neural networks [31], and diffusion model priors [32] has significantly advanced the reconstruction quality and real-time performance of NLOS systems. Real-time NLOS video at 4 fps for room-sized dynamic scenes has been demonstrated [33], marking a landmark towards practical deployment. Beyond the classical optical setting, NLOS imaging has been extended to mmWave radar [34] and ultrasound [35], opening new application domains. New NLOS scenes have also emerged, including human pose estimation [36] and polarization-based scanning-free single-pixel imaging [37]. This updated survey incorporates these advances and provides an up-to-date reference for researchers in the field.

II. Active Methods

Active NLOS imaging employs a controllable light source (usually a narrow-band laser) and a detector to obtain reflected information, which is then used to reconstruct hidden scenes. This section introduces recent advances of active NLOS imaging from three parts: hardware devices, physical light transport models, and reconstruction algorithms. Besides, the challenges and prospects of active NLOS imaging are analyzed at the end of the section. Table I summarizes active imaging systems, where each row of the table lists the light source, sensor, information, and task for different NLOS imaging techniques.

A. Hardware Devices in Active Methods

A variety of detectors, from professional interferometers [46] and single-photon counters [39], [147] to ordinary cameras and even cell-phone cameras, have been used in NLOS imaging. Different detectors need to be combined with corresponding illumination sources to complete specific NLOS tasks. Here, we introduce such combinations to describe what hardware is used in active NLOS imaging.

1) Pulsed laser and high temporal resolution detector: NLOS imaging was first proposed in a time-resolved imaging work by Raskar and Davis [148], first theoretically evolved by Kirmani et al. [149] and experimentally demonstrated by Velten et al. [3]. All of these works [3], [148], [149] were in the context of active time-resolved transient imaging, also called "light-in-flight imaging" or "freezing light in motion" [23].

Transient imaging uses an ultrafast pulsed laser as the light source and a high time-resolved detector as the camera to measure the time of the photon arrival event in each pulse period, and then obtain the time distribution of photon events through the accumulation of multiple pulse periods, which then is converted to transient images. Considering that transient imaging is based on pulsed lasers and high-time-resolved detectors, the combination of pulsed laser and high-time-resolved detector is one kind of hardware used in active NLOS imaging.

Streak cameras When NLOS imaging was first experimentally achieved by Velten et al. [3], the time-resolved detector was a streak camera with extremely high temporal resolution. As an important optical time characteristic measurement detector, streak camera has been widely used in the experimental research of ultrafast physical processes such as laser fusion and high energy density physics [150]. When the ultrafast light signal passes through the slit of the streak camera, the photoelectric conversion is completed. These photoelectrons are accelerated and focused under the high-voltage electric field's action and enter the scanning system. Within the linear range of the voltage, the vertical distance between the photoelectron's position and the original movement direction is proportional to the time of which the photoelectron enters the scanning plate, hence completing the conversion of time information to space information. Finally, using the function between the optical signal's spatial information and the scanning speed of the streak camera, the temporal information of the optical signal is obtained, which can be reconstructed to get transient images.

The temporal resolution of the streak camera is remarkably high. In [3], the theoretical resolution could reach 2ps (limited by a finite temporal-point spread function of the camera, the effective temporal resolution was 15ps), and the most advanced streak camera at present can reach a temporal resolution of $200 \sim 300$ fs. However, streak cameras are too expensive (typically more than 70,000£), limiting the practical application of NLOS imaging. Therefore, later works attempt to use an inexpensive time-resolved detector to complete active NLOS imaging.

SPAD (Single Photon Avalanche Diode) In 2015, Buttafava et al. [39] demonstrated that SPAD (Single Photon Avalanche Diode) is feasible for NLOS imaging. In Geiger mode, a single photon may trigger an avalanche of about 10^8 carriers, which can be used as a single photon counter and get pretty accurate photon timing. The current commercial SPAD covers the wavelength range from $300nm$ to $1700nm$. Among them,

TABLE I
ACTIVE NLOS IMAGING

Reference	Illumination	Sensor	Information	Task
[3], [38]	Pulsed laser	Streak camera	Time of fight	3D reconstruction
[1], [2], [4], [6], [7], [14], [16]–[18], [30], [39]–[100]	Pulsed laser	SPAD	Time of fight	3D reconstruction
[12], [101]	Pulsed laser	SPAD	Integrated photon counts and occlusion	2D reflectivity reconstruction
[102], [103](long range), [104], [105](single point), [106], [107](few/time-multiplexed channels)	Pulsed laser	SPAD	Time of fight	Detection/ Tracking/ Identification
[108]–[111]	Pulsed laser	SPAD	Time of fight	Pose estimation
[15], [112]–[116]	Pulsed laser	SPAD array	Time of fight	3D reconstruction
[117]	Pulsed laser	SPAD array	Time of fight	Detection/ Tracking/ Identification
[46], [118], [119]	Pulsed laser	Interferometer	Coherence	3D reconstruction
[120]–[123]	Frequency-swept laser	Coherent FMCW interferometer	Beat frequency, phase, and Doppler	3D imaging / tracking / vibrometry
[5], [124], [125]	Modulated laser	ToF camera	Time of fight	3D reconstruction
[126]	Integrated	LiDAR	Depth and intensity	3D reconstruction
[20], [127]–[129]	Continuous laser	Conventional camera	Intensity	3D reconstruction
[130]–[132]	Pulsed / patterned multi-wavelength illumination	Single-pixel detector / time-resolved single-pixel camera	ToF, correlation, and colour channels	2D / 3D RGB reconstruction
[133]–[135]	Shaped pulsed / continuous laser	SPAD / camera	ToF, intensity, and polarization	High-resolution 2D/3D reconstruction
[136]	Continuous laser	Polarization-sensitive camera	Speckle correlation / Fourier amplitude	2D reconstruction
[137]	Spectrally modulated broadband illumination	Camera	Spectral speckle separation and phase retrieval	Multispectral 2D reconstruction
[138]	Polarization-multiplexed illumination	Polarization-sensitive camera	Polarization-component extraction	Multi-target 2D reconstruction
[139]	Polarization-encoded coherent illumination	Polarization-sensitive camera	Multiplexed speckle correlations	Multi-target 2D reconstruction
[140]–[142]	Continuous laser	Conventional camera	Intensity	Detection/ Tracking/ Identification
[143], [144]	Projector / incoherent illumination	Conventional camera	Intensity	Localization / posture classification
[145]	Incoherent light source (imaging side)	Conventional camera	Intensity	2D reconstruction
[146]	Speaker	Microphone	Acoustic "ToF"	3D reconstruction

Si SPAD corresponds to $300 \sim 1100nm$, Ge SPAD corresponds to $800 \sim 1600nm$, and InGaAs SPAD corresponds to $900 \sim 1700nm$ [151]. Therefore, most existing methods with visible light (e.g., $675nm$ in [1]) used Si SPAD (e.g., PDM series from Micro Photon Devices), while the recent $1.43km$ NLOS imaging [16] with infrared spectrum ($\sim 1550nm$) used InGaAs/InP negative-feedback SPAD.

Combined with time-correlated single-photon counting technology, SPAD can be used to obtain a histogram of photon arrival time of different points, i.e., transient images. In addition to the relatively lower cost (about 10000£), compared to streak cameras, SPAD has the following advantages. First, as a subset of APDs (avalanche photodiode), although the temporal resolution($\sim 20ps$) is lower than streak cameras, the SPAD has a higher quantum efficiency($\sim 70\%$) which means that it is more suitable for NLOS imaging scenes with weak effective signals. The work in [16] which used SPAD to complete an amazing $1.43km$ NLOS imaging is a typical example. Moreover, SPAD has been widely used in commercial LiDAR systems, and the SPAD array, which can avoid the mechanical raster scan process, has the potential to save scanning time and realize real-time data collection for active NLOS imaging.

Gradient-gated SPAD arrays for transient capture. Zhao et al. demonstrated a reconfigurable 256×128 SPAD imager for single-photon counting, LiDAR, and NLOS imaging [152]. Its 16-phase spatially graded time gates (with 0.12–3.8 ns intervals) and 39 ps timestamp quantization support selective capture of weak delayed returns. Unlike algorithmic transient inversion, this hardware contribution controls the photon arrival windows recorded and demonstrates NLOS acquisition experimentally.

Acquisition-parameter sensitivity for practical SPAD NLOS. Beyond accelerating only the inverse solver, Christ-

nacher et al. [94] systematically examined the measurement parameters that determine how quickly a usable transient data cube can be collected. Their study considers SPAD-pixel calibration, exposure time, the number of images accumulated per frame, camera field of view, and focus adjustment, and evaluates how these choices trade acquisition time against transient quality and ultimately reconstruction fidelity. Because the downstream examples are framed around established back-projection, f-k migration, and phasor-field reconstruction, this work complements later detector-timing calibration [93]: the former asks which acquisition settings should be spent in the first place, while the latter corrects timing-grid errors once SPAD-array data have been captured.

Time-sequential first-photon acquisition. Most transient NLOS work optimizes the forward model or inverse solver after accumulating a photon-arrival histogram. Li et al. instead examined the detector process and represented each measurement by time-sequential first-photon events [91]. Their TSFP likelihood uses the earliest detection in successive laser periods and was validated on synthetic and measured hidden scenes, obtaining reconstruction quality comparable to conventional histogram acquisition with substantially less collection time. This detection-aware branch is complementary to LCT, f-k migration, and phasor-field propagation: it reduces the photon and acquisition budget before those inverse operators are applied, and therefore provides an early link between single-photon statistics, photon-starved operation, and later real-time NLOS systems.

Detector-response inversion and compressive transient sampling. Two complementary studies made the acquisition bottleneck itself part of the reconstruction pipeline. Zhu et al. trained a convolutional network entirely on simulated transients and reconstructed 32×32 hidden images from only 8×8 scan points on measured data, corresponding to 6.25% of the dense spatial samples [97]. This result is an early precursor to later transient-completion and under-scanning networks, although its output is two-dimensional and its learned prior remains category-dependent. Wang et al. instead addressed the temporal response of the hardware: they modeled the measured histogram as a Poisson convolution with the calibrated instrument response function and applied iterative deconvolution before LCT or f–k migration [98]. Simulations and experiments showed successful recovery even when total timing jitter reached 1200 ps, a regime previously associated with roughly 200 ps timing. Together, these works clarify that faster NLOS need not rely only on a new back end: spatial scan density and detector timing blur can be reduced or inverted upstream while retaining established physical solvers.

From few-channel inference to time-multiplexed imaging. Li et al. showed that dense relay scanning is not necessary when the desired output is a coarse target descriptor: a rectangle-based multilayer perceptron maps only two or three transient channels to hidden-target width and height [107]. Zheng et al. then introduced delayed multi-beam illumination so that echoes from two relay points occupy separable windows in one single-pixel single-photon histogram, enabling centimeter-precision positioning and motion tracking with a single time-correlated channel [106]. The subsequent imaging system generalized this coding idea from target state estimation to hidden-scene reconstruction: delayed illumination channels multiplex several relay samples onto one detector timeline and halve the demonstrated scan count [92]. This trajectory distinguishes three levels of information recovery—coarse size, position and motion, and reconstructed appearance—while showing that acquisition complexity can be reduced by optical temporal coding rather than only by learned transient completion or detector arrays.

SPAD-array timing calibration and intrinsic time referencing. Parallel SPAD arrays avoid relay-wall raster detection but introduce detector-dependent temporal grids: manufacturing tolerances change the mean time-bin width across pixels and produce differential nonlinearity within each pixel. Spaett et al. measured approximately six-percent pixel-to-pixel mean-bin-width variation and 0.6-percent bin-level fluctuation, then corrected the resulting aggregation blur with a precomputed lookup table [93]. For NLOS reconstruction, they additionally infer a scan-specific absolute reference from the midpoint of intrinsic diffraction-wave and lens-flare peaks, removing unknown LOS path delays without an external calibration target. Experiments with two non-confocal SPAD-array systems and simulated confocal data show sharper phasor-field reconstructions, placing sensor calibration and temporal alignment upstream of the conventional inverse operator.

Poisson optimization and semantic scan allocation at low sampling rates. Two proceedings studies further reduce acquisition and reconstruction cost from complementary directions. Yang et al. formulate low-count transient inversion with a Poisson likelihood and solve it using GPU-parallel Nesterov-accelerated proximal gradients [95]; on measured $64 \times 64 \times 4096$ transients, the method retains favorable reconstruction quality at 1.56% sampling and reports approximately one second per frame, narrowing the speed gap between iterative reconstruction and direct LCT-like operators. Wang et al. instead use a target semantic contour to restrict the relay-wall scan area to one quarter and then apply neighborhood sparse sampling at one-fifth spatial density [96]. The latter is explicitly task-prior dependent, but it demonstrates that under-scanning can be optimized jointly in spatial support and density rather than treated only as uniform grid decimation.

Pulsed laser In transient imaging, the laser plays the role of illumination, triggering, and synchronization. Compared with time-resolved detectors, lasers have relatively low requirements and are more affordable than detectors. Nevertheless, it should still meet the following requirements. First, the pulse period of the laser should be adjustable to ensure that it can meet the requirements of different scene sizes without causing excessive noise. Second, in order to achieve accurate synchronization, the pulse width of the laser should be narrow with low jitter. Besides, its wavelength should be consistent with the response frequency of the detector.

Although the combination of pulsed lasers and time-resolved detectors suffer from long scanning time when collecting data, it is still the most popular active NLOS imaging camera method. It can obtain accurate time information and

the potential applications of scanning-free technologies, such as SPAD array [15], [112], [153].

Coherent FMCW NLOS ranging, imaging, tracking, and vibrometry. A coherent branch of optical NLOS replaces pulsed photon-arrival histograms with frequency-swept interferometric beat signals. Huang et al. calibrated FMCW LiDAR sweeps with an optical frequency comb, obtaining subpicosecond effective temporal resolution and sufficient coherent gain for submillimeter hidden-scene localization and three-dimensional imaging under strong ambient illumination [120]. Because the coherent measurement also preserves frequency and phase, the same platform recovers hidden-object vibration with dozen-hertz accuracy, extending NLOS sensing beyond static geometry. Ye et al. subsequently used comb calibration for snapshot multi-object tracking, reporting 2 mm three-dimensional position accuracy and 2 mm s^{-1} velocity accuracy without the accumulation required by many ToF or radar trackers [121]. Chen et al. pursued a lower-complexity vector-enhanced sensitive FMCW design based on laser-feedback interferometry, polarization-vector enhancement, and K-domain resampling, demonstrating better-than-32-μm absolute ranging and millimeter-level hidden three-dimensional imaging at an approximately 2.8 m NLOS distance [122]. The later dual-path, fixed-delay-fiber calibration and dynamic temporal phase subdivision system [123] further improved cost and sweep-linearity compensation while retaining 450-μm ranging resolution, millimeter-scale imaging, and operation beyond 8 klux. Together, these works establish coherent FMCW LiDAR as a complementary trajectory to SPAD-based transient NLOS: it directly measures range, phase, Doppler, and vibration with fine resolution, while trading the simplicity of photon counting for interferometric calibration and coherence control. A related coherent-ranging hardware direction uses multi-soliton optical microcombs. Zhi et al. demonstrated dual-microcomb precision ranging, photon-level detection, and a non-line-of-sight imaging proof of concept [154]. Their results emphasize power efficiency and interferometric ranging precision; the NLOS demonstration is an enabling hardware result rather than evidence of general volumetric hidden-scene reconstruction.

Scan-free SPAD-array acquisition and sub-pixel resolution recovery. Zhang et al. first removed relay-wall raster scanning by combining parallel SPAD-array acquisition with non-confocal time-to-space boundary migration [113]. The system reports 151-fps transient acquisition and 19-fps end-to-end hidden-scene imaging, while a plug-in super-resolution stage reduces the detector requirement from 32×32 to 8×8. Their subsequent PSM-NLOS method uses DMD pixel-shift modulation to synthesize sub-pixel relay samples and explicitly models spatial and temporal signal broadening [114]; the reported lateral resolution improves from approximately 7 cm to 1 cm while the measurements remain compatible with LCT, virtual-wave, and boundary-migration reconstruction back ends. The later SCBSF-NLOS formulation models scan-free non-confocal measurements using a three-dimensional blur kernel and spatial-correlation resampling [115], reporting approximately 2 cm lateral resolution and 5-fps dynamic reconstruction with a 16×16 detector. Together, this lineage shifts real-time transient NLOS from acquisition parallelism, through physical sub-pixel sampling densification, to model-based computational resampling.

Edge-resolved transient imaging. Rapp et al. combined active transient sensing with a ubiquitous vertical edge occluder [155]. A pulsed laser scans a short semicircular arc around the edge while a gated SPAD records photon-arrival histograms; differencing adjacent histograms suppresses the approximately static visible-scene and background terms and isolates the response of successive angular wedges. The resulting edge-resolved transient measurements provide range from time of flight and azimuth from occlusion coding, enabling a 2.5D plan-view-plus-height reconstruction over a 180° field of view from only 45 illumination positions. This hybrid of active ToF sensing and corner-camera occlusion substantially reduces the aperture and number of measurements required for room-scale hidden-scene recovery. Saunders et al. [156] subsequently analyzed acquisition-time, illumination, background, reflectivity, range, and azimuth effects, optimized angular dwell time for deeper hidden directions, and quantified coaxial-approximation error.

Foveated acquisition combining transient and correlation cues. Dave et al. proposed a hybrid NLOS capture concept combining transient imaging with correlography to mitigate the field-of-view versus spatial-resolution tradeoff [157]. This early result appeared as a COSI conference digest rather than a full journal article, and is a precursor to later adaptive and multi-resolution capture strategies.

Coherent indirect-imaging correlography and its resolution operator. Balaji et al. derived a scene-dependent imaging operator for indirect imaging correlography (IIC), which recovers the hidden object's albedo autocorrelation from coherent indirect measurements [158]. Their analysis explicitly relates attainable spatial resolution to target position and pose, and a self-supervised autocorrelation inversion reconstructs features on the order of 250 μm at a 1-m standoff. Unlike pulsed ToF geometry inversion, this coherent branch measures correlation structure and must resolve autocorrelation ambiguities during image recovery.

Recent SPAD Advances. Specialized SPAD array designs tailored to NLOS imaging have emerged in recent years to address detector saturation from strong direct-path reflections. Riccardo et al. developed a fast-gated 16×16 SPAD array with 16 on-chip 6 ps time-to-digital converters that can selectively gate out the direct-bounce photons while recording the weaker NLOS signal [159]. Taking this further, gradient-gated SPAD arrays apply a spatially varying gate delay across the pixel array, providing improved suppression of saturating photons with minimal hardware overhead [26]. Zhu et al. used nonlinear frequency conversion for picosecond-gated single-photon detection, isolating weak around-corner returns for position, surface-profile, and vibration-frequency sensing [160]. Beyond silicon SPADs, Feng et al. demonstrated active NLOS imaging at 1560 and 1997 nm using a fractal superconducting nanowire single-photon detector (SNSPD), with wavelet-domain denoising before light-cone-transform reconstruction [27]. These developments highlight complementary strategies for low-photon

acquisition: temporal rejection of background and broader infrared sensitivity.

Compact eye-safe nanosecond SPAD-array localization. Albert et al. replace bulky picosecond sources with inexpensive nanosecond laser diodes and observe the relay wall in parallel using a SPAD array with integrated timing [116]. Two off-axis illumination positions mitigate first-photon saturation, while matched temporal filtering and ellipsoidal backprojection localize the hidden target despite the extended pulse width. This result complements miniaturized TCSPC and scan-free arrays by showing that deployment cost and eye safety can be improved at the illumination source, provided the acquisition geometry and temporal filtering are co-designed.

Miniaturized TCSPC electronics. Compact timing electronics are becoming a deployment bottleneck rather than only detector sensitivity. Wu et al. integrated a four-channel TCSPC module around a commercial TDC ASIC, providing a 10 ps bin size and 27.4 ps minimum RMS timing resolution [54]. In a 1550 nm confocal single-photon NLOS experiment at a 5 m sensor distance and 1 ms pixel dwell time, the module supported 6.3 cm lateral and 2.3 cm depth resolution. This work moves the Velten–phasor-field sensing chain toward compact handheld or vehicle-mounted implementations by miniaturizing the timing backend rather than changing the inverse model.

Timing super-resolution by laser-pulse multiplexing. Detector jitter and histogram bin width ordinarily impose a hard axial-resolution floor. Miao et al. instead exploit deterministic offsets between repeated laser pulses to build a sub-bin modulation matrix and recover fine transient histograms with non-negative least squares [78]. Their laser-pulse-multiplexing system reconstructs 64 ps transients from hardware with a native 704 ps single-photon timing resolution, remains useful when only about 5% of relay positions are sampled, and extends to non-confocal arrays. This shifts part of the temporal-resolution burden from specialized detectors to calibrated computational modulation.

Weak-return extraction without photon accumulation. Li et al. addressed the low-photon NLOS tracking regime in which naively accumulating transient histograms also accumulates noise [161]. They evaluated two temporal-correlation approaches: dual-detector weak-echo extraction and a sequential-histogram pipeline for following moving hidden targets. Their experimental trajectories demonstrate detection and tracking rather than dense 3D surface reconstruction, and anticipate later arrival-window selection and detector-aware low-SNR NLOS processing.

Adaptive photon-arrival windowing under high ambient light. Miao et al. address the acquisition-side failure mode in which a weak three-bounce response is overwhelmed by temporally broad background detections [62]. AW-NLOS first models the arrival statistics of signal and noise, then amplifies correlated weak events through spatio-temporal pixel clustering so that a matched filter can estimate a target-dependent temporal support. Per-pixel short-duration range windows reject most ambient photons before total-variation-regularized hidden-volume reconstruction. Measured experiments report recovery at an SBR of 2.12 and only 0.02 signal photons per pixel, together with an approximately 0.5 SSIM improvement over conventional processing at 42 mW illumination. This work forms an acquisition-aware bridge between photon-efficient single-photon statistics and the subsequent detector/regularizer co-design for all-day, long-range NLOS.

All-day SPAD NLOS imaging. Yin et al. co-designed detector selection and reconstruction for active NLOS under extreme daylight [63]. A SPAD model incorporating photon-detection efficiency and timing jitter motivates the use of a Si-SPAD, while phase-congruency-based structured ϵ-regularization separates coherent object structure from Poisson and background noise in the virtual phasor field. The resulting system reports an 18-fold SBR increase over InGaAs-SPAD capture and reconstructs targets at 200 m under 94,314 lx ambient illumination, connecting phasor-field robustness with hardware-level detector optimization for continuous outdoor operation.

Multi-surface sub-resolution transient decomposition. Dense hidden geometry can produce temporally overlapping returns whose separation is below the native impulse-response width. Wei et al. modeled multi-surface transient reflections as Gaussian mixtures, combined L_1–L_2 waveform deconvolution with constrained modified Levenberg–Marquardt fitting, and deposited the separated components into individual transient volumes before LCT reconstruction [67]. The reported centimeter-scale discrimination substantially improves both axial and lateral separation of nearby surfaces, showing that resolution can be increased in the transient-signal domain before applying a conventional volumetric inverse.

Compact kilometer-range NLOS imaging. Long-range outdoor NLOS is limited jointly by the cubic-bounce photon budget, daylight background, detector dead time, and the bulk of laboratory timing hardware. Zeng et al. co-designed optimized transmit/receive optics, scan-synchronized adaptive gating, a gated InGaAs/InP SPAD, and fast control electronics in a compact prototype [69]. The system reports daylight imaging at a relay-wall distance of approximately one kilometer, roughly 4 cm spatial resolution, and 2 fps reconstruction of simple moving targets. This result advances the earlier kilometer-range demonstration from proof of range toward a portable, high-efficiency instrument and shows that practical progress can come from hardware-level photon collection and gating as much as from a new inverse algorithm.

Scanning-free laser reflective tomography. Long-range transient systems usually trade spatial resolution against the number of relay-wall scan positions. Wang et al. instead use the diffuse wall as a natural beam expander, record third-bounce light with a single detector, and recover the hidden target from angular measurements generated by laser reflective tomography [72]. Indoor experiments report roughly twice the resolution and $91\times$ the acquisition speed of point scanning; an outdoor experiment reconstructs targets at 3.3 km with better than 3 cm resolution in about three minutes. This result removes relay-wall raster scanning from the kilometer-scale branch rather than merely accelerating its inverse.

Active Focusing for High Resolution. An entirely different

paradigm uses wavefront shaping rather than time-of-flight to realize NLOS imaging. Cao et al. proposed the UNCOVER (Unseen Non-line-of-sight Casted Optical aperture Visibility-Enhanced Return) method, which actively focuses light through the scattering relay wall onto the hidden target using iterative wavefront shaping, then reconstructs by raster scanning the focus [162]. This approach achieves a distance-to-resolution ratio of approximately 900, surpassing conventional ToF-based NLOS imaging by more than an order of magnitude. At a target distance of 0.55 m, UNCOVER achieves a spatial resolution of ~ 0.6 mm, demonstrating the potential of active-optics-based NLOS imaging for millimeter-scale resolution.

From planar aperture expansion to vectorial digitelligent focusing. A complementary high-resolution lineage modifies the physical illumination and relay aperture before reconstruction. Zhao et al. inserted a liquid-crystal planar angle-magnification element into a transient NLOS system, enlarging the effective relay-wall scan aperture while using measurement correlation for sparse sampling that reduced acquisition time by more than 20% without sacrificing the demonstrated resolution [133]. Wang et al. then formulated a vector-optical-field reflection model that selects illumination angle, incident polarization, and received polarization, converting polarization diversity into multi-view measurements with stronger echoes for hidden-object reconstruction and recognition under low SNR [134]. Guo et al. closed the loop with vectorial digitelligent optics: feedback jointly optimizes phase and polarization to refocus through a random scattering wall, after which the focus is raster-scanned within the optical-memory-effect range [135]. Their measured 0.40 mm resolution at 0.35 m approaches the diffraction limit. Together, these papers shift part of the NLOS inverse problem into optical co-design—aperture magnification, polarization-aware view selection, and closed-loop vector wavefront control—rather than relying only on a more powerful LCT, f–k, or phasor-field back end.

From time-resolved RGB to compressive multi-wavelength single-pixel NLOS. Single-pixel NLOS has developed along a complementary coding trajectory to relay-wall raster scanning. Musarra et al. first combined high-efficiency time-resolved single-pixel detection with spectral channels to demonstrate sub-second three-dimensional RGB imaging of a hidden scene [130]. Li et al. later considered an active patterned-illumination system in which second-order correlation supplies the initial hidden estimate and a Fourier-domain inverse-diffraction stage compensates propagation blur [131]. Their Hadamard-coded implementation improved simulated and measured reconstruction and reported an SSIM of 0.9286 at full sampling. The subsequent multi-wavelength formulation used TV-regularized compressive sensing in RGB space to recover spatial and colour information at a reported 29% compression rate [132]. Together, these studies move single-pixel NLOS from time-resolved full-colour proof of concept toward inexpensive spatial coding and undersampled multispectral recovery; the later polarization-speckle method replaces explicit spatial masks with polarization-generated random illuminations.

Polarization-speckle single-pixel imaging. Zhou et al. introduced polarization as an illumination-coding degree of freedom for scanning-free NLOS imaging [77]. Different incident polarization states produce diverse speckle patterns after interaction with the rough relay surface, and a single-pixel detector measures the corresponding hidden-scene responses. The system achieves millimeter-scale keyhole reconstruction and uses the angular memory effect for noninvasive calibration, expanding single-pixel NLOS beyond spatial-mask and temporal-coding designs.

2) Modulated light source and ToF camera: In addition to the pulse-based ToF measurement, by encoding the ToF into phase measurement, a ToF camera combined with a modulated light source has also been proposed to obtain the light travel time to complete NLOS imaging [5], [124]. Compared with pulse-based photon-level detectors, ToF cameras have some obvious advantages. First, it can complete data collection without scanning, thereby reducing data collection time. Second, its price is much lower (about 1000£) than streak camera and SPAD. However, due to the influence of frequency aliasing, the measurement distance of the ToF camera is limited (no more than 10m). Besides, due to the longer exposure time, the ambient noise of the ToF camera is usually greater, limiting the temporal resolution to ns level with cm imaging resolution. However, in practical active NLOS imaging scenarios, the modulated light source and ToF camera are on the same side, which would increase the direct bounce signal from the wall and decrease the hidden signal (the third-bounce light), resulting in poor SNR. Therefore, in general, the ToF camera is more suitable for NLOS scenes that require real-time performance without high imaging quality.

Fang et al. examined non-Lambertian relay surfaces using a lower-cost modulated ToF camera [163]. A genetic algorithm estimates an approximate scattering kernel from calibration targets, after which Lucy–Richardson deconvolution restores hidden-object depth from degraded measurements through polypropylene and acrylic relay surfaces. This calibration-driven alternative reduces hardware requirements, although its estimated kernel is material-specific and the genetic optimization can be time-consuming.

Caligiuri et al. demonstrated that the lower-cost modulated-light branch can be pushed to commodity indirect-ToF hardware: NIGHT uses an off-the-shelf iToF sensor without additional time-resolved instrumentation and learns a virtual-mirror representation that maps multipath measurements to hidden-scene depth [125]. Together with a purpose-built synthetic dataset, the method shifts consumer-ToF NLOS from analytic phase-based models toward learned reconstruction, although its reported validation remains primarily synthetic.

3) Active light source and interferometry: Willomitzer et al. demonstrated high-resolution, wide-field NLOS reconstruction using synthetic-wavelength holography, recovering hidden-scene information from coherent optical interference rather than requiring picosecond-resolved photon histograms [164]. This method extends earlier superheterodyne interferometric demonstrations [118] toward practical coherent hidden-scene imaging.

Pulse-based detectors and coherent-based ToF cameras are

the two main active NLOS imaging cameras. However, their resolution is strongly restricted, difficult to break through the $\sim mm$ level. The resolution of the pulse-based detector is limited by the pulse width of the laser ($\sim ps$), while the coherent ToF camera is limited by the modulation frequency ($\sim ns$). To achieve higher resolution ($\sim \mu m$), interferometers have also been applied to NLOS imaging. Unlike those ToF-based cameras that can only be used for active imaging as mentioned above, the interferometer can not only be combined with narrowband LEDs or coherent light for active imaging, but also be used with ambient light for passive imaging. For active methods, Xin et al. used the imaging device in [165] to complete the NLOS imaging of coin-sized scenes and achieved femtosecond scale resolution [46]. Willomiitzer et al. utilized lasers with two wavelengths to complete high-resolution NLOS imaging based on superheterodyne interferometry (SHI) with a resolution of about $50\mu m$ [118]. Some studies performed passive NLOS imaging by applying narrowband or ambient illumination to hidden objects, which will be introduced in Section III. Compared with streak cameras, SPAD, and ToF cameras, the interferometers can achieve higher resolution but with the disadvantages of higher hardware complexity and cumbersome calibration.

Cost-effective FMCW interferometry. Liang et al. developed a coherent NLOS ranging and imaging system based on frequency-modulated continuous-wave interferometry rather than pulsed photon timing [123]. A polarization-maintaining optical-delay fiber supplies a fixed Mach–Zehnder calibration path, and dynamic temporal phase subdivision estimates and compensates laser-sweep nonlinearity without an optical frequency comb. The reported 1.8 ps effective temporal resolution supports 450 μm ranging resolution, 1 mm axial and 2.4 mm lateral imaging resolution at a 110 cm hidden distance, including measurements above 8 klux ambient illumination. This adds a cost-conscious coherent branch between superheterodyne interferometry and SPAD ToF systems.

4) Laser and conventional camera: In this paper, a conventional camera refers to a camera that uses conventional intensity sensors, such as CCD or CMOS array, which cannot record spatial/temporal coherent information or ToF information. Since transient images cannot be measured, imaging using traditional cameras is often called steady-state imaging. Because conventional cameras can only record intensity information, they are mainly used in passive NLOS imaging. However, because the conventional camera has the advantages of not requiring scanning and low cost, Chen et al. completed RGB active NLOS imaging using lasers with different wavelengths for illumination and exploiting conventional cameras to collect intensity information [127]. Due to the lack of distance information, it is not easy to complete high-precision three-dimensional reconstruction only using conventional cameras.

Low-cost NIR steady-state raster scanning. Roueinfar and Salmanian demonstrated a deliberately simple active NLOS acquisition baseline using a 500 mW, 808 nm laser mounted on a pan–tilt unit to raster scan the relay wall and a conventional NIR camera to record the three-bounce steady-state response [128]. Experiments on three hidden targets quantify image error with MSE and RMSE. The method does not recover transient depth or introduce a new inverse operator, but it shows how near-infrared illumination and commodity spatial capture can lower hardware complexity. Although uploaded to arXiv in July 2026, the work had already appeared in the 2025 IEEE International Conference on Electrical Engineering; the survey therefore cites the final conference version.

Single-shot polarization differential correlography. Liu et al. introduced PDC-NLOS, a steady-state active system that illuminates the relay wall with independently polarized laser speckles and forms a polarization-difference correlogram before phase retrieval [129]. Polarization modulation replaces repeated mechanical scanning, while differential speckle statistics add redundancy against vibration and coloured perturbations. Experiments on horizontal and vertical stripe targets demonstrate millimeter-level resolution, and the reported average SSIM of about 0.76 improves on the authors' earlier chromato-axial single-shot correlography baseline. This work connects steady-state NLOS, speckle correlation, and polarization coding, providing a low-complexity high-resolution alternative to transient time-of-flight inversion.

Ptychography-enhanced correlography. Liu et al. subsequently addressed the incomplete Fourier-amplitude sampling and phase ambiguity that limit conventional NLOS correlography by introducing PEC-NLOS [136]. A continuous-wave laser scans overlapping indirect speckle probes across the relay wall, while a polarization-sensitive camera records local correlation measurements; ptychographic phase retrieval jointly refines the hidden object and probe. A geometry-derived three-dimensional data-correction stage uses non-invasive positional priors to compensate probe and scan distortion. Unlike the scanning-free polarization methods above, PEC-NLOS deliberately trades additional overlapping measurements for redundancy and conditioning, extending steady-state correlography from simple sparse targets to high-spatial-frequency, high-entropy hidden scenes and establishing a bridge between NLOS speckle imaging and coherent-diffraction-style ptychographic reconstruction.

5) LiDAR: Although most existing works have used separate light sources and detectors, recent work [126] exploited a commercial LiDAR, which integrated light source and detector, to complete NLOS imaging. For the imaging area on the relay surface that is close to LiDAR, most of the collected information is the direct reflection from the relay surface. However, for the area on the relay surface that is closer to the hidden object, the collected information mainly encodes the shape of the hidden object, which can be utilized to restore 3D shapes. The existing commercial LiDAR can only provide point cloud output, instead of transient images. Therefore, the classic active NLOS reconstruction methods, such as FBP [3], LCT [1] and phasor field [2], cannot currently be used in LiDAR-based systems. [126] used deep learning to fuse point cloud information and reflection intensity information to complete the reconstruction. Compared with separate laser and time-resolved detectors, integrated LiDAR have lower cost and faster imaging speed, with the cost of poor imaging detail.

To replace the separate laser and cameras in NLOS systems, existing commercial LiDAR needs to solve the problem of crosstalk between multi-channel lasers under NLOS conditions and provide original temporal signals rather than point cloud after processing.

B. Forward propagation model

The forward propagation model of active NLOS imaging aims to establish the imaging model of measurement. The current imaging models are mainly divided into ToF-based imaging models and wave-based imaging models. The ToF-based imaging model [1], [3] uses geometric optics to establish the model with the flight distance and surface albedo or normal direction as constraints. The wave-based imaging model [6], [7] mainly uses wave optics to construct the propagation and reflection of waves with boundary conditions.

SPAD transient forward-data simulation. Scholes and Leach developed a path-tracing-inspired forward simulator for direct time-of-flight SPAD NLOS measurements [166]. Rather than reconstructing hidden geometry, the framework samples a room-scale environment in Unreal Engine and combines path lengths, material reflectance, occlusion, detector response, and secondary scattering to predict time-dependent photon-detection likelihoods. This provides a sensor-aware complement to analytical three-bounce models and differentiable transient renderers, supporting controlled studies of multipath contamination, acquisition geometry, and simulated training data without requiring new physical captures.

1) ToF-based model: The transient-based forward model established using time of flight as a constraint is a classic active NLOS imaging model. This type of model is based on point-by-point scanning, leading to a long scanning time. Therefore, there are many methods proposed to improve the scanning mechanism, represented by the confocal setting [1]. Besides, 3D reconstruction is an ill-posed problem, and using partial occlusion to add additional constraints is also a representative improvement [12]. Therefore, we review the general ToF-based NLOS image formation model, followed by the confocal imaging and occlusion-based models.

General NLOS imaging model. The imaging model fits the optical transport process by expressing the measurement data τ as a function of the hidden object space $\Omega \in \mathbb{R}^3$. As shown in Fig. 1-(a), the emitted laser first irradiates at the illumination point μ on the wall, causing the first bounce. Then, the second bounce occurs on the surface of the hidden object. Finally, the detection point μ' on the wall is collected after the third bounce. For a given illumination point μ, the collected signal τ by the detection point μ' at time t can be expressed as [1]

$$\tau(\mu,\mu',t) = \int_{\Omega} \rho(s) F(\mu \to s \to \mu') R(\mu \to s \to \mu') B(\mu \to s \to \mu') ds \tag{1}$$

where s represents a point on the hidden object, and $\rho(s)$ represents the albedo of point s. F refers to the optical transfer process from the illumination point μ to the detection point μ', reflected at the surface of hidden object s. R means the amplitude attenuation. B represents the bidirectional reflectance distribution function (BRDF).

According to geometric optics, with the speed of light c, the optical transport process F should satisfy $\|\mu - s\|_2 + \|\mu' - s\|_2 = ct$, which can be represented by delta function $\delta()$. Please note that this $\delta()$ function is established based on two assumptions: (1) Only three-bounce light is considered; (2) There is no interreflection in the hidden scene. The attenuation is inversely proportional to the square of the distance. Therefore, assuming that light scatters isotropically, Eq. (1) can be re-written as

$$\tau(\mu,\mu',t) = \frac{1}{\|\mu - s\|_2^2 \|\mu' - s\|_2^2} \int_{\Omega} \rho(s) \delta(\|\mu - s\|_2 + \|\mu' - s\|_2 - ct) ds \tag{2}$$

Equation (2) can be discretized into a linear transform of equations

$$\tau = \mathrm{A}\rho \tag{3}$$

where A is the optical transport matrix, which maps the hidden scene ρ to measurement τ. In the general NLOS setup, ρ has three spatial dimensions of x, y, and z, and τ has 5 dimensions (4 spatial dimensions and 1 time dimension). Therefore, the calculation complexity of matrix A is extremely high. Besides, it is an under-constrained problem inherently. In addition, too many scanning points lead to a long scanning time. In order to solve these problems, many new methods have been proposed (see Sec. II-E2 for details). The confocal setup [1] and adding additional obstacles [12], [41] are frequently used methods in recent studies, as introduced in the following.

Confocal NLOS imaging model Through a beam splitter, the laser and the detector can be placed coaxially, which means "confocal", as shown in Fig. 1-(b). In the confocal NLOS imaging model, the illumination and detection points are same during each scan, i.e., $\mu = \mu'$, which can be substituted into Eq. (2)

$$\tau(\mu,t) = \frac{1}{\|\mu - s\|_2^4} \int_{\Omega} \rho(s) \delta(2\|\mu - s\|_2 - ct) ds \tag{4}$$

O'Toole et al. proved that under confocal conditions, the imaging model could be further converted to the convolution of the hidden object albedo ρ after resampling with the system response function, which can be solved quickly by Fourier transform. However, under the confocal setting, the SPAD detector receives direct reflections, which may aggravate the pile-up effect [167]. In order to alleviate the influence of direct reflections, [16] and [1] illuminate and image two slightly (to avoid affecting the confocal imaging model) different points on the relay wall.

Occlusion-based active NLOS imaging model For the occlusion-based imaging model, a visibility term $V(\mu,\mu',s)$ describes whether the illumination and observation paths connecting relay-wall points to a hidden point remain unblocked. Thrampoulidis et al. showed that the binary masks cast by natural hidden-scene occluders make the rows of the optical transport matrix substantially more diverse, improve its

conditioning, and can therefore replace fine time-of-flight resolution when recovering a static hidden reflectivity map [12]. Partial-occlusion formulations further allow $V(\mu, \mu', s)$ to vary continuously while jointly estimating surface geometry and normals [41].

Xu et al. translated this principle into a photon-efficient room-scale experiment [101]. Their forward model accounts for three-bounce attenuation, known occluder geometry, background light, and the binomial statistics of a non-photon-number-resolving SPAD. By integrating the third-bounce detections rather than using their arrival-time structure, the system reconstructs two-dimensional hidden-wall reflectivity with roughly sixteen times fewer detected photons than the preceding Gaussian-noise treatment. This branch of active NLOS thus established environmental occlusion as a computational aperture rather than merely a nuisance and anticipated later edge-resolved and passive computational-periscopy methods.

It should be noted that Eq. (2) only considered 3-bounce diffuse reflections. Since the number of photons decreases exponentially with the order of diffuse reflections, 3-bounce reflections is sufficient for current NLOS imaging. The limitations of 3-order reflections are discussed in detail in [13].

Time-gated polarization beyond the missing cone. Feature-visibility analysis shows that time of flight alone cannot distinguish all hidden surface positions and orientations, leaving a null or "missing-cone" region even when photon timing is ideal. Pueyo-Ciutad et al. augmented picosecond transient capture with polarization-resolved measurements and formulated a polarized NLOS transport model whose relay-wall Fresnel response encodes propagation direction [85]. Their inversion prunes candidate hidden points using this directional information and reconstructs surfaces that LCT, f–k migration, and conventional phasor fields cannot observe. Simulated and measured experiments also retain fine detail from a fraction of the relay samples. This work changes polarization from a passive contrast cue or noise filter into an additional active transport dimension that directly improves geometric observability.

Polarization-speckle single-pixel NLOS. Zhou et al. introduced a complementary scanning-free route in which changing the incident polarization produces diverse speckle illuminations after reflection from a rough relay wall [77]. A single-pixel detector records the hidden-scene responses, while the angular memory effect supports noninvasive calibration. The demonstrated keyhole system recovers millimeter-scale detail. Unlike time-gated polarimetry, which augments transient transport with directional observability, polarization-speckle modulation uses polarization diversity as the illumination code itself and opens an active single-pixel branch without spatial relay scanning.

2) Wave-based model: The wave-based NLOS imaging model regards the forward imaging model as the propagation of waves from hidden objects to detectors in 3D space. Lindell et al. recorded the light field in space as $\Psi(x, y, z, t)$, then the forward model can be converted to wave propagation from $\Psi(x, y, z, t = 0)$ to $\Psi(x, y, z = 0, t)$. Specifically, The time-dependent field Ψ can be written as a superposition of plane waves [7]

$$\Psi(x, y, z, t) = \iiint \Phi\left(k_x, k_y, k_z\right) e^{2\pi i(k_x x + k_y y + k_z z - ft)} \, \mathrm{d}k_x \, \mathrm{d}k_y \, \mathrm{d}k_z \tag{5}$$

where the wave vector $\mathbf{k} = 2\pi \cdot (k_x, k_y, k_z)$ indicates the direction of propagation of the independent plane wave [7]. $f = c\sqrt{k_x^2 + k_y^2 + k_z^2}$ means the relationship between the wave vector $\mathbf{k}$ and frequency f when the speed of light is c. The function Φ represents the amplitude and phase of each plane wave at $t = 0$. After that, f-k migration is used to solve the problem in the frequency domain through fast Fourier transform (FFT) and Stolt interpolation (see Sec. II.D.II-D2 for details).

The phasor field methods [2], [6], [119], which have attracted widespread attention recently, regard the NLOS imaging as a diffraction-based LOS (line-of-sight) optical imaging problem. The projector function and diffraction function are determined by selecting a suitable LOS template, thereby directly reconstructing the hidden scene. Although based on wave propagation, these methods are all suitable for ToF measurement, making it easy to collect data and apply the model to public NLOS imaging datasets.

From paraxial theory to physical and nonparaxial phasor optics. Dove and Shapiro formalized phasor-field imaging under paraxial wave optics and introduced two-frequency spatial-Wigner-distribution (TFSWD) propagation primitives to carry directional information associated with occlusion and specular transport that is not represented by the P-field alone [168]. Reza et al. then experimentally demonstrated P-field interference, focusing, and imaging with modulated optical sources and detectors, validating that the virtual-wave analogy is observable rather than only a post-processing construction [169]. Two subsequent results extended this foundation in complementary directions: ordinary lenses were shown to physically focus and project P-fields through intervening diffusers [170], while Rayleigh–Sommerfeld formulas for both P-fields and TFSWDs relaxed the Fresnel approximation for wide-angle reflective NLOS geometries [171]. Together, this lineage connects Liu's computational phasor-field camera to a broader physical-optics and phase-space account of what virtual waves preserve, how they propagate, and where paraxial reconstruction ceases to be valid.

Forward and inverse diffraction in phasor fields. Garcia-Pueyo and Muñoz revisited the apparent paradox that phasor-field reconstruction solves an inverse hidden-scene problem with a forward Rayleigh–Sommerfeld propagation operator [84]. By exploiting diffraction reciprocity and the unitary structure of the propagation operator, they interpret conventional phasor fields as inverse diffraction and derive an explicit Inverse Phasor Fields formulation. The analysis identifies conditions under which diffraction inversion becomes well posed and introduces a matrix-rank quality metric linked to the Rayleigh lateral-resolution criterion. This contribution shifts the virtual-camera metaphor toward an operator-theoretic account of conditioning, explaining when phasor propagation can

recover a hidden scene and how relay aperture, wavelength, and propagation distance govern information loss.

Confocalizing SPAD-array non-confocal measurements. A practical route to faster transient acquisition is to replace rastered point detection with a SPAD array, but fixed point illumination and array detection naturally produce non-confocal ellipsoidal measurements for which the fastest confocal inverses are not directly applicable. Zheng et al. addressed this acquisition–inversion mismatch with sectionalized-ellipsoid interpolation (SEI) [73]. The method partitions each non-confocal ellipsoidal constraint into spatial slices and maps those slices into an emulated semi-confocal histogram, using spatial information to refine the effective temporal sampling beyond the detector-bin spacing. Simulation and measured experiments show improved separation of closely spaced structures relative to filtered backprojection and conventional normal-moveout correction. More importantly, the conversion makes high-throughput SPAD-array measurements compatible with mature confocal back ends such as LCT and f–k migration. This work therefore bridges acquisition-side parallelism and reconstruction-side fast transforms, and provides a complementary 2024 route to the direct non-confocal frequency-domain inversion discussed next.

Non-confocal spherical-slice and reference-function frequency-domain inversion. Yu et al. introduced a spherical-slice transform that maps non-confocal transient measurements from spatial and temporal frequency to space and time [74]. The formulation directly addresses the lower resolution and scene dependence of earlier non-confocal reconstructions, suppressing artifacts, shape distortion, and position offset while reaching several-hundred-millisecond reconstruction on a GPU for a 32×32 PF32 photon-array measurement. This 2024 result forms a useful frequency-domain precursor to two later developments from the same non-confocal branch: virtual modulated range migration uses super-resolved histograms to recover resolution from coarse timing hardware [100], whereas reference-function phase compensation transfers a single-input/multiple-output frequency-domain formulation to optical NLOS and further reduces phase-induced distortion and artifacts [75]. Together, the sequence shows how non-confocal transient NLOS has moved from direct spherical-slice inversion toward timing-resolution recovery and explicit phase compensation without reverting to dense confocal scanning.

Nonuniform and scaled Fourier sampling. FFT-accelerated LCT, f–k migration, and phasor-field methods conventionally couple acquisition and reconstruction to uniform Cartesian grids. Sultan et al. showed that practical relay measurements are often spatially oversampled and introduced modified Fourier operators that preserve fast wave-based inversion while relaxing this coupling [59]. Their NUFFT formulation reconstructs from sparse, irregularly sampled relay regions of arbitrary shape and evaluates the hidden volume at arbitrary locations, while a scaled FFT enlarges the reconstructed volume without increasing the number of stored samples. This development shifts Fourier-domain NLOS from fixed-grid acceleration toward acquisition-aware and memory-aware sampling without sacrificing FFT-like asymptotic complexity. Sultan et al. subsequently introduced trapezoidal-grid reconstruction using scaled Rayleigh–Sommerfeld diffraction (RSD) [172]. By varying lateral voxel spacing with depth, Scaled RSD better matches the distance-dependent spatial resolution of phasor-field imaging and avoids oversampling distant hidden regions on a uniform Cartesian lattice. This ICCP 2026 extension links nonuniform acquisition to resolution-aware volumetric reconstruction while retaining FFT-style per-voxel efficiency.

Unified transport and operator formulations. Sultan et al. revisited the transient image-formation model and derived a more complete light-transport formulation that places time-of-flight and shadow/occlusion NLOS measurements in a shared framework [86]. Redo-Sanchez et al. then used Dirac notation to express backprojection, f–k migration, and phasor-field propagation as related linear operators, clarifying when Rayleigh–Sommerfeld diffraction acts as the virtual-wave propagator and which differences arise from filtering or sampling rather than distinct physical assumptions [87]. Together, these works strengthen the theoretical bridge between geometric transient inversion and wave-based reconstruction instead of treating each fast solver as an isolated algorithm. A complementary mathematical treatment by Qiu et al. studies directional confocal NLOS as recovery of a vector-valued albedo field [173]. The measured spherical means determine the field's divergence but leave its divergence-free component unobservable; an explicit Fourier–sine inversion reconstructs the irrotational component, and partial relay-aperture data admit local uniqueness under the stated model assumptions. This identifiability analysis clarifies which hidden-scene information is recoverable from directional transient measurements, rather than relying on reconstruction fidelity alone.

Zero-phase geometry and virtual transport matrices. Luesia-Lahoz et al. showed that zero crossings of virtual phasor-field phase provide a direct hidden-depth observable with high precision and substantial robustness to measurement noise [88]. This phase-geometric cue complements amplitude-based virtual-wave focusing and inverse-diffraction conditioning. In a related expansion of the phasor-field camera, Marco et al. constructed virtual light-transport matrices between hidden projector and camera positions, enabling relighting and separation of direct, first-order indirect, and higher-order transport rather than recovering only a static hidden albedo volume [89]. The latter work was formally published at ICCV 2021; the repository therefore uses the final conference record rather than its earlier arXiv-only label. Royo et al. later developed Cascaded NLOS [174], extending phasor-field virtual transport from a single relay surface to virtual projector–camera systems on hidden secondary walls. The resulting fourth- and fifth-bounce pathways enable reconstruction of challenging target orientations and scenes hidden around two corners. Alongside virtual mirrors and iterative TLTM methods, this shifts higher-order light transport from a discarded multipath component to an additional imaging resource.

Iterating measured transient light-transport matrices. Sultan et al. expanded the virtual-transport-matrix idea from a reconstructed projector–camera operator to a full-dimensional

measured first-order transient light-transport matrix (TLTM-1) between relay-wall illumination and detection patches [90]. A dense laser scan and gated 16×16 SPAD array sample this matrix, after which phasor-field beamforming independently focuses virtual illumination and detection within the hidden scene. The resulting hidden-to-hidden TLTM-2 exposes indirect shadows, interreflections, and volumetric scattering and supports direct–indirect separation, virtual relighting, and dual photography. This work shifts active NLOS beyond recovering only geometry: iterating TLTM-n treats the hidden scene as a synthetic time-of-flight camera whose internal transport can itself be measured computationally.

Two-dimensional Quasi-Fresnel inversion. Most fast transient inverses still allocate a three-dimensional hidden volume even when the target is effectively a surface. Wei et al. represent both aggregated measurements and the hidden scene as two-dimensional functions and derive a direct Quasi-Fresnel transform between them [79]. The reduction in representation dimension lowers runtime and memory by orders of magnitude while preserving reconstruction quality, creating a route toward high-resolution active NLOS on mobile or embedded hardware.

Unified reconstruction across NLOS scenario classes. Most active inverses are specialized either to time-resolved around-corner transport or to transmission through a scattering medium. Ji et al. propose a single scale-modulated and jointly regularized formulation for both categories [175]. The method recovers hidden albedo and depth under diverse measurement patterns and is accompanied by a dataset spanning the two scenario families. This cross-scenario formulation complements operator-specific acceleration such as LCT, f–k migration, phasor fields, and the Quasi-Fresnel transform: rather than deriving another fixed inverse for one geometry, it regularizes a broader family of forward models. The work is currently a Research Square preprint, because no final journal or conference publication could be verified.

Wave propagation on non-planar relay surfaces. The fast convolutional forms of LCT, f–k migration, and conventional phasor-field propagation usually assume that illumination and detection samples lie on a planar relay wall. Gu et al. relaxed this restriction with 3D Rayleigh–Sommerfeld diffraction (3D RSD), decomposing propagation from a measured non-planar relay surface to the hidden volume into two stages connected by a planar proxy [58]. This construction retains efficient wave-based propagation while accounting for curved or irregular relay geometry, and forms an important intermediate step between planar virtual-wave solvers and later arbitrary-relay inverse-rendering and Gaussian-transient formulations.

Direct arbitrary-relay back projection on GPUs. Sun et al. returned to the geometry-general back-projection model and made it practical for irregular relay surfaces and arbitrary nonuniform source–detector samples [83]. Rather than flattening or resampling the measurements onto a planar grid, the method accumulates each measured ellipsoidal constraint in its native geometry, applies a frequency-domain bandpass filter to reduce noise and multipath artifacts, and parallelizes the full reconstruction with CUDA. Measured non-planar-wall experiments report at least two orders of magnitude acceleration over CPU execution while remaining competitive with CC-SOCR, Virtual Scanning, and 3D RSD. This complements proxy-plane wave propagation: geometric generality can also be obtained by retaining the original back-projection operator and removing its computational bottleneck.

Real-time two-dimensional transient testbeds. Peña et al. reformulated the transient path integral for self-contained two-dimensional worlds and coupled it to phasor-field NLOS camera models [176]. The resulting end-to-end pipeline performs progressive transient simulation and hidden-scene reconstruction in real time while exposing controlled variations in filtering, reflectance, geometry, and acquisition parameters. Reported computational savings reach five orders of magnitude relative to equivalent three-dimensional experiments, with substantial memory reduction. Flatland is therefore not a replacement for measured 3D benchmarks, but a complementary instrument for rapidly testing assumptions and failure modes along the Velten–LCT–phasor-field trajectory before committing to expensive capture or rendering.

C. Reconstruction topology

The reconstruction algorithms of active NLOS imaging have been developed rapidly in recent years [1], [6], [46]. Here, we review these reconstruction methods from two aspects: topology and methodology. Reconstruction topology is classified according to different reconstruction targets (Volumetric, Surface and High-level representation), and reconstruction methodology is based on different reconstruction methods (Inverse method and Wave-based methods).

1) Volumetric: The volumetric NLOS methods reconstruct the hidden scene by estimating the albedo values of the hidden scenes, which is the most naive idea in NLOS reconstruction. In 2012, Velten et al. used filter back projection (FBP) to estimate the albedo of the hidden object surface based on the measurement data and completed the NLOS reconstruction for the first time [3]. Many classic algorithms, such as light cone transform (LCT) [1], which exploited confocal settings to convert NLOS reconstruction into a 3D deconvolution problem, and wave-based algorithms [6], [7], all aim to restore the albedo of voxels.

The volumetric methods usually have a faster recovery speed (less than 1s for 64×64 spatial resolution [49]). However, because they do not estimate the surface parameters (e.g., normals and BRDFs) of the hidden scene, these methods are not good at recovering details [44], [49].

2) Surface: Unlike the above methods that use volumetric albedo to represent NLOS scenes, a line of work aims to reconstruct the surface of hidden objects. Such methods usually use special photon measurements (rather than using all photon measurements in volumetric albedo methods [1], [3]) to get detailed geometry. Tsai et al. found that the first-returning photons contain the shortest length information to the hidden object, from which the boundary and the surface normal vector of the hidden object can be reconstructed [177]. Xin et al. showed that the discontinuities of ToF measurement are produced by special light paths (Fermat Paths), which

contain the surface information of the hidden scene [46]. Since these discontinuities are independent of photon intensity, this approach is robust to different BRDFs.

Implicit occlusion fields beyond Fermat-visible surfaces. Grau et al. linked recoverability analysis with neural implicit surface reconstruction by representing the hidden object as the decision boundary between points that are visible from at least one relay-wall sample and points occluded behind the target [56]. In contrast to memory-intensive voxel albedo and the conservative specular Fermat-path criterion, the learned occlusion field can be converted into an adaptively tessellated mesh, recovers surface portions outside the Fermat-visible set, and remains robust when the target self-occludes across parts of the relay aperture. This direction shifts the role of learning from directly regressing a predefined shape representation toward learning which regions of hidden space are geometrically recoverable, providing an early bridge between physical visibility constraints and later neural implicit NLOS surfaces. Besides, by introducing surface normal into the previous transport matrix in Eq. (8), the surface normal reconstruction and volumetric albedo reconstruction can be combined to achieve good results [41]. Similarly, fusing the surface reconstruction into the convolution kernel of the LCT [1] can also achieve better reconstruction results than using only the volumetric albedo representation [49].

Geometric constraints on hidden surface normals. Liu et al. extended the LCT/DLCT line toward sparse and short-exposure surface reconstruction by jointly optimizing transient consistency, volumetric albedo, and projected depth/albedo maps [55]. Instead of estimating a free three-dimensional normal field, their model regularizes the Frobenius norm of the shape operator of the depth map, directly constraining spatial variation of hidden-surface normals while retaining geometric detail. This reduces the dimensional burden of joint albedo–normal estimation and, on measurements acquired within 15 seconds, was reported to recover more accurate surfaces with roughly a thirty-fold speedup over a prior surface-reconstruction baseline. The method marks a model-based trajectory complementary to neural implicit surfaces: stronger geometric priors can make under-sampled transient inversion both more detailed and computationally practical.

The above methods restore the surface of the hidden object directly, while other methods adopt the concept of reverse rendering, i.e., to reconstruct the hidden surface by finding the surface parameters (such as BRDF and surface normal) of the hidden object that can fit the measurement data. Considering that accurate reverse rendering methods are time-consuming, a differential renderer can be used to speed up rendering [44].

3) High-level representation: Besides volumetric and surface-based methods, with the rapid development of deep learning in recent years, some works have exploited high-level representation to complete NLOS reconstruction [17], [18]. Specifically, in a data-driven manner, an encoder is used to extract high-dimensional features of the measurement data, while a decoder is used to map them to the hidden object space to complete the reconstruction. Due to the lack of suitable data sets and the incomplete exploration of the network structure, such methods usually have limited generalization capabilities and reconstruction effects. However, compared to traditional reconstruction methods [1], [6], [46], [49], the high-level representation methods based on deep learning have faster inference speed (less than 100ms) and stronger feature extraction capabilities, which is a kind of promising methods.

To emphasize the application prospect of deep learning in NLOS imaging, this paper puts all deep learning methods into Sec. IV. Therefore, please refer to Sec. IV for detailed discussion and challenges of the high-level representation methods.

D. Reconstruction methodology

Active reconstruction algorithms can be divided into two categories according to reconstruction methodology: inverse methods and wave-based (forward) methods.

1) Inverse methods: It can be seen from the forward propagation model (Eq. 1) that what the active NLOS imaging completed is the inverse process, that is to recover the hidden scene ρ from the measurement $\tau(\mu, \mu', t)$. Under the assumptions of isotropic reflection, no interreflection and only three-bounce light, Eq. 1 degenerates into a linear model, which can be solved by a variety of inverse methods (e.g., back projection [3], [4], [43] and matrix inverse [1], [5]).

Back projection For general NLOS imaging scenes, the Dirac function in Eq (3) essentially represents an ellipsoidal constraint

$$r + r_l = ct \tag{6}$$

The reconstruction of this model is similar to the ellipsoidal Radon transform used in CT. In confocal NLOS imaging, the ellipsoidal constraint degenerates into a spherical constraint, and the corresponding reconstruction also becomes the spherical Radon transform in CT. Therefore, the back projection algorithm can be directly used to complete the NLOS reconstruction task. In 2012, Velten et al. used the back projection algorithm to complete the NLOS imaging experiment for the first time [3]. Arellano et al. reformulated back projection as the voxelization of ellipsoidal space–time manifolds induced by individual three-bounce measurements [4]. Rather than evaluating every candidate voxel against every captured sample, the algorithm projects only signal-bearing ellipsoids into the reconstruction volume using the GPU rasterization pipeline. This ellipsoidal-manifold voxelization lowers the dominant computation from voxel-wise accumulation toward measurement-driven processing, scales far more gently with output resolution, and reports speedups of up to three orders of magnitude with negligible reconstruction error. Subsequent iterative error-backprojection methods further refined the residual between simulated and measured transients [43]. Jin et al. separated overlapping ellipsoid modes for multiple hidden objects [42], then used Richardson–Lucy deconvolution of measured time histograms before backprojection to recover finer details under noise [179]. Shi et al. reweighted scanning contributions to estimated edge voxels before re-projection [180], whereas Feng and Gao introduced aperture-apodization and spatial-coherence weighting to suppress streaks and photon noise in filtered backprojection [181]. Zhou et al. introduced

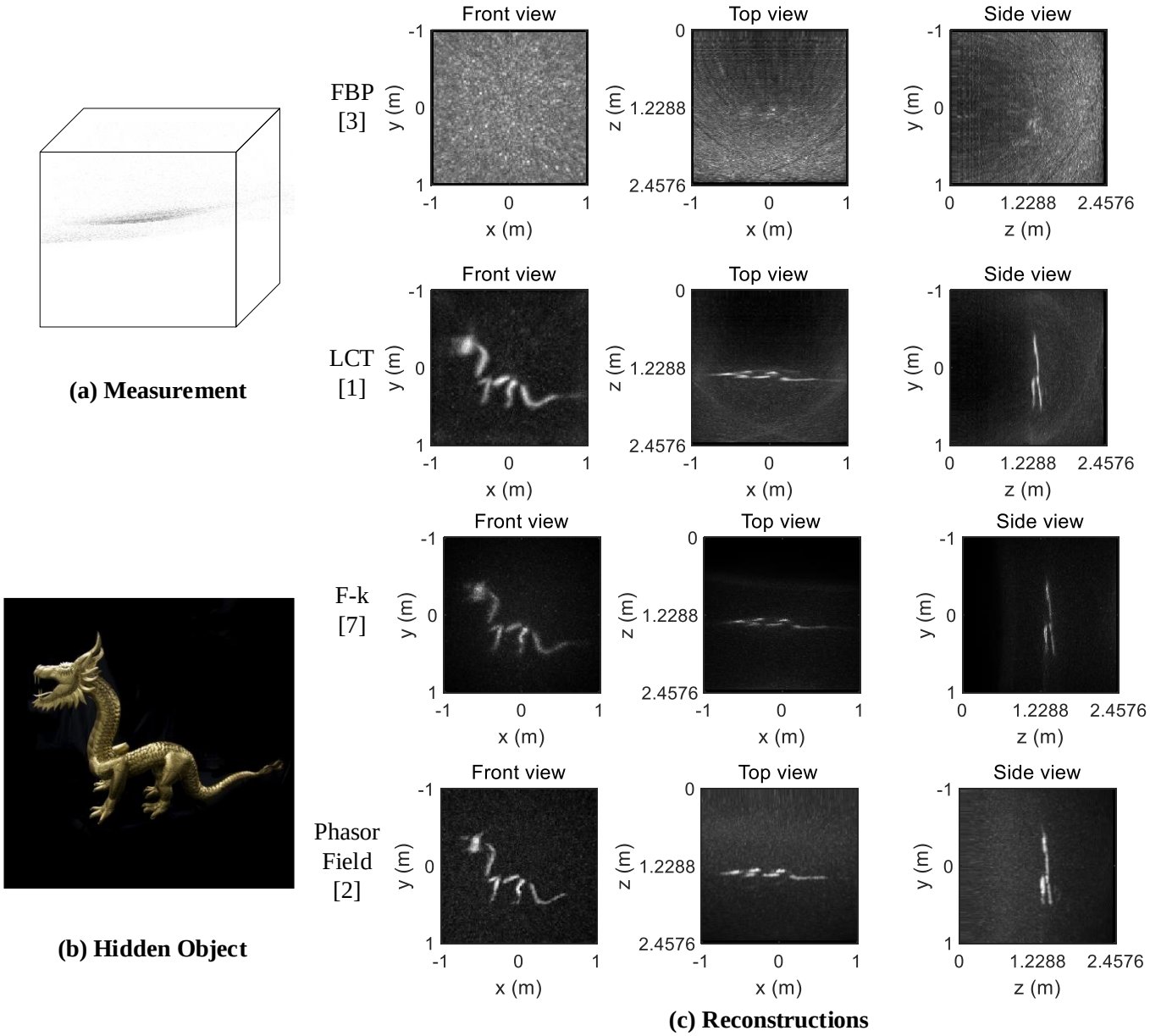


Fig. 3. Reconstructions completed by different methods. (a) Measurement data; (b) Hidden object (dragon); (c) Reconstruction results under several typical methods, including FBP [3], LCT [1], f-k migration [7] and phasor field [2]. This figure is finished based on the public code and data of [1], [7], [41], [178].

absorption backprojection (ABP), drawing low-confidence reconstruction voxels toward high-confidence object regions to suppress diffuse ellipsoidal artifacts in simulations and laboratory measurements [182]. These approaches show how geometry-aware inverse refinements preceded later learned transient operators. Since it has been widely used in CT, the back projection algorithm is simple, easy to understand, and has a low space complexity ($O(N^3)$), which can effectively reconstruct simple hidden scenes. However, the back projection algorithm's time complexity is relatively high ($O(N^5)$), and it is difficult to use other prior information, unable to reconstruct a complex optical transport process.

Matrix Inverse. In addition to the back projection algorithm based on the ellipsoidal constraint, NLOS reconstruction can also be completed by directly solving the inverse process of Eq. (3), that is

$$\rho = A^{-1}\{\tau\} \tag{7}$$

One straightforward approach is to directly solve the pseudo-inverse of the optical transport matrix A, such as using the singular value decomposition. Taking singular value decomposition as an example, since the optical transport matrix A is with the size of $N^3 \times N^3$, the time and space complexity of the algorithm to find the matrix inversion directly is $O(N^{3\times 3=9})$ and $O(N^6)$ respectively, as discussed in [1]. Another obvious disadvantage of direct inversion is that it is difficult to use the inherent non-negativity, sparsity, and other priors (see Sec. IV-IV-A for details) in NLOS imaging, leading to bad results. A common alternative is to solve the following optimization

problem [5] iteratively

$$\rho_{opt} = \underset{\rho}{\operatorname{argmin}} \frac{1}{2} \|\mathbf{A}\rho - \tau\|_2^2 + \Gamma(\rho) \tag{8}$$

Among them, $\Gamma(\rho)$ enables this type of iterative inverse method to utilize multiple constraints. Although iterative methods [38], [39] can further reduce errors and achieve good reconstruction results, they require multiple iterations, which is still difficult to apply in real-time. O'Toole et al. [1] found that under confocal conditions, the forward imaging process can be expressed in the form of three-dimensional convolution, thus using efficient deconvolution algorithms (such as Wiener filtering) to perform the efficient reconstruction. It reduced the time complexity to $O(N^3 log(N))$, which greatly improved the reconstruction speed.

Circular confocal acquisition and transient sinograms. Isogawa et al. asked whether confocal NLOS could reduce the relay-wall scan itself rather than only accelerate inversion [53]. Their C^2NLOS acquisition samples a one-dimensional circle instead of a dense two-dimensional raster. After the light-cone-transform time reparameterization, each hidden scatterer traces a sinusoid whose amplitude, phase, and offset encode its three-dimensional location; Hough voting, inverse-Radon reconstruction, or compact linear inversion then recover hidden positions and images. This direct extension of confocal LCT and f–k migration established structured scan-path design as a route toward rapid NLOS capture, anticipating later sparse, irregular, and learning-based sampling methods.

Super-field-of-view reconstruction by translated PSFs. A finite relay scan conventionally restricts reconstruction to the normal space associated with that detection region. Li et al. observed that translating the transient point-spread function equivalently shifts the virtual reconstruction region, and introduced spatial-domain encoding to suppress the circular-convolution artifacts that would otherwise corrupt this extension [183]. Numerical experiments recover multiple targets over nine times the physical detection area, while measured data reconstruct targets beyond twice the single-axis detection range and support stitched coverage equivalent to approximately $19.6\times$ the original area. This work treats field of view as a computational degree of freedom, complementing sparse scanning and non-planar relay methods without requiring a second acquisition aperture.

Adaptive spiral relay-wall sampling. Zhang et al. introduced fixed Archimedean-spiral confocal relay-wall sampling, extending earlier circular transient-sinogram acquisition to a trajectory that also covers the central relay region [184]. Fixed circular and spiral trajectories reduce acquisition relative to dense raster scanning, but they spend the same sampling budget regardless of where a localized hidden target returns useful photons. Oyama et al. proposed Dynamic Archimedean Spiral Confocal NLOS, which updates the spiral center from sequential estimates of relay-wall return intensity and concentrates later measurements on informative regions [71]. Voronoi density compensation corrects the nonuniform sampling induced by this adaptive path before transient-sinogram reconstruction. The method directly continues C^2NLOS from predefined structured paths toward closed-loop measurement allocation.

Model-decomposition reconstruction from sparse transients. Sparse transient acquisition reduces scan time but usually shifts the computational burden to large regularized inverse problems. Yang et al. formulated MD-NLOS as a spectrally filtered non-negative LASSO problem and decomposed it into smaller updates suited to parallel GPU execution [70]. Their experiments reconstruct 128×128 hidden images from only 64 relay measurements in 4.6 s. In the development from analytic LCT and structured scan paths toward learned completion, this work occupies a complementary optimization route: it preserves an explicit linear transport model and sparsity prior while restructuring the solver for practical low-sample latency.

Temporal encoding beyond detector jitter. Miao et al. introduced temporal-encoding NLOS (TE-NLOS) [185], combining high-frequency temporal modulation and spatial–temporal correlations to computationally recover fine transient structure from coarse single-photon timing responses. In their evaluated setting, the method reconstructed 20-ps transients using a detector with nanosecond-scale single-photon timing resolution, enabling subsequent light-cone-transform recovery of hidden geometry. Rather than improving the detector's physical timing jitter, this line trades structured modulation and inverse computation for effective temporal resolution and anticipates later pulse-multiplexing and temporal-decoupling approaches.

Temporal decoupling for transient super-resolution. Shi et al. investigated temporal super-resolution of transient NLOS measurements through temporal decoupling [186]. This complements structured relay-wall sampling and Bayesian deconvolution by treating temporal information recovery as a separate computational bottleneck. Quantitative performance comparisons require further full-text verification.

Sparse Bayesian transient restoration. Analytic inverses cannot recover temporal detail that has already been buried by photon noise or broadened by the instrument response. Tian et al. introduced sparse Bayesian learning as a geometry-independent deconvolution stage before LCT or f–k migration [76]. Adaptive sparsity inference suppresses background fluctuations while retaining physically plausible multipath peaks, improving boundary sharpness and robustness without changing the acquisition geometry or downstream reconstruction. This signal-domain route complements spatial undersampling methods by treating transient fidelity itself as a deployment bottleneck.

Frequency-domain regularization experts. A single spectral prior is rarely optimal across low-frequency shape, high-frequency boundaries, and scene-dependent noise. Zhang et al. formulate active NLOS reconstruction as a frequency-domain mixture of regularization experts and derive expert weights from optimization dual variables rather than heuristic or learned gating [80]. The resulting solver retains direct-method computational efficiency while improving real-data albedo fidelity and robustness over analytic, iterative, and representative learned baselines.

Reference-free artifact identification and correction. Chen et al. address backprojection artifacts without ground-truth images or a perfectly matched forward model [81]. Canny edges and morphological closing separate likely object and artifact regions; Tenengrad sharpness and residual energy define a reference-free genetic-algorithm objective; and an optimized blur kernel corrects the measured transient before backprojection. Tests on confocal and non-confocal simulated and measured data show a complementary robustness route based on measurement correction rather than a stronger scene prior.

Specular-flight-path regularization for nonconfocal multi-orientation scenes. Shi et al. examined why nonconfocal measurements can become poorly conditioned when a single illumination point and a finite detector aperture observe tilted or mutually occluding surfaces [82]. Local transport-matrix analysis shows that relay samples associated with approximate specular flight paths have lower column similarity and less redundant hidden-voxel responses than nonspecular regions. Their SFPR formulation converts this geometry into voxel-wise specular weights and alternates albedo and weight updates with total-variation regularization in an ADMM solver. Simulated scenes achieve reported SSIM above 0.85, while public and measured experiments improve completeness for multiple objects with different orientations. The result complements frequency-domain phase compensation and artifact correction by using hidden-surface orientation to improve the conditioning of the inverse problem itself.

Inverse Rendering. Back projection [3], [43] and matrix inverse methods [1], [5] are both based on voxel representation in reconstruction topology. The voxel-based representation is easy to solve mathematically, but the accuracy is usually low due to ignoring factors such as BRDF and non-Lambert reflections [44]. On the other hand, by representing the surface of the hidden object, a more accurate forward imaging model (see [44]) can be established, based on which the optimization methods can be used to solve it to complete the surface reconstruction. Since the imaging models based on surface parameters (such as BRDF and surface normal) rely on rendering, the corresponding reconstruction methods are also referred as inverse rendering, or analysis-by-synthesis. Compared with voxel-based inverse methods, inverse rendering can reconstruct more details, but often requires higher temporal complexity [41], [49].

Beyond volumetric albedo: direct surface optimization. Tsai et al. introduced a foundational analysis-by-synthesis framework that optimizes hidden surface geometry and reflectance directly, rather than first recovering an unoriented volumetric albedo field [44]. Their radiometric rendering formulation efficiently computes derivatives of transient measurements with respect to both NLOS shape and reflectance while retaining the underlying multi-bounce transport physics. Coupled with stochastic optimization and geometry-processing regularization, this surface representation reconstructs substantially finer geometric detail than earlier volumetric inversions. It established the explicit surface-optimization branch that later progressed toward physically based transient renderers, analytical differentiable rendering, neural implicit surfaces, self-calibration, and Gaussian transient representations.

Geometry-constrained joint normal and albedo reconstruction. Surface orientation supplies geometric and illumination cues that are discarded by a scalar albedo volume, but jointly estimating albedo and normals substantially enlarges the inverse problem. Liu et al. regularized hidden normal-field variation through the shape operator and jointly reconstructed surface normals and albedo from transient measurements [55]. The method improves geometric detail and robustness on synthetic and measured data while reporting approximately $30\times$ faster optimization than an earlier surface-reconstruction baseline. Its final IEEE TVCG publication formalizes a trajectory from direct surface optimization toward explicit differential-geometric priors for hidden-scene recovery.

Physically based transient analysis-by-synthesis. Iseringhausen and Hullin established an early physically grounded inverse-rendering formulation for transient NLOS surface reconstruction [50]. Instead of backprojecting measurements into an unoriented voxel volume, their method represents the hidden object as an implicit level-set surface, renders wall–object–wall transport with a custom GPU transient renderer, and globally optimizes the scene to match measured transients. The forward model explicitly includes surface orientation, BRDF-dependent scattering, shading, visibility, and self-occlusion, allowing detailed recovery under substantial noise and non-diffuse reflectance. This analysis-by-synthesis milestone supplied the conceptual and computational bridge from surface optimization to later differentiable transient renderers, neural transient fields, self-calibrating inverse rendering, and Gaussian transient scene representations.

Fast differentiable transient rendering. Plack et al. introduced a physically based differentiable transient renderer tailored to NLOS reconstruction [187]. By making gradients through time-resolved light transport efficient enough for consumer hardware, their analysis-by-synthesis pipeline reduces inverse-rendering runtimes from hours to minutes while improving optimization stability. The same renderer also supports self-supervised learning, forming an important bridge between explicit transient inverse rendering and later self-calibrating or learned NLOS reconstruction pipelines.

Self-calibrating differentiable NLOS inverse rendering. Choi et al. closed a complementary deployment gap by coupling phasor-field volumetric reconstruction in the frequency domain with differentiable path-space transient rendering in the temporal domain [57]. Starting only from measured transients, the end-to-end pipeline ray-marches the reconstructed volume to extract hidden surface points and normals, then jointly optimizes virtual-illumination and diffraction-filter parameters, the combined laser–sensor temporal response, and per-surface albedo by minimizing the discrepancy between rendered and measured transients. This self-calibration removes much of the manual filter and hardware-response tuning required by earlier wave-based pipelines, improves geometry and reflectance recovery under substantial noise, and forms a direct methodological bridge from efficient differentiable transient rendering to later neural-field and Gaussian transient

representations.

2) Wave-based methods: The theoretical basis of the three types of algorithms mentioned above is geometric optics. Besides that, wave-based methods have also achieved rapid development in recent years. Lindell et al. described NLOS imaging as a wave propagation problem in three-dimensional space. Inspired by inverse methods used in seismology, it completed NLOS imaging using f-k migration [7]. In that work, referring $\Psi(x, y, z, t)$ as the field in 3D space as a space-time function, the NLOS reconstruction is converted as

$$\Psi(x, y, z = 0, t) \quad \Rightarrow \quad \Psi(x, y, z, t = 0) \tag{9}$$

where $\Psi(x, y, z = 0, t)$ and $\Psi(x, y, z, t = 0)$ can be regarded as the available measurement data and the hidden target scene, respectively. With $t = 0$, the functions Φ and Ψ in Eq. 5 are related by a Fourier transform. By replacing dk_z with d_f and using Stolt interpolation, Eq. 5 can be converted to another representation. In the new representation, when z = 0, functions Φ and Ψ are again a pair connected by Fourier transform. In this way, taking $\Psi(x, y, z = 0, t)$ as input, the hidden scene $\Psi(x, y, z, t = 0)$ can be easily reconstructed through three steps: 3D Fourier transform, Stolt interpolation, and inverse 3D Fourier transform. Strictly speaking, f-k migration [7] is an inverse method (reconstruction through the reverse process of forward propagation). However, compared to other typical inverse methods such as back projection [3], [4], matrix inverse [5] and inverse rendering [44], f-k migration does not have an obvious inversion process (e.g., ADMM [1]), but is more like a forward process completed in the frequency domain.

Fast shift-variant inversion and low-resolution histogram enhancement. Two works extend frequency-domain NLOS reconstruction without assuming that acquisition already provides a dense, picosecond-resolved confocal transient. Xu et al. approximate the general non-confocal ellipsoidal forward operator and its adjoint by spatially local product convolutions, then use FFT evaluation and low-rank matrix decompositions to accelerate iterative inversion [99]. Validation on the Zaragoza and Wisconsin datasets shows reconstruction quality comparable to phasor-field processing at substantially reduced runtime, providing an efficient operator-level alternative for shift-variant transport. Tian et al. address temporal resolution upstream of migration: deconvolution-modified iterative backprojection first estimates a super-resolution transient histogram, after which virtual modulated range migration reconstructs the hidden volume [100]. Their experiments recover a 50-fold temporally super-resolved histogram from 1 ns measurements and support both confocal and non-confocal configurations with less data. Together, these methods broaden the LCT/f–k/phasor trajectory from closed-form confocal transforms toward efficient shift-variant operators and computational compensation for commodity-scale timing resolution.

Besides, Liu et al. [2], [6] and Reza et al. [119] introduced virtual wave phasor to NLOS imaging. Based on the phasor field, NLOS imaging can be transformed into a LOS optical imaging problem and then solved by the existing algorithm in diffraction imaging. Different from other methods, phasor field methods [2], [6] transform the relay wall into a virtual aperture (or lens of any LOS system). The reconstruction is the diffraction integral of the wavefront of the virtual aperture, which is equivalent to the forward propagation process of the measurement data. Therefore, phasor field methods [2], [6] do not need to reverse the forward process like inverse methods [1], [3] – they can directly complete the reconstruction through wavefront propagation.

Virtual light-transport analysis. Marco et al. extended phasor-field virtual wave optics beyond hidden-geometry reconstruction by constructing computational projector–camera pairs and estimating a virtual light transport matrix of the hidden scene [89]. Probing this matrix enables virtual relighting and separates direct, first-order indirect, and higher-order indirect illumination in cluttered NLOS scenes, showing that transient NLOS measurements can support light-transport analysis as well as geometric imaging.

Visibility limits and recoverability. Beyond developing faster inverses, Liu et al. analyzed which hidden-scene features are physically visible in transient NLOS measurements [13]. Their study connects recoverability to surface orientation, finite relay-wall aperture, and acquisition geometry, explaining why some surfaces remain weak or absent even under accurate reconstruction. This visibility perspective established an important limit-aware complement to LCT, f–k migration, and phasor-field propagation, and motivates later methods that explicitly model normals, occlusion, and adaptive capture.

Finite-aperture yaw observability. Romanelli et al. characterized how a finite sampled relay wall limits orientation recovery in confocal transient NLOS [110]. A geometric switch-line criterion identifies when the endpoint ordering that supports unique planar reconstruction leaves the measured aperture, while a Fisher-information analysis shows that yaw sensitivity decays continuously rather than disappearing abruptly. Simulated and experimental f–k and backprojection results connect this loss of angular information to edge-like or spatially ambiguous reconstructions. The study turns the qualitative missing-cone and feature-visibility limitation into quantitative guidance for relay-wall size, placement, and pose estimation.

Stereo relay-wall acquisition. The missing cone is not only an algorithmic artifact: with one finite relay aperture, surfaces whose orientations direct little transport toward that aperture remain intrinsically weak. Luesia-Lahoz et al. introduced a stereo NLOS configuration with two distinct relay walls and used phasor fields to treat both as generalized virtual camera apertures [68]. The reconstruction combines measurements that illuminate and detect on the same wall with cross-wall paths that illuminate one wall and observe the other. These complementary angular views improve conditioning, diminish missing-cone artifacts, and provide surface-orientation cues from the relative wall contributions, extending phasor-field imaging from a single virtual aperture to coordinated multi-aperture capture.

Reference-function phase compensation for non-confocal capture. Non-confocal acquisition improves photon use and capture speed but produces bistatic path geometry that is less convenient for confocal Fourier inverses. Yu et al. trans-

ferred a single-input/multiple-output frequency-domain imaging formulation from millimeter-wave sensing and derived a reference-function phase compensation method for optical NLOS [75]. FFT implementation reduces computation while measured and simulated results show fewer artifacts and less geometric distortion. The work provides an explicit RF-to-optical algorithmic bridge rather than using radar only as an application analogy.

Specular-flight-path regularization. Non-confocal measurements provide flexible capture but can become highly redundant for complex tilted objects. Shi et al. use local matrix analysis to identify specular-flight-path regions with lower column similarity and information redundancy, then regularize inversion around those paths [82]. The method improves simulated and measured reconstruction for multi-orientation hidden scenes, showing that path selection can condition the inverse problem before generic spatial regularization is applied.

The wave-based approaches have two attractive advantages: (1) It is more robust to the material of the relay surface; (2) It can easily combine NLOS imaging with other related fields, such as LOS imaging and seismic imaging. These advantages, as well as the phase acquisition problem in wave-based approaches, are explained in detail in Faccio et al. 's review [23].

Extensions to Non-Planar Relay Surfaces. The classical ToF and wave-based models assume a planar relay wall, restricting applicability to flat, vertical surfaces. Gu et al. proposed a fast algorithm that generalizes confocal NLOS reconstruction to arbitrary non-planar relay surfaces by transforming the measurements to an equivalent planar configuration, enabling efficient FFT-based reconstruction even when the relay geometry deviates from a plane [188]. Similarly, Liu et al. introduced a Bayesian framework with confocal-complemented signal-object collaborative regularization (CC-SOCR) that handles arbitrary illumination and detection patterns on the relay surface without requiring a regular grid [189], greatly broadening the applicability of NLOS systems to irregular or partial relay surfaces encountered in real-world scenarios.

Moving from arbitrary relay sampling toward extreme measurement sparsity, Liu et al. introduced signal-surface collaborative regularization (SSCR), which jointly regularizes the photon-event signal, a three-dimensional voxel representation, and a two-dimensional hidden-surface representation [190]. The method supports both confocal and non-confocal acquisition and reconstructs complex hidden geometry from as few as 5×5 confocal measurements, showing how mixed-dimensional physical priors can replace much of the dense raster scan required by earlier transient NLOS systems.

Higher-Order Bounce Imaging. Standard transient NLOS reconstruction usually truncates transport at the third bounce, which limits visibility for unfavorably oriented surfaces and confines most systems to a single corner. Royo et al. showed that planar diffuse surfaces behave specularly at the computational wavelengths of the phasor-field domain and can therefore act as "virtual mirrors" [191]. By propagating measured transients to these surfaces and constructing secondary virtual apertures, their method recovers limited-visibility geometry and objects hidden behind a second corner without requiring physical specular reflectors. This result shifts higher-order light transport from a discarded nuisance to an additional imaging resource.

Phasor Fields under Scattering Conditions. The phasor field framework assumes a clean relay surface, but in practice the relay surface may be contaminated by scattering media (e.g., dust, fog). Luesia et al. systematically analyzed the behavior of phasor-field NLOS reconstruction under such scattering conditions, identifying the mechanisms that degrade performance and proposing modifications to improve robustness [192]. This study provides important design guidelines for practical NLOS systems operating in non-ideal environments.

Spectral and polarization component separation for steady-state NLOS. Wei et al. extend speckle-correlation imaging from grayscale recovery to multispectral hidden scenes by treating wavelength-decorrelated speckles as the vertices of a multispectral simplex [137]. Random spectral-intensity modulation, wavelength-count estimation, vertex-component initialization, constrained non-negative matrix factorization, and phase retrieval separate and reconstruct as many as six wavelength channels; reflective experiments verify multispectral imaging around a diffuse corner. Zhao et al. subsequently replace spectral diversity with polarization-component extraction to recover multiple targets that occupy distinct optical memory-effect regions from one exposure [138]. This conference precursor establishes the channel-separation principle later generalized by polarization-encoded spatial multiplexing, connecting spectral-content recovery and beyond-memory-effect field-of-view expansion within a common steady-state speckle framework.

Single-shot imaging beyond the optical memory-effect range. Speckle-correlation NLOS imaging is normally limited to one optical memory-effect region, so targets separated beyond that angular range produce mutually decorrelated speckles that cannot be inverted together. Wei et al. encode different regions into polarization-dependent channels and formulate a linear spatial-multiplexing model that demultiplexes their speckles from one camera exposure [139]. Their two-stage recovery combines endmember estimation with truncated-Cauchy non-negative matrix factorization, local-neighborhood weights, and graph-Laplacian regularization. Reflective around-corner and transmissive scattering experiments recover multiple hidden targets across distinct memory-effect regions, reporting 32 dB demultiplexing fidelity and a 3.75-fold field-of-view expansion. This result extends steady-state correlography from single-region hidden imaging toward single-shot multi-target recovery without relay scanning, while remaining distinct from time-resolved transient inversion.

Non-confocal reference-function phase compensation. Yu et al. transfer a frequency-domain single-input/multiple-output imaging formulation from millimeter-wave sensing to optical non-confocal NLOS [193]. Reference-function phase compensation and FFT reconstruction reduce artifacts and shape distortion without requiring a dense iterative inverse.

It should be noted that not all of the above algorithms are suitable for confocal settings. Specifically, FBP [3], [4], phasor

field [2], [6], and matrix inverse methods [5] are applicable to all NLOS systems. In contrast, LCT [1] and f-k migration [7] are only applicable to confocal NLOS systems. [7] proposed a conversion method from non-confocal data to confocal data, so that confocal algorithms can also be used to reconstruct non-confocal data. From the perspective of time complexity, for confocal settings, a series of algorithms after LCT [1] (including f-k migration [7]) can reach the time of $O(N^3 logN)$ complexity due to the efficient calculation of FFT in the frequency domain, while only the phasor field algorithm [6] can achieve the same temporal complexity for non-confocal NLOS scenes. Figure 3 illustrates the measurement with a dragon as the hidden scene, and the reconstruction results obtained by different reconstruction methods [1]–[3], [7]. It can be seen that existing methods can already reconstruct complex hidden scenes. In general, the wave-based methods [2], [7] are more suitable for hidden scenes with rich details.

3) Transformer and Neural Representation-Based Methods: Motivated by the success of transformers and implicit neural representations in computer vision, recent works have brought these advances into NLOS reconstruction.

Physics-prior fusion with dual-branch attention. Liu and Li [194] combined physical priors with channel/spatial attention and dilated multi-scale large-kernel features for active hidden-target reconstruction. Their synthetic and real-measurement experiments provide an attention-based alternative to dense volumetric neural inversion, distinct from later transient-transformer and neural-operator approaches.

Transformer-based reconstruction. Li et al. proposed NLOST, the first transformer-based network for NLOS reconstruction [28]. NLOST extracts shallow features with physics-based priors, then employs two spatial-temporal self-attention encoders to capture both local and global correlations within 3D NLOS measurement volumes at different scales. A cross-attention decoder integrates these features, enabling superior reconstruction of complex hidden scenes on both confocal and non-confocal real-world data. Building on this line, Yu et al. proposed enhancing physics-based NLOS reconstruction by introducing a learnable inverse kernel in the Fourier domain combined with an attention mechanism, allowing the network to better capture high-frequency details that are difficult to recover due to spectral bias [195]. On the sensor side, Cho et al. proposed to enhance the aperture phasor field representation using a learned network, achieving state-of-the-art performance [196], while Shim et al. introduced a domain reduction strategy that identifies the most likely occupied regions in the 3D volume and focuses reconstruction effort there, reducing computational cost by an order of magnitude [197].

Geometry-constrained hidden surfaces. Liu et al. regularize the variation of hidden-surface normals with a shape-operator prior while jointly recovering reflectance and geometry [198], extending transient reconstruction beyond volumetric albedo priors toward differential-geometric surface constraints.

Neural implicit surface reconstruction. Fujimura et al. proposed NLOS-NeuS, extending neural transient fields to neural implicit surfaces with a signed distance function (SDF) representation [199]. By incorporating SDF constraints derived from first-returning photon geometry, NLOS-NeuS enables smooth yet detailed 3D surface reconstruction in NLOS scenes, overcoming the voxel resolution limitation of volumetric methods. This represents the first neural implicit surface approach with volume rendering for NLOS scenes.

Joint LOS/NLOS holistic shape reconstruction. Huang et al. introduced Omni-LOS, which combines direct LOS transients and diffuse-wall NLOS transients in a unified neural level-set representation [200]. The method derives differentiable rendering models for straight LOS rays and spherical NLOS wavefronts, jointly optimizing visible front surfaces and self-occluded back geometry from a fixed scan position. With nearby walls acting as diffuse virtual mirrors and a SPAD–laser prototype, Omni-LOS recovers near-360° object shape without a surrounding camera or mirror rig, extending neural transient imaging from hidden-scene reconstruction toward holistic single-position 3D scanning.

Diffusion model-based reconstruction. Su et al. proposed a model-guided iterative diffusion sampling framework for NLOS reconstruction [32]. Starting from the back-projection result, the diffusion model iteratively refines the reconstruction while respecting the NLOS physical forward model as a consistency constraint. This approach achieves physically consistent reconstructions with high visual quality, benefiting from the strong generative prior of diffusion models while avoiding hallucination artifacts.

4) Detection, location and identification: Under some scenarios, we do not necessarily need to complete the difficult but low-level task of imaging/ reconstruction. On the contrary, some high-level vision tasks, such as detection, localization, and recognition of hidden objects, are needed. For the detection task, we only need to judge whether there is a peak in the measurement data. For localization tasks, a naive but effective algorithm is back projection [102], [117]. Through back projection, the distribution probability of objects in three-dimensional space can be established to determine the object's position. Compared with the reconstruction task, the localization task can greatly reduce the number of scanning points and the resolution of discrete voxels with much less reconstruction time and space complexity.

For recognition tasks, traditional algorithms are limited by their ability to understand high-level semantics, making it difficult to map from experimental data to label results directly. Therefore, NLOS recognition is typically based on data-driven methods (see Sec. IV for details).

Real-measurement hidden-human detection. Celebi and Turkoglu built an active SPAD–TCSPC NLOS sensing system and collected transient waveforms for hidden people across multiple poses, orientations, and rubble-like object configurations [103]. CNN, GRU, and random-forest models all detected the human-present class with full sensitivity, while the random forest achieved the strongest specificity and weighted F1 under the available photon statistics. This result complements image reconstruction and tracking by showing that low-cost transient hardware can support a high-level search-and-rescue decision directly from measured histograms, without first recovering a detailed hidden scene.

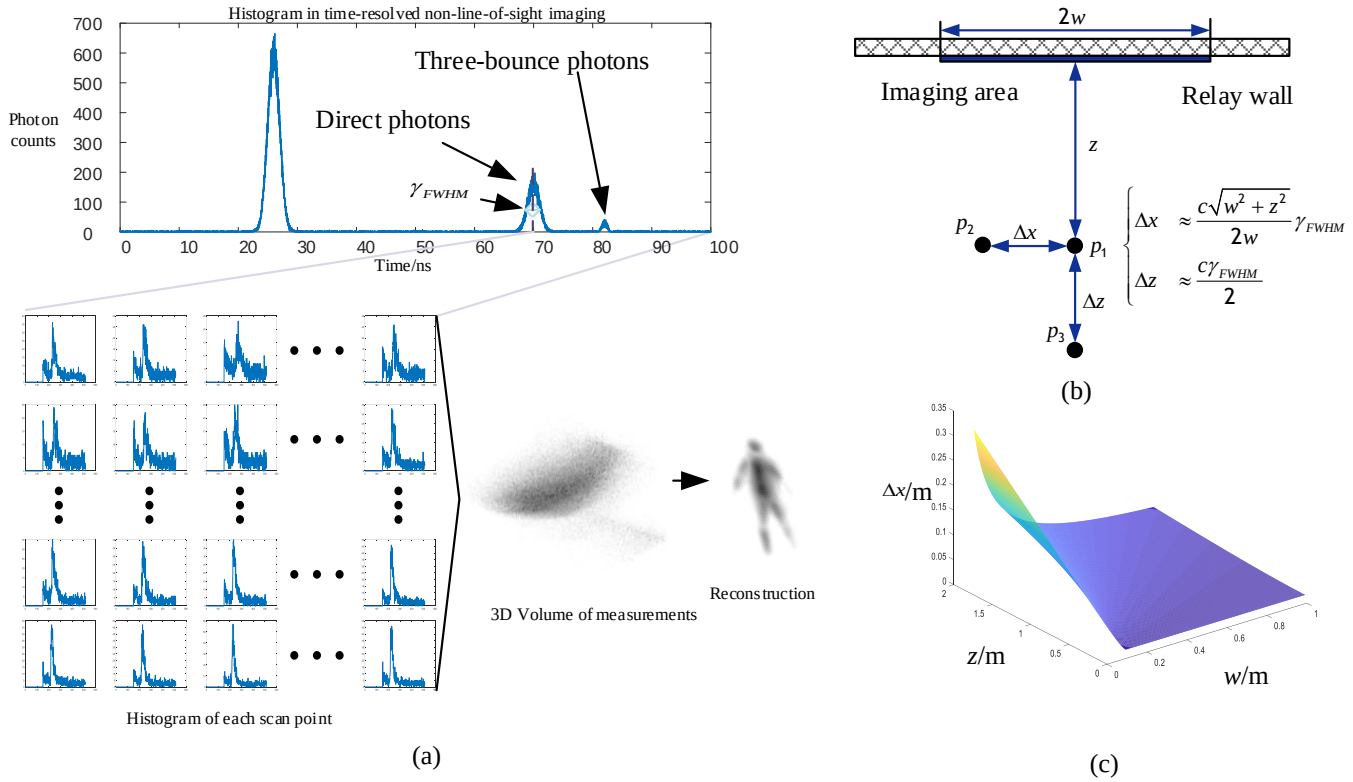


Fig. 4. The principle and spatial resolution of active (confocal) NLOS imaging. (a) The detector receives the photon arrival time histogram at each point, and these histograms form a 3D Volume of measurements τ, which can be used to recover the hidden object (albedo) ρ. The measurement data is from [1]. (b) The resolution of the imaging system. (c) The influence of the distance between the object and the wall z and the scan area size w on the horizontal resolution Δx.

Measurement-adaptive light-cone inversion. Wang et al. extend the original confocal LCT with measurement-conditioned blending of an analytical and learned Fourier response, plus frequency-dependent regularization [201]. Their MALCT architecture uses spatio-temporal attention and is evaluated on synthetic and real transients, including 32×32 spatial sampling. Unlike methods for arbitrary relay surfaces, the current formulation retains the confocal LCT assumptions.

Learned Stolt calibration and dimension-reduced inversion. Wang et al. retain the speed of f–k migration but replace fixed Stolt interpolation with a lightweight Grid Offset Network that predicts local frequency-domain resampling offsets under timing, propagation-speed, and sparse-sampling perturbations [202]. Wei et al. take an orthogonal efficiency route with the Quasi-Fresnel transform, representing typical hidden surfaces and aggregated measurements as two-dimensional functions and deriving a direct inversion with substantially lower computational and memory complexity [203]. These developments show that practical acceleration can come from calibrating the migration grid or reducing the inverse problem's intrinsic dimensionality, rather than only pruning relay samples.

Common-model benchmarking of ToF NLOS inverses. Marco et al. place representative ToF NLOS methods under a shared forward model and controlled hardware and photon-count assumptions [204]. Their analysis relates Radon-style, frequency-domain, and phasor-field formulations and shows that apparent method differences must be separated from common aperture, visibility, resolution, and noise limits. This benchmark-oriented viewpoint complements algorithm-by-algorithm summaries and provides a reproducible basis for evaluating future inverse operators.

Irregular, aperture-limited, and under-scanned relay measurements. Recent work increasingly relaxes the dense planar-wall assumption. DO-NLOS [205] combines convolution approximation and optimization for strongly under-scanned measurements; sub-pixel resolving modulation [114] extracts spatial detail beyond the native relay sampling grid; and CUDA-accelerated reconstruction on irregular relay surfaces [206] makes geometry-aware filtered backprojection practical on modern GPUs. These developments complement self-supervised Virtual Scanning and diffusion-based transient completion by improving the physical solver and acquisition model rather than relying only on learned interpolation.

E. Challenges and Prospects

Despite considerable progress in recent years, active NLOS imaging still faces many challenges, including ill-posedness with low SNR, limited resolution, and long data acquisition time.

1) Ill-posedness with low SNR: Since the collected effective signal is three-bounce of reflected light, and the signal intensity has an attenuation of $r^2 \sim r^4$ [1] (varies with different reflective materials) with distance r, the signal strength is

weak. In many NLOS scenes (such as long-distance imaging [16], [207]), the echo can reach the order of a single photon. On the other hand, there are various kinds of noise. All of the dark count and after-pulse of SPAD, the pile-up effect of TCSPC, and the ambient light can cause noise. Therefore, NLOS imaging tasks have a very low SNR, which makes NLOS imaging very challenging. Moreover, hidden objects in different locations may contribute the same measurement value [13], which further exacerbates the problem's ill-posedness.

The methods of improving the SNR have been discussed in [16], [207], [208], including improving the receiving device to increase the detection efficiency and thus the signal strength, combining spatial filtering with multimode fiber, spectral filtering with narrow-band filters and temporal filtering through gate-mode SPAD, and using polarizers to minimize noise. Based on the above methods, an amazing 1.43km NLOS imaging can be achieved. In addition to improving the signal-to-noise ratio from the hardware, it is also important to improve the reconstruction algorithm to alleviate the ill-posedness through various prior constraints, such as the sparsity, non-negativity, surface normal, partial occlusion (as discussed in Sec. II-B2) and the recently developed data-driven scene prior (see in Sec. IV).

2) Long data acquisition time: Active NLOS imaging requires a full raster scan of a relay wall. This point-by-point scanning mechanism leads to long data acquisition time, which hinders real-time NLOS imaging applications. Current work usually uses a multifunction I/O device (e.g., NI DAG USB-6343) to control the galvanometer mirrors (e.g., Thorlabs GVS012) to scan point by point and completes the event synchronization with the photon counter(e.g., TCSPC, PicoHarp 300). Limited by the scanning speed of the galvanometer and the minimum number of photons at each point, the scanning frequency is only 1Hz to 10Hz (depending on the imaging conditions) in state-of-the-art NLOS systems, such as LCT [1] and f-k migration [7]. When the number of scanning points is 64×64, the scanning speed of less than 10Hz is difficult to complete real-time data acquisition.

Reducing scanning points. In theory, each scan point in NLOS imaging contains the whole hidden object's shape information. Therefore, although reducing the scan points would inevitably reduce the imaging quality, it is possible only to scan a few points to complete the imaging task. Liu et al. studied the impact of randomly removing some test points on the reconstruction results [209]. Ye et al. applied compressed sensing to NLOS imaging, and only 5×5 points can achieve a resolution of 64×64, thereby greatly reducing the time required for scanning [14]. Isogawa et al. further changed the scanning methods, using the circular scanning path, and proposed C^2NLOS(Circular and confocal NLOS) [210]. C^2NLOS improved the limit of the scanning speed of the galvanometer through circular scanning and reduced the number of scanning points, thereby reducing the data acquisition time.

More recently, CVPR 2023 established two complementary routes to extreme sparse acquisition: SSCR uses mixed-dimensional optimization priors, while Wang et al. proposed a signal superresolution network that maps sparse scanning measurements to dense ones before reconstruction, achieving a $16\times$ reduction in acquisition time while maintaining comparable quality [190], [211]. Li et al. then proposed the first deep learning framework specifically for NLOS reconstruction from under-scanning measurements (USM), using a transient recovery network (TRN) to recover full-density transients from sparse inputs, followed by a physical-model-based volume reconstruction network (VRN) that achieves $50\times$ faster reconstruction than iterative methods [212]. Cui et al. introduced Virtual Scanning, an unsupervised framework for NLOS reconstruction from irregularly undersampled transients without requiring paired ground truth, leveraging a physics-guided denoiser to handle the noise inherent in low-photon conditions [213].

Chang et al. further coupled acquisition design with reconstruction by learning the compressive sensing measurement matrix jointly with an end-to-end active NLOS network [214]. Rather than treating under-sampling as a fixed mask followed by post-hoc transient completion, their PRCV 2023 approach optimizes the coded measurements themselves and reports at least 1.4 dB higher depth-map PSNR than partial Hadamard sensing matrices. This complements sparse-scanning and virtual-scanning methods by moving the learned prior into the sensing operator.

From the detection side, Li et al. proposed FPES (First Photon Event Stamping), which records only the first detected photon at each scan point [215]. By exploiting the Poisson statistics of single-photon arrival, FPES achieves comparable reconstruction quality with up to $10\times$ fewer total photons, enabling faster acquisition in photon-starved scenarios. Similarly, Liu et al. developed a photon-efficient NLOS approach that achieves robust reconstruction with fewer than one photon per pixel on average [216].

SPAD array. In addition to reducing the scanning points, the development of SPAD array to achieve scannerless NLOS imaging is a very promising direction. There have been the latest researches that develop algorithms suitable for SPAD array [6], [112] or directly use 32×32 SPAD camera to complete scannerless NLOS imaging [15]. Multiple lasers simultaneously illuminating would cause crosstalk problems, which is unacceptable. Therefore, the current NLOS imaging systems based on SPAD array all adopt the structure of "one laser, multiple pixels". In essence, this is equivalent to multiple parallel conventional non-confocal NLOS imaging systems. Therefore, any algorithm suitable for non-confocal reconstruction, including but not limited to back projection [4], f-k migration [7] and phasor field, can be used in NLOS imaging systems based on SPAD array. [15] verified that the reconstruction quality using commercial SPAD array based on phasor field [6] is close to the early single-pixel NLOS imaging [3].

Zhang et al. demonstrated a real-time scan-free NLOS imaging system achieving up to 19 fps by combining single-pixel detection with structured illumination (SPAD array) and computational reconstruction, and further demonstrated robustness to outdoor daylight conditions [217]. Towards

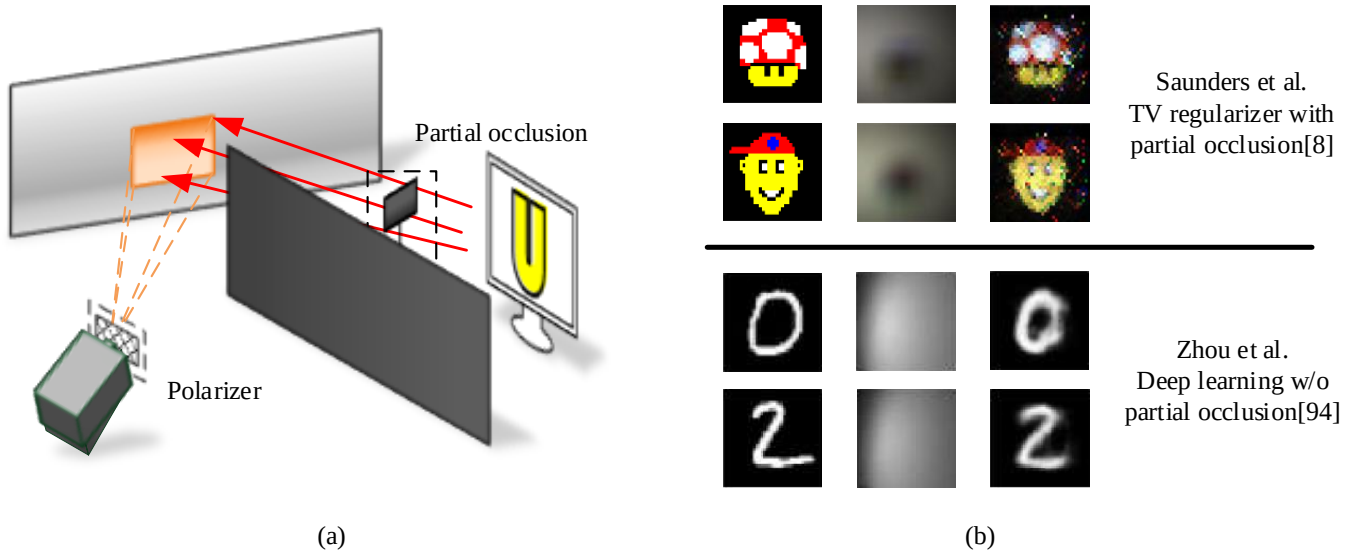


Fig. 5. Passive NLOS imaging: (a) Common constraints in passive NLOS imaging; (b) Passive NLOS imaging results.

real-time NLOS video of room-sized dynamic scenes, Ye et al. proposed a framework of spectrum filtering and motion compensation, achieving 4 fps NLOS video reconstruction of real-life dynamic scenes through denoising in the frequency domain and interleaved scanning [33]. This landmark result marks a major step towards practical deployment. Further, Li et al. developed TransiT, a transient transformer architecture for NLOS videography that reconstructs 64×64 resolution videos at 10 fps from 16×16 sparse transient measurements via transfer learning to bridge the synthetic-to-real domain gap [29].

Before these later dynamic systems, Nam et al. combined two gated 16×1 SPAD arrays (28 active pixels), sparse relay-wall scanning, virtual-aperture remapping, and fast Rayleigh–Sommerfeld phasor-field propagation to acquire and reconstruct diffuse, non-retroreflective hidden scenes at 5 fps with approximately 1 s latency [112], providing an early full-pipeline real-time NLOS video milestone.

Steady-state imaging. Another possible alternative is to replace the pulsed laser with a steady-state laser [127] or even an ordinary condensing light source (such as a projector) [143] and use a conventional camera to collect the imaging area directly. These steady-state methods usually rely on data-driven deep learning methods for reconstruction, which will be discussed in Sec. IV. Zhu et al. proposed a two-step deep remapping framework that first remaps sparse non-confocal transients to dense confocal equivalents, then applies standard reconstruction algorithms, achieving a $4\times$ speedup over FK migration while maintaining quality [218]. Mu et al. combined a physics-based forward model with deep learning to enable high-speed NLOS reconstruction from significantly fewer and noisier measurements, demonstrating reconstruction at up to 5 fps, a $10\times$ improvement over prior approaches [219]. Snapshot NLOS acquisition—capturing hidden scene information from a single camera frame without scanning—has also advanced. Seidel et al. demonstrated a snapshot corner camera that reconstructs both hidden foreground objects and background scenes without laser scanning from a single optical response model [220].

Long-range tomography and FMCW acquisition. Chen et al. explored long-range learned reconstruction from projected images acquired under multiple light fields [221]; the experiments include sensor-to-relay distances above 50 m and use a learned mapping to suppress reconstruction artifacts. This provides a complementary multi-light-field route to long-range sensing before recent transient and coherent LiDAR prototypes. Wang et al. replace dense relay-wall scanning with laser reflective tomography, using the wall as a natural beam expander and target rotation for tomographic projections [222]. Their 3.3-km outdoor demonstration extends active NLOS range, subject to the target-rotation assumption. Qian et al. explore predistorted frequency-modulated continuous-wave LiDAR for high-speed NLOS capture in a conference demonstration [223].

3) Limited resolution: Like LiDAR and Radar, the resolution of NLOS imaging is also divided into axial resolution Δz and transverse resolution Δx. Time jitter γ, the time domain response of the imaging system greatly influences Δx, and it also determines Δz. γ mainly depends on the detector's temporal resolution, laser pulse width, laser spatial divergence, and other components. In the recent long-distance NLOS imaging research [16], the time jitter γ has been formulated. Another factor of Δx is the size of the scanning area (equivalent to the size of the aperture). Usually, we use w to represent the radius of the scanning area, as shown in Fig. 4-(b). For a confocal system, the resolution of the system in two directions Δz and Δx are [1]

$$\begin{cases} \Delta x & \approx \frac{c\sqrt{w^2+z^2}}{2w}\gamma_{FWHM} \\ \Delta z & \approx \frac{c\gamma_{FWHM}}{2} \end{cases} \tag{10}$$

Among them, γ_{FWHM} refers to the full width at half maximum (FWHM) of the time jitter γ, as illustrated in Fig. 4-(a). From Fig. 4-(c), it can be seen that reducing the time jitter γ and increasing the size of the scanning area w are effective ways to improve the confocal system resolution. For other active NLOS scenes, [13], [39], [124] discussed the limits of spatial resolution. [13] also analyzed how the position and the normal direction of the hidden object affect the imaging results from the Fourier domain.

Romanelli et al. quantify finite-aperture yaw observability for a planar confocal target using a relay-wall switch-line criterion and Fisher information [224], distinguishing the geometric ambiguity threshold from gradual loss of angular sensitivity.

Beyond improving temporal resolution, recent works have tackled the challenge of long-range NLOS imaging. The groundbreaking active-focusing approach UNCOVER [162] achieves $\sim 0.6\,\text{mm}$ resolution at $0.55\,\text{m}$, demonstrating that diffraction-limited NLOS imaging is attainable via wavefront shaping. At the system level, coherent sensing with frequency-comb calibration [225] enables NLOS imaging under strong ambient illumination conditions where incoherent ToF methods fail, additionally demonstrating the novel capability of NLOS vibrometry—remotely measuring vibrations of hidden objects. These advances substantially broaden the conditions under which active NLOS systems can operate in real-world outdoor environments.

a) Acquisition, aperture, and implementation co-design.: Recent active NLOS work increasingly treats acquisition geometry, virtual-aperture design, detector timing, and reconstruction implementation as a coupled systems problem rather than optimizing the inverse solver alone. Tian *et al.* co-design a quasi-constant-delay circular relay-wall trajectory with quasi-constant-delay super-resolution, exploiting deliberately shifted transients to improve the effective timing resolution from 200 ps to about 20 ps while preserving compatibility with conventional f–k and phasor-field reconstruction backends [226]. This direction transfers part of the temporal-resolution burden from specialized detector hardware to acquisition geometry and computational recovery, which is particularly relevant to lower-cost SPAD systems. Complementing arbitrary-relay-surface formulations, Luesia-Lahoz *et al.* use two relay walls as a generalized stereo phasor-field aperture and incorporate both same-wall and cross-wall transport paths, improving hidden-surface visibility under the missing-cone limitation while turning view-dependent visibility into a cue for surface orientation [68]. At the implementation level, López-Ruiz and Royo re-engineer both f–k migration and phasor-field pipelines for GPU execution using fused CUDA kernels, warp-level photon binning, batched transforms, CUDA graph replay, selective half-precision storage, and memory-efficient phasor kernels [227]. Their results highlight a newer systems bottleneck: as parallel SPAD arrays and sparse/scan-free acquisition raise measurement throughput, memory movement, kernel materialization, and scheduling can dominate the latency of real-time NLOS cameras.

III. Passive Methods

Passive NLOS imaging aims to see hidden scenes without using a controllable external light source, which is a very challenging problem. This section has a similar structure with Sec. II. It reviews passive NLOS imaging from three aspects: data acquisition devices, physical models, and reconstruction algorithms. Finally, we discuss the challenges and prospects of passive NLOS imaging. Table II summarizes the existing passive NLOS imaging technologies, as well as their lighting conditions, sensors, available information, and target tasks.

A. Cameras in passive methods

1) Conventional camera: In the absence of a controllable light source, the light field remains steady. Therefore, the most useful information is intensity, which can be recorded by conventional cameras, i.e., charge-coupled device (CCD) and complementary metal-oxide-semiconductor (CMOS). Most passive NLOS imaging research used conventional cameras to collect data. However, their illumination was uncontrollable and different, including ambient light [8], [228] and active incoherent light [229][1]. When using ambient light, the hidden scene produces shadows (or speckles) on the wall. The only information available at this time is the intensity information of the shadows. However, if a narrowband spatially-incoherent source is used for illumination, the obtained speckles can encode the hidden scene's information, which can be reconstructed by the optical memory effect [230]–[232].

Conventional cameras are the most inexpensive but have at least two limitations among all the cameras discussed in this article. First, the shadow is very sensitive to ambient light intensity, i.e., SNR decreases as the ambient light intensity increases. Second, for broadband illumination, the optical memory effect is no longer valid, due to which the problem is extremely ill-posed and usually requires additional constraints and priors.

From edge apertures to multi-depth and background-robust computational periscopy. Ordinary-camera computational periscopy first showed that a partial occluder can turn steady-state wall intensity into an informative coded measurement of a hidden scene [8]. Seidel et al. then specialized this idea to the ubiquitous edge of a wall: a single photograph of its floor penumbra supports joint estimation of floor albedo and a one-dimensional angular hidden-scene representation [233]. Saunders et al. extended the same ordinary-camera setting to multiple hidden depths, recovering two hidden images together with their distances from the visible wall from one photograph [234]. Seidel et al. subsequently added range to the edge-camera model, producing a two-dimensional angle–range hidden-scene estimate with radial falloff modeling and statistical performance analysis [11]. Finally, Saunders and Goyal replaced simple background cancellation with optimized preconditioning that jointly suppresses plausible ambient components and improves inverse conditioning, making computational periscopy more practical in challenging illumina-

[1]Although active illumination is used, it is still considered as passive NLOS imaging because the light source is on the side of the hidden object and is uncontrollable.

tion [235]. This sequence fills the transition from the original passive periscope to later learned, diffusion-based, long-range, spectral, and geometry-aware occluder-aided methods.

Consumer-camera and doorway-aperture precursors. Murray-Bruce et al. developed occlusion-based computational periscopy with consumer cameras [236], and Czajkowski and Murray-Bruce subsequently reported a preliminary doorway-frame passive 3D imaging demonstration [237]. These conference precursors preceded the full two-edge-resolved 3D reconstruction in *Nature Communications* [238].

Transmissive-window apertures and multi-object passive reconstruction. Beyond edge occluders, Shi et al. considered a large visible transmissive window that passes overlapping light from multiple hidden objects onto a diffuse receiving wall [239]. They modeled window-dependent visibility and light transport, separated objects and their depths with projection operators and residual analysis, and reconstructed the individual images using TVAL3 and Split-Bregman regularization. This extends ordinary-camera passive NLOS to broader apertures and simultaneous hidden targets without requiring ultrafast transient sensing.

White-light and single-shot speckle correlation. A complementary steady-state trajectory replaces ultrafast timing with statistical structure in wall speckle. Zheng et al. first demonstrated ordinary-camera NLOS under broadband white-light illumination with a two-step learning strategy: a speckle-correlation model supplies the physical representation, one DNN regularizes the scattered-pattern autocorrelation, and a second DNN maps that representation to the hidden image [240]. This result established the learned white-light speckle branch without requiring coherent illumination. Zhou et al. used Zernike-polynomial envelope fitting and low-pass correction before autocorrelation and phase retrieval, maintaining white-light reconstruction under ambient background and detector misalignment [241]. Fu et al. then embedded the speckle-correlation model and an explicit denoising prior inside a physics-enhanced network, improving ordinary-camera scalability without coherent active illumination [242]. A distinct single-shot formulation estimates object-spectrum amplitude from the covariance of one random speckle pattern, removing multi-frame stitching and extending the field of view beyond the conventional optical-memory-effect range while operating at negative SNR [243]. Together, these methods shift speckle-based NLOS from controlled darkroom demonstrations toward broadband, ambient-light, and reduced-measurement operation.

Single-shot speckle imaging beyond the optical-memory-effect range. Speckle-correlation NLOS is attractive because it avoids ultrafast timing, but its usable field of view is normally bounded by the optical memory effect. Wei et al. addressed this limitation with polarization-encoded spatial multiplexing, establishing a linear mapping between the polarization-resolved camera measurement and speckles associated with distinct memory-effect regions [139]. A two-stage N-FINDR and robust non-negative matrix-factorization demultiplexer separates those components before conventional speckle reconstruction. The method is demonstrated in both transmissive scattering and reflective around-corner configurations within a single exposure, extending earlier multi-frame component-separation approaches toward fast large-field passive NLOS.

Room-scale real-time passive reconstruction. Li and Zhang targeted a deployment regime that is only weakly represented by small laboratory passive-NLOS datasets: moving people in room-scale hidden scenes observed through diffuse relay surfaces under ambient illumination [244]. Their unseen-scene encoding enhancement network (USEEN) is a compact convolutional reconstruction model designed to preserve hidden-scene structure while suppressing relay-wall and illumination interference. The reported 12.2 ms inference time makes reconstruction latency an explicit systems objective, complementing computational-periscopy methods that emphasize inverse conditioning and later attention, diffusion, spectral, and thermal models that emphasize reconstruction fidelity or robustness. This work therefore supplies an intermediate step from ordinary-camera passive inversion toward practical real-time indoor sensing.

Rough relay surfaces as passive cameras. Li et al. reframed realistic relay-wall roughness from a source of spatial mixing into a useful physical encoding [245]. By modeling the wall's microscale scattering properties, they obtain a well-conditioned passive inverse and recover hidden scenes with an ordinary monochrome camera, reporting sub-millimeter spatial resolution, 25-fps temporal resolution, a field of view approaching 90°, full-color recovery, and millimeter-scale keyhole imaging. The same formulation supports non-invasive wall calibration without manipulating the hidden scene. This result changes the passive-NLOS trajectory from compensating for rough relays to deliberately exploiting their microscopic structure, and it supplies an optical precursor to later rough-wall thermal reconstruction.

Joint passive imaging and localization from speckle. Wang et al. showed that steady-state wall-mediated speckle can encode not only hidden appearance but also object position [246]. Their speckle-based position and image recognition network (SPIR-Net) combines modified LeNet, U-Net, and conditional-GAN components to recover a hidden image and classify its spatial location from a single-shot CCD speckle measurement. Because the system does not require a controllable pulsed source or time-gated detector, it introduces a complementary localization route to transient ToF: position becomes a learned output of the passive scattering pattern. This work therefore bridges early passive deep reconstruction/recognition and later passive tracking and semantic sensing by explicitly coupling image recovery with spatial localization.

Diffuse-aware attention encoding for passive NLOS. Recent ordinary-camera methods increasingly encode the relay-wall transport structure inside the network rather than relying on a generic image-to-image backbone. Wang et al. introduced diffuse-aware attention-enhanced encoding for passive NLOS reconstruction [247]. By explicitly emphasizing features that survive diffuse relay transport, the method represents a further step from early U-Net mappings toward attention mechanisms designed around the conditioning of the passive forward

process.

Task-oriented detection and enhanced U-Net reconstruction. Wang et al. reported a passive NLOS formulation centered on hidden-target detection rather than full image recovery [248], marking an early task-oriented branch of learned passive sensing. Yu et al. later published an enhanced-reflection U-Net reconstruction method for passive NLOS [249]. The latter retains the supervised encoder–decoder paradigm while extending the sequence from early passive U-Net/GAN mappings toward the more recent attention-, diffusion-, and light-transport-aware models.

Light-transport-aware dynamic channel selection. More recently, Wang et al. targeted the channel-aliasing failure mode of low-SNR passive NLOS reconstruction with a dynamic channel enhancement encoding mechanism (DCEEM) [250]. Rather than deriving channel attention only from latent feature statistics, DCEEM combines those statistics with light-transport descriptors through Bayesian multiplicative fusion, then follows the signal-to-noise evolution of the encoder with low-dimensional denoising, high-dimensional refinement, and vector-quantization optimization. This direction moves learned passive inversion from increasingly expressive backbones toward explicit physics-aware selection of which latent channels should be trusted under weak indirect illumination.

Lightweight fusion and polarization-guided diffusion. Chen et al. introduced LMS-NLOS [251], combining multi-scale encoder–decoder features with a detail-enhanced Transformer, asymmetric shallow/deep fusion, contour-aware supervision, and spatial-shift feed-forward units. Its compact variant reduces model size by nearly one half while retaining competitive passive reconstruction, making efficiency an explicit design objective rather than a post-hoc compression step. Jin et al. subsequently proposed MSPDiff [252], which uses polarized long-wave-infrared relay observations and a coarse-to-fine diffusion schedule. Polarization supplies a physically meaningful guide to source and surface structure, while progressive diffusion restores increasingly fine hidden detail. Together, these works trace a transition from generic attention to deployable multi-scale networks and then to physics-guided generative priors for passive NLOS.

Untrained neural priors for occluder-aided passive NLOS. Wu et al. combined the calibrated occluder-aided forward model with an untrained deep decoder whose weights are optimized separately for each observed wall image [253]. The network therefore acts as an implicit image prior rather than a supervised mapping and requires neither paired hidden scenes nor a pre-collected training corpus. Simulated and measured results under strong ambient illumination improve detail and robustness over conventional inversion. This work supplies an early zero-shot bridge between ordinary-camera computational periscopy and later physics-guided passive networks: the light-transport operator enforces measurement consistency, while network architecture regularizes the hidden image.

Supervised occluder-aided passive reconstruction. Alongside physics-constrained and untrained priors, early supervised networks explored direct mappings from passive relay observations to hidden images. He et al. used a dual-input U-Net for passive NLOS reconstruction [254], while Yu et al. introduced CAGAN, a channel-aware adversarial model that emphasizes informative feature channels and suppresses background interference on NLOS-Passive data [255]. Kim and Jang subsequently targeted reliability across acquisition geometry with a standard digital camera and an occluder, showing learned hidden-object reconstruction designed to tolerate changes in both occluder and target positions [256]. Together, these studies fill the transition from calibrated computational periscopy and dataset-driven passive inversion toward later attention, diffusion, multimodal, and geometry-aware passive NLOS models.

From separable soft-shadow inversion to diffusion. Raji and Murray-Bruce generalized ordinary-camera computational periscopy from low-dimensional hidden images to joint three-dimensional occluder geometry and two-dimensional non-occluding radiosity. Their ICASSP formulation separates these unknowns into a nonlinear least-squares block and a conditionally linear block, enabling alternating optimization from a single soft-shadow photograph [257]. The subsequent Soft Shadow Diffusion framework retains this transport decomposition but learns a physics-inspired solver from simulation, transferring to unseen simulated classes and real measurements while remaining robust to ambient illumination and noise [258]. Together, these works provide the methodological bridge from computational periscopy to the later multimodal MDUNet architecture.

Fisher-equalized edge-resolved depth sampling. The two-edge-resolved doorway camera gains depth information because orthogonal penumbrae encode complementary hidden coordinates, but uniform range discretization wastes variables where the measurement is insensitive. Czajkowski and Murray-Bruce used Fisher information to equalize the discretization of the hidden range axis [259], providing a compact conditioning-oriented companion to their full-colour TERI system. This analysis clarifies how computational-aperture geometry should determine the numerical representation rather than only the forward model.

Doorway apertures and volumetric computational periscopy. Czajkowski and Murray-Bruce demonstrated that two orthogonal doorway edges encode complementary angular and depth information, enabling three-dimensional hidden-scene reconstruction from an ordinary intensity photograph [238]. Their subsequent EUSIPCO study treats visible doorway occluders as opportunistic computational apertures and extends the reconstruction to a wide 180° horizontal field of view with greater depth extent [260]. Together with Fisher-equalized range sampling, these methods turn ordinary architectural edges into both sensing apertures and reconstruction constraints without requiring active illumination.

Long-range passive NLOS under severe background. The practical range of ordinary-camera passive NLOS is commonly limited by geometric attenuation and ambient illumination. Zhou et al. combined pattern-based calibration with low-rank matrix decomposition to enhance the wall-encoded indirect component and separate slowly varying background

contributions [261]. Their experiment placed the camera 10 m from the relay wall and the hidden object 4.5 m from the wall, retaining centimeter-scale resolution at an SBR as low as -13 dB. This result extends the computational-periscopy trajectory from laboratory-scale conditioning improvements toward long-range passive sensing with inexpensive hardware.

Passive action recognition from relay-wall video. Passive computational periscopy has primarily reconstructed static hidden intensity, whereas many safety and assistive applications require only the hidden activity class. Sun et al. introduced AME-Net, which recognizes actions from standard RGB videos of a visible relay wall by adaptively amplifying weak frame-to-frame motion while reducing environment-specific variation [262]. Its alternating recurrent feature extractor combines raw and enhanced frames, and the accompanying NLOS-Action benchmark contains 2,000 synthetic and more than 500 measured videos. This work extends passive NLOS from image recovery and tracking toward semantic temporal understanding without active illumination.

Event-camera passive NLOS for moving objects. An earlier event-camera formulation explored passive NLOS recovery of moving targets from asynchronous measurements rather than conventional frame sequences [263]. Wang et al. used an event camera to isolate asynchronous motion-induced changes in relay-wall diffusion patterns and combined them with a physics-embedded reconstruction network [264]. Simulation-based pretraining models dynamic light transport and sensor behavior, followed by fine-tuning on limited measured data. This extends frame-based passive NLOS toward neuromorphic acquisition, emphasizing weak temporal changes that would otherwise be attenuated or aliased in conventional video.

Speckle-centroid tracking of deformable hidden objects. Zhai et al. introduced a speckle-correlation centroid localization method (SCCLM) for tracking moving, deformable objects through scattering media and around corners [265]. Rather than locating the peak of the speckle cross-correlation, the method estimates the relative displacement from the centroids of consecutive auto- and cross-correlations, allowing trajectory recovery despite changes in hidden-object shape and motion beyond a single optical memory-effect region. The around-corner experiments, using an emitting target or indirect LED illumination, extend steady-state speckle sensing toward shape-robust motion estimation; importantly, the output is a trajectory rather than a reconstructed hidden image.

Structure-guided adaptive total variation. A fixed TV weight cannot simultaneously preserve weak boundaries and suppress spatially varying passive-NLOS noise across different scenes. Zhang et al. derive a guidance map from a preliminary reconstruction and use it to assign spatially varying regularization strengths [266]. The resulting parameter-free SG-ATV solver removes manual tuning while retaining the conventional-camera light-transport model, showing how classical computational periscopy can remain competitive through scene-adaptive optimization rather than learned priors alone.

Doorway and occlusion-aided computational periscopy. Following ordinary-camera computational periscopy, Krska et al. investigated doorway geometry as an additional passive indirect-imaging configuration [267]. Raji and Murray-Bruce later studied occlusion-aided three-dimensional NLOS inference without time-resolved measurements [268]. These conference contributions extend the use of natural boundaries as informative apertures and complement the edge-occluder and learned soft-shadow reconstruction methods discussed below.

Passive plenoptic light fields from visible scattering to thermal and THz sensing. A distinct passive branch preserves both spatial and angular information instead of collapsing the wall observation to a single intensity image. Sasaki and Leger first formulated reconstruction from wall-scattered plenoptic data as a BRDF-aware system of Fredholm integral equations and analyzed regularization together with fundamental limits on retrievable hidden-scene information [269]. They then connected the light field to a full spatial-frequency representation, deriving scattering- and noise-limited depth/transverse resolution and a mixed-space-frequency filter for NLOS object localization [270]. Moving to naturally emitted radiation removed the need for visible illumination: Sasaki et al. scanned an LWIR camera to build a scattered light-field cube and demonstrated passive 3-D localization of human-temperature objects in a life-size diffusive hallway [271]. At submillimeter wavelengths, Grossman et al. showed that ordinary building materials retain enough specular reflection at 336 GHz for passive one- and two-dimensional hidden-human imaging with uncooled direct-detection hardware [272]; spatial–angular THz plenoptic measurements and refocusing subsequently enabled explicit 3-D localization with a room-temperature sensor [273]. More recently, Sasaki et al. unified the scattering-limited geometric-optics regime and diffraction-limited physical-optics regime using BRDF and Wigner descriptions, respectively, yielding a design analysis that predicts how roughness, wavelength, and focusing determine recoverable passive NLOS information from LWIR through THz [274]. This lineage broadens passive NLOS from RGB wall intensity to wavelength-dependent light-field sensing and provides a physical bridge to recent rough-wall thermal reconstruction.

Parallax-assisted thermal NLOS denoising. Hashemi et al. used a four-dimensional thermal light field and geometrical parallax to suppress spatial sensor noise and the relay surfaceś own radiation, improving the localization of weakly contrasting hidden people behind inhomogeneous scattering surfaces [275]. This physics-driven multi-view approach predates recent learned rough-wall thermal inversion and illustrates how angular diversity can improve NLOS conditioning without active illumination.

Thermal NLOS through rough relay surfaces. Thermal cameras remove the need for external illumination by sensing radiation emitted by hidden targets, but rough relay walls strongly mix the measured spatial signal. Ye et al. formulated this transport as a convolution and proposed NLOSFormer, which explicitly estimates the corresponding kernel and uses it to guide an end-to-end reconstruction network [276]. The accompanying ThermalNLOS dataset and geometry-aware augmentation improve generalization across wall materials.

Experiments further demonstrate relative depth estimation and dynamic thermal NLOS reconstruction at 4 fps, extending passive thermal imaging beyond the reflective-surface assumptions of earlier systems.

Long-range infrared transport decomposition. Chen et al. extended passive NLOS pedestrian imaging to long-wave infrared sensing and explicitly decomposed the light-transport matrix into illumination and reflection components [277]. The illumination component restores gross brightness, while the reflection component sharpens contours; an illumination-oriented Transformer restricts contextual modeling to informative regions instead of propagating low-SNR background. The accompanying IF-NLOS dataset contains 33,000 images across distances, relay angles, and illumination conditions, and experiments reach hidden-pedestrian ranges of 20 m. This work connects computational-periscopy-style transport estimation with self-emitted thermal cues and task-specific long-range perception.

Dual-spectral passive NLOS at the edge. Shi et al. combined synchronized visible and long-wave-infrared relay observations with attention-based dual-branch fusion and INT8 inference [278]. The CPU-oriented implementation reports 78.4 ms latency and sub-10 W power on an embedded Jetson platform. Although the conference study emphasizes deployment rather than a new light-transport inverse, it marks a useful systems trajectory: complementary spectral cues can be fused under strict latency and energy constraints instead of assuming a laboratory GPU and a single sensing band.

Joint hidden thermal-source and environment recovery. Most passive thermal NLOS methods assume that the relay geometry or its transport parameters are calibrated before reconstruction. Matsubara et al. instead represent a moving hidden infrared emitter as a set of three-dimensional luminous points and jointly optimize those points with unknown scene parameters from wall- and floor-reflected far-infrared images [279]. A reprojection loss supplies self-supervision on real measurements, while synthetic scenes provide supervised geometric guidance. This work extends thermal NLOS from reconstructing through a learned but fixed relay model toward inverse rendering in which the hidden target and its environment are recovered together.

Specular vehicle bodies as opportunistic NLOS mirrors. Kozawa et al. use reflections on curved vehicle bodies to detect pedestrians hidden from an in-vehicle camera and estimate their three-dimensional road positions [280]. A temporal detector first identifies strongly distorted human reflections; monocular depth supplies the reflection point and local surface normal, and a specular-reflection loss constrains the predicted hidden position to the physically valid reflected ray. Unlike diffuse-wall computational periscopy, this setting exploits uncontrolled glossy objects already present in traffic. The output is task-oriented passive NLOS localization rather than a complete hidden image or surface reconstruction.

Vehicle-body reflections as opportunistic automotive NLOS sensors. Takatori first used stereo observations of virtual obstacle images reflected by neighboring vehicles or roadside glass to geometrically recover hidden road-obstacle positions [281]. Nakamura and Takatori extended the formulation to transparent relay surfaces, analyzing stereo parallax, incidence angle, and polarization filtering [282]. Oyama and Takatori subsequently moved from localization toward semantic detection on weak and color-distorted painted-surface reflections, combining pseudo-reflection augmentation with direct-view normalization and pretrained object detectors [283]. Sheng and Takatori then evaluated reflection availability in real traffic and microscopic simulation and validated stereo virtual-image unfolding across reflector orientations [284]. Together, these studies establish an application-facing passive/specular NLOS branch for intelligent transportation; their outputs are obstacle position or identity rather than hidden appearance or complete geometry.

2) Interferometer: Considering the limitations of intensity information, some studies employ interferometry to obtain phase information and use coherence to solve it, as illustrated in Tab. II. Batarseh et al. used the Dual-Phase Sagnac Interferometer (DuPSaI) to measure the spatial coherence of the light field [9], thereby completing the detection and positioning of the hidden broadband ambient light source. Beckus et al. combined spatial coherence and intensity information to complete multi-modal passive NLOS imaging [285]. In addition to spatial coherence, temporal coherence can also be used to obtain depth information. Boger-Lombard et al. utilized a diffuser and an ordinary camera to provide passive ToF information through the time coherence of the measurement [10]. The interferometer enables depth information measurement in passive NLOS scenes and can be fused with intensity information to improve imaging quality. However, it requires a precise and complex calibration process, and the depth information it contains is very limited, which can only be used for discrete scenes.

3) Hyperspectral and Multispectral Cameras: Capturing richer spectral information beyond a single visible-light band can substantially improve passive NLOS imaging quality. Chen et al. proposed Hyper-NLOS, which uses a hyperspectral camera and a dedicated fusion network to combine complementary wavelength-dependent cues [286]. Liu et al. further combined visible, short-wave infrared, and long-wave infrared measurements in a deep-fusion passive NLOS framework [287], improving hidden-person reconstruction at distances exceeding 5 m from the relay wall. This multispectral route complements polarized LWIR imaging through cross-band rather than within-band polarization diversity. Hashemi et al. instead used spectral content to separate desired wall-mediated radiance from clutter: multispectral unmixing handles unknown uniformly colored sources, whereas a convex known-spectrum formulation requires fewer bands and remains effective when clutter is much stronger than the target signal [288]. HSBS-Net then made band selection part of the reconstruction objective. Entropy-rich wavelengths are selected with differentiable constrained measurement learning, while a spectral-energy key-area Transformer suppresses poorly illuminated regions and a robust sparse loss preserves colour and structure [289]. The accompanying HP-NLOS dataset covers physical full-colour targets, 256 spectral bands, multiple distances, and changing external conditions. This

progression moves spectral passive NLOS from adding channels indiscriminately toward clutter-aware unmixing and task-driven acquisition.

Active–passive fusion for hidden 3D color. Whereas passive spectral methods exploit differences across wavelengths, Shi et al. combine active and passive NLOS observations to recover complementary hidden-scene attributes [290]. Their amplitude–color decomposition (ACDA) couples geometric information from active transient measurements with color cues from passive relay-wall images in a joint optimization, separating hidden amplitude and color components. Simulated and measured experiments indicate improved 3D shape recovery with limited active acquisition and reduced color aliasing without additional passive scene priors. This early-posting *Photonics Research* study illustrates a transition from modality-specific hidden shape or appearance estimates toward joint 3D color reconstruction, while retaining the different measurement assumptions of active and passive NLOS.

B. Forward imaging model

Considering the classic intensity-based passive NLOS imaging scene, as shown in Fig. 1-(c), the goal is to reconstruct the hidden scene based on the speckle area's intensity information d on the wall. Assuming that each point on the hidden scene is an independent point light source, the measured intensity information is

$$I_\tau(d) = \int\int_{s\in S} A(s,d) I_\rho(s) ds \tag{11}$$

Here I_τ is the observed projection image of resolution $p \times q$, $I_y(\tau)$ is the light intensity on the projection area d. I_ρ is the hidden target image of resolution $m \times n$ displayed on the screen, and $I_\rho(s)$ is the intensity of the pixel point s of I_x. Besides, $A(s,d)$ is the optical transport from the point light source s to area d on the relay wall, and S denotes all pixels on the whole screen. The model can be discretized as

$$\mathbf{I}_\tau = \mathbf{A}\mathbf{I}_\rho + \mathbf{N} \tag{12}$$

where $\mathbf{N}$ is an additional background term.

It can be seen that Eq. (3) and Eq. (12) look very similar. Because passive NLOS imaging usually only collects intensity information, the condition number of $\mathbf{A}$ in Eq. (12) is larger, meaning more ill-posed. In addition to the above-mentioned intensity-based forward model, imaging models based on other principles and settings (including the Fresnel model [285], partial occlusion [8], [11], spatial coherence [9], [285], and polarizers [291]) can all be discretized to inverse optimization problems similar with Eq. (12). Passive methods based on speckle coherence have a special imaging model and reconstruction process, which will be explained in the following (see Eq. (13)).

C. Reconstruction with constraints

Passive NLOS imaging is extremely ill-posed, and it is difficult to complete high-quality reconstruction only with conventional cameras. Existing researches reduce the condition number of the transport matrix $\mathbf{A}$ by adding additional constraints, thereby improving the imaging quality, as shown in Tab. II. This section will discuss reconstruction algorithms under three common constraints, including partial occluder [8], [292], coherence [285] and polarizer [291]. Data-driven scene priors are another common constraint, which will be explained in Sec. IV.

1) Partial occluder: In the most primitive cameras, large-area obstructions can be used to form pinholes to tighten the beam and complete imaging [300]. In other early computational imaging applications, such as stereo vision [301], light field recovery [302], [303] and image synthesis [304], partial occlusion also played an important role. In the field of passive NLOS imaging, many state-of-the-art works also exploit partial occlusion. Among them, some tasks need to know or estimate the prior knowledge of occlusion (such as position and shape) before they can reconstruct hidden scenes using the prior as constraints [8]. Others only utilized partial occlusion to improve the conditioning of the problem without knowing its specific information [11], [228], [292]–[294], as illustrated in Fig 5-(a).

Passive NLOS imaging modelled by Eq. (12) can be solved as an optimal problem

$$\mathbf{I}_{\rho,\mathbf{opt}} = \underset{\mathbf{I}_\rho}{\operatorname{argmin}} \frac{1}{2}\|\mathbf{A}\mathbf{I}_\rho - \mathbf{I}_\tau\|_2^2 + \Gamma(\mathbf{I}_\rho) \tag{13}$$

where Γ represents regularization terms, such as total variation (TV) or other sparsity constraints. When the position of the partial occlusion p is known, the optical transport matrix $\mathbf{A}$ can be estimated, after which the inverse problem can be solved through Eq. (13). The results of [8], a representative work using TV regularization, are shown in Fig. 5-(b). When the partially occluded position p cannot be obtained, the constraints at different times can also be used to estimate the optical transport matrix $\mathbf{A}$ and the corresponding hidden scene I_ρ. For example, Aittala et al. performed matrix decomposition through an unsupervised learning method to complete passive NLOS imaging [228]. This is essentially similar similar to blind deconvolution [305]–[307], as discussed in [292].

Fully learned multimodal computational periscopy. Recent occluder-aided passive methods recover both the hidden occluding geometry and the radiance distribution of non-occluding content. Raji and Murray-Bruce proposed MDUNet, whose shared encoder and specialized 3D-SDF and 2D-radiosity decoder paths jointly infer these representations from a single soft-shadow photograph [299]. Compared with the preceding hybrid diffusion and physics pipeline, the fully trained model substantially reduces inference time while transferring from simulation to real measurements under changing ambient illumination, marking a shift toward direct multimodal passive reconstruction.

Multi-object passive reconstruction. Most learned passive systems are trained and evaluated with one dominant hidden target. Chen et al. introduced a nested U-Net multi-attention architecture for infrared scenes containing one to three hidden objects [298]. Channel, spatial, and self-attention emphasize complementary transport features, while multiple decoder branches and joint supervision stabilize reconstruction at different semantic scales. The work broadens passive deep

TABLE II
PASSIVE NLOS IMAGING

Ref	Illumination	Sensor	Information and constraints	Task
[279]	Hidden-object far-infrared emission	Thermal camera	Reprojection consistency; jointly estimated scene parameters; synthetic supervision	3D reconstruction
[280]	Ambient visible illumination	In-vehicle RGB camera	Curved specular vehicle relay; monocular depth and reflection geometry	3D localization
[10]	Incoherent light source (object side)	Conventional camera	Temporal coherence	Detection/ Tracking/ Identification
[229]	Incoherent light source (object side)	Conventional camera	Speckle coherence	2D reconstruction
[241]–[243]	White light / coherent speckle	Conventional camera	Speckle statistics with physics-enhanced or single-shot inversion	2D reconstruction
[288], [289]	Ambient/incoherent light	Multispectral or hyperspectral camera	Spectral unmixing and learned band selection	Full-colour 2D reconstruction
[9]	Incoherent light source (object side)	Interferometer	Spatial coherence	Detection/ Tracking/ Identification
[285]	Ambient light	Interferometer + Conventional camera	Spatial coherence + Intensity	2D reconstruction
[8], [11], [228], [233]–[235], [292], [293]	Ambient light	Conventional camera	Intensity with partial occluder	2D reconstruction
[291]	Ambient light	Conventional camera	Intensity with polarizer	2D reconstruction
[294]	Ambient light	Conventional camera	Intensity of moving object	2D reconstruction
[294]	Ambient light	Conventional camera	Intensity	Detection/ Tracking/ Identification
[262]	Ambient light	Conventional RGB camera	Temporal relay-wall motion	Action recognition
[264]	Ambient/incoherent illumination	Event camera + conventional camera	Asynchronous diffusion-pattern motion with physics-embedded learning	Dynamic 2D reconstruction
[295]	Infrared radiation	Infrared camera	Infrared intensity	Detection/ Tracking/ Identification
[296]	Ambient/incoherent light	Conventional camera	Intensity via optimal transport	2D reconstruction
[297]	Ambient light	Conventional camera	Modulated light transport intensity	2D/3D reconstruction
[261]	Ambient light	Conventional camera	Pattern calibration and low-rank background separation	Long-range 2D reconstruction
[266]	Ambient light	Conventional color camera	Intensity with structure-guided adaptive TV	Parameter-free 2D reconstruction
[298]	Infrared radiation	Infrared camera	Learned multi-attention transport features	Multiple-object 2D reconstruction
[257], [258]	Ambient/incoherent illumination	Conventional camera	Soft-shadow intensity with separable occluder/radiosity model	Joint 3D geometry and 2D reconstruction
[259]	Ambient light	Conventional camera	Two orthogonal edge penumbrae with Fisher-equalized depth discretization	3D reconstruction
[299]	Ambient light	Conventional camera	Soft-shadow occlusion with coupled learned 2D/3D decoding	Joint 2D radiosity and 3D occluder reconstruction

reconstruction from isolated silhouettes toward scenes with variable object count and mutual signal mixing.

2) Polarizer: Although partial occlusion is inevitable in many NLOS scenes, adding occlusion may still change the hidden scene. However, in practical applications, it is not the hidden scene but the imaging system that can be changed. A demonstrated alternative method is to add a polarizer in front of the camera, as shown in Fig. 5-(a). In this case, a small angle difference between the light paths leads to a large intensity change, and the optical transport matrix $\mathbf{A}$ becomes [291]

$$A'(s,d) = A(s,d)\lambda\left(\boldsymbol{\omega}_{\mathbf{i}}, \boldsymbol{\omega}_{\mathbf{o}}, \mathbf{p}\right)$$

where λ indicates the light leaking/blocking effect caused by the polarizer. $\omega_{\mathbf{i}}$,$\omega_{\mathbf{o}}$ and $\mathbf{p}$ indicate viewing vector, incident vector and the polarizer axis, respectively. λ decreases the condition number of matrix $\mathbf{A}$, thereby improving the image quality.

Polarization-conditioned passive transport. Tanaka et al. showed that a camera-mounted polarizer can improve passive NLOS reconstruction by exploiting polarization-axis rotation along oblique indirect paths [291]. The added directional cue lowers the effective conditioning of the wall-to-hidden-scene transport matrix and improves recovery both with and without a known occluder, complementing computational periscopy without requiring a controllable active source.

Polarized infrared passive NLOS. Liu et al. extended polarization-conditioned passive imaging to long-wavelength infrared measurements in PI-NLOS [308]. A dual-channel network fuses polarized LWIR intensity with degree-of-polarization contours to reconstruct hidden people at relay-wall distances of approximately 6 m. This complements physics-based polarizer transport with learned thermal-band feature fusion.

Learned polarization feature fusion. In the same pas-

sive polarization-reconstruction lineage, Liu et al. proposed DFAR-Net, which jointly processes infrared polarization intensity and degree-of-polarization images through a dual-input, three-branch attention-fusion network [309]. Split-kernel channel attention and multi-scale losses are evaluated on the PI-ND dataset, showing how polarization cues can be fused in learned feature space rather than only by an explicit transport inverse.

3) Coherence: Three types of coherence, speckle coherence, spatial coherence and time coherence, have been used in passive non-line-of-sight imaging. Among them, speckle coherence does not require a special interferometer, but needs a narrowband incoherent light source as illumination. The speckle coherence model is based on the optical memory effect [231], which completes NLOS imaging through the scattering medium and corner objects by analyzing the coherence information between hidden object intensity and measured speckle

$$[\mathbf{I}_\tau \star \mathbf{I}_\tau](\theta) = [\mathbf{I}_\rho \star \mathbf{I}_\rho](\theta) \tag{14}$$

where $\star$ represents the autocorrelation operation. Using the phase recovery algorithm [310], [311], hidden object's diffraction-limited image $\mathbf{I}_\tau$ can be recovered from its autocorrelation [229]. The field of view of speckle autocorrelation imaging depends on the range of memory effect, which is generally very small. This is the bottleneck that limits the speckle correlation based on the memory effect.

Reconstruction algorithms based on spatial and temporal coherence are no longer subject to the viewing angle limitation caused by the optical memory effect [231], but require a very sensitive and costly interferometer. Since the interference effect is sensitive to the light source's number and size, spatial coherence and temporal coherence can usually only be used to complete the positioning, detection of discrete points, and very simple shape restoration. The combination of coherence and intensity information can effectively overcome this limitation. For example, multi-modal passive NLOS imaging that combines intensity information and spatial coherence can achieve promising results [285].

4) Optimal Transport and Transport Modulation: **Optimal transport formulation.** Going beyond standard ℓ_2 and TV regularization, Geng et al. formulated passive NLOS reconstruction as an optimal transport (OT) problem [296]. The method minimizes a Wasserstein-type distance between the predicted and measured light distributions on the relay wall, providing a geometrically meaningful and noise-robust objective. By casting the ill-posed passive measurement model (Eq. (12)) in the OT framework, the approach handles non-Gaussian measurement noise more gracefully than conventional least-squares methods. Validated on both synthetic and real passive NLOS scenes, the OT-based reconstruction yields measurably improved quality over TV-regularized baselines published in IEEE Transactions on Image Processing.

Light transport modulation. Zhang et al. proposed to improve passive NLOS reconstruction quality by modulating the light transport between the hidden scene and the relay surface [297]. Rather than treating the transport matrix $\mathbf{A}$ as fixed, the modulation strategy adaptively adjusts the imaging conditions to reduce the condition number of the inverse problem, thereby enabling more stable and accurate reconstruction from passive measurements captured with ordinary cameras. The method achieves state-of-the-art reconstruction performance on IEEE Transactions on Image Processing benchmarks.

5) Real-Time Passive NLOS Tracking: A practically important task for passive NLOS imaging is real-time tracking of moving persons. Wang et al. proposed PAC-Net (Propagate-and-Calibrate), a purely passive method that tracks a person walking in an invisible room by observing only a relay wall [312]. PAC-Net uses difference frames as carriers of temporal-local motion information and alternates propagation and calibration modules to leverage both dynamic and static cues at frame-level granularity. The authors also released NLOS-Track, the first large-scale dynamic passive NLOS tracking dataset containing thousands of video clips with corresponding trajectories, enabling systematic evaluation of passive NLOS tracking methods.

3D passive NLOS reconstruction has also advanced significantly. Czajkowski and Murray-Bruce demonstrated full-color, three-dimensional passive NLOS imaging with an ordinary camera by exploiting two orthogonal edges of an occluding structure for transverse resolution and an information-orthogonal scene representation for range resolution [313]. This work achieves accurate 3D reconstruction from inexpensive, ubiquitous hardware without calibration images, opening passive NLOS to practical indoor applications such as reconnaissance and search-and-rescue.

6) Deep Learning: Data-driven scene priors are another common constraint, which can be exploited by deep learning methods. Although data-driven passive methods [145], [314], [315] also belong to passive NLOS imaging, this article will explain it in Sec. IV together with active data-driven NLOS methods to highlight the application of deep learning in this field. Besides, in Sec. IV, unsupervised passive NLOS imaging algorithms [228] based on physical constraints (matrix factorization) are also introduced.

Adaptive regularization for passive NLOS. Zhang et al. proposed structure-guided adaptive total variation for passive NLOS reconstruction [316]. By deriving spatially varying regularization weights from an initial hidden-scene estimate, the method adapts the inverse prior to the recovered structure rather than relying on a manually selected global TV penalty. This complements the existing shadow-transport decomposition and diffuse-aware attention methods, which respectively encode geometric and learned light-transport constraints.

D. Challenges and prospects

Because there is no controllable light source, passive NLOS imaging can only obtain intensity information and limited coherent information, leading to low reconstruction quality. The two main challenges for passive NLOS imaging are high ill-posedness and limited measurement.

1) Ill-posedness: Passive NLOS imaging is an extremely ill-posed problem. The reasons can be summarized as follows. (1) The collected data contains little effective information. Since most of the information collected by passive NLOS imaging is intensity information, it can be regarded as a

projection of a hidden scene on a two-dimensional plane, without any available temporal information. (2) There is no effective space coding. The problem of passive NLOS imaging is somewhat similar to the recent lensless imaging, but there are no encoders such as moiré fringes [317], optical phased array [318] and known scattering medium [319], which results in the data collected on diffuse reflection surfaces (without valid BRDF encoding) being messy and the degree of ill-posedness being high. (3) The influence of ambient light. When there is ambient light, it is difficult to separate the ambient light from the effective speckle, which also leads to an increase in the ill-posedness [22].

Adding constraints. Adding constraints is an effective way to improve the condition. When the additional constrained parameters are unknown, accurate estimation of constrained parameters (such as the position and shape of partial occlusion) can also improve the imaging effect. Besides partial occlusion [8], [292] and polarizer [291] discussed above, the data-driven methods have a good application prospect, which will be discussed in Sec. IV.

2) Limited measurement: In practical application scenes, adding appropriate constraints between the hidden object and the observation wall may not be allowed, which leads to the limitation of passive NLOS measurement. Therefore, passive NLOS imaging only exploiting intensity information is extremely challenging [8]. Existing research can obtain coherence through the interferometer, which contains some depth information but is limited by the shape and size of hidden objects and cannot complete high-quality reconstruction.

Multimodal measurements fusion Beckus et al. combined spatial coherence information and intensity information to complete multimodal passive NLOS imaging, which provides a path to improve the imaging quality [285]. The basic principle of passive NLOS imaging is to collect visible light emitted/reflected from hidden objects. However, in addition to visible light, hidden objects also emit/reflect electromagnetic waves on other wavelengths, such as infrared ($\sim 780nm$) and radio signals everywhere in the environment. On the other hand, NLOS imaging based on infrared wave [295] and radio [323] has also been explored in recent years. Therefore, if visible light can be fused with other types of electromagnetic waves to obtain multimodal measurement information, it will hopefully break through the bottleneck of passive NLOS imaging and achieve high-quality reconstruction without occlusion and scene constraint.

Spectral and Polarization Fusion. Building on multimodal measurement, recent works have leveraged spectral and polarization information as additional passive cues. Hyper-NLOS [286] established hyperspectral wavelength diversity as an explicit conditioning variable for passive reconstruction. Li et al. then introduced a hyperspectral autoencoder network (HAENet) [324], combining spatial–spectral separation reconstruction with spatial–spectral residual attention to reduce redundant processing while preserving cross-band context. Together with later band-selection and multispectral-unmixing approaches, these works show that wavelength diversity can supply an additional coding dimension when steady-state spatial measurements alone are severely ill-conditioned. On the polarization side, the PI-NLOS framework [308] combines polarization imaging with long-wavelength infrared (LWIR) sensing to improve reconstruction under challenging lighting conditions, exploiting the geometric cues encoded in the polarization state of diffusely reflected light. Turning rough surfaces into relay surfaces is another recent direction: Li et al. showed that rough surfaces with mixed diffuse-specular reflectance can serve as NLOS relay surfaces by developing a modified light transport model that accounts for the specular component [325], greatly expanding the set of surfaces usable for NLOS sensing in uncontrolled environments.

IV. Deep Learning Methods

Conventional physics-based reconstruction algorithms have been discussed above. In recent years, with the successful application of data-driven algorithms (e.g., deep learning [326]) in blind deconvolution [327], depth estimation [328], and LOS imaging [329]–[331], recent works have also explored the possibility of deep learning in NLOS imaging. This section first explains the significance of using deep learning algorithms from a priori perspective, then introduces the end-to-end learning algorithms, as well as the hybrid algorithms that combine deep learning with physical models. Finally, we discuss the challenges and prospects of deep learning algorithms. Table III summarizes the existing deep learning-based NLOS imaging technologies and their network, input, output, and training datasets.

A. Priors in NLOS imaging

Some priors have been effectively used in NLOS imaging, as reviewed below.

Non-negative prior. The non-negative prior includes two parts. First, for the optical transport matrix $\mathbf{A}$, by definition, the value of each element is non-negative; Second, for the reconstruction model based on the albedo ρ, the albedo of each voxel is also non-negative. The non-negativity of the light transport matrix and albedo ρ together constitute non-negative prior.

Sparsity prior. Since the measurement data only contains the surface information of the hidden scene, which is sparse in the entire 3D space, the albedo (target scene) ρ has a sparsity prior. A common sparse constraint is to directly use the L_1-norm constrain, i.e., $\|\rho\|_1$, and [5] added different weights to each voxel. Another sparse constraint is a gradient constraint, which limits the gradient in the depth axis is sparse. The last type of sparsity constraint is that there should be only one non-zero value on the z-axis corresponding to each (x, y) coordinate [5]. Most NLOS reconstruction algorithms used sparse constraints as the main component of the regularization term $\Gamma(\rho)$ [41], [49].

Total variational (TV) prior. TV represents the sum of discrete gradients, and the TV regularization term is widely used in edge-preserving denoising [332], [333] and sparse

TABLE III
NLOS IMAGING BASED ON DEEP LEARNING

Ref	Network	Input	Output	NLOS Scene	Train set
[314]	End-to-End: U-Net	Scattering image	2D Image	Ambient light + conventional camera like [8]	Experimental and synthetic data
[315]	End-to-End: a variant of the U-Net	Speckle image	2D Image	Incoherent light source + conventional camera like [229]	Experimental data
[127]	End-to-End: a variant of the U-Net	Scattering image	3D reconstruction	CW laser + conventional camera	Synthetic data
[17]	End-to-End: a variant of the U-Net	Transient images	3D reconstruction	pulsed laser + SPAD like [1]	Synthetic data
[145]	End-to-End: CNN/ VAE	Scattering image	2D Localization/ identification/ reconstruction	Incoherent light source + conventional camera like [143]	Synthetic data
[104], [105]	End-to-End: CNN	Temporal histogram	Detection/ Identification/ Tracing	Pulsed laser + SPAD like [1]	Experimental data
[142]	End-to-End: ResNet	Steady image	Identification	CW laser + conventional camera like [127]	Synthetic data
[228]	Deep matrix factorization	Scattering image/ video	2D image/ video	Ambient light + conventional camera like [8]	Experimental data
[19]	Deep-inverse correlography	Speckle image	2D Image	CW laser + conventional camera like [127]	Synthetic data
[18]	Learned feature embeddings	Transient images	3D reconstruction	Pulsed laser + SPAD like [1]	Synthetic data
[126]	Learned feature embeddings	Depth and intensity images	3D reconstruction	Commercial LiDAR	Synthetic data
[108]	Physics-based deepRL framework	Transient images	Pose estimation	Pulsed laser + SPAD like [1]	Synthetic data
[320]	Hybrid physics–data dual path	Scattering image	2D image	Ambient-light passive NLOS with or without an occluder	Synthetic and experimental data
[298]	End-to-End: nested U-Net with multi-attention	Infrared scattering image	Multiple hidden-object image	Passive infrared NLOS with variable object count	Experimental and synthetic data
[321]	End-to-End: Parallel Encoder	Scattering image	2D Image	Ambient light + conventional camera	Synthetic data
[322]	End-to-End: Alternate R&R	Scattering image	2D reconstruction + recognition	Ambient light + conventional camera	Synthetic data

reconstruction of low-sample data(e.g., compressed sensing) [334]. For 3D NLOS reconstruction, TV prior represents the sparsity of hidden objects, especially sparsity on the z-axis, which is the gradient constraint discussed above. For two-dimensional NLOS imaging, the TV prior is mainly to restore the image from the inverse problem and maintain the edge information. Due to the effectiveness of the total variation constraint, many NLOS imaging studies have used the TV prior as the regularization term [8], [16], [19], [291].

Other priors. The above three types of priors apply to most NLOS imaging problems. Besides, there are some priors suitable for specific NLOS imaging. Specifically, for ToF-based NLOS imaging, the ellipsoid constraint [1], [3], as well as the surface parameter constraint (e.g., the discontinuity of the ToF measurement contains the information of the surface normal) [46], [177] are two important priors. For NLOS imaging with partial occluders, the prior knowledge of the occluder's position and shape are also significant for high-quality reconstruction [12], [292]. Besides, the BRDF prior also plays an important role in improving imaging results [124].

1) Unexploited prior – scene prior: However, there are still some priors that have not been effectively used. Therefore, using this undeveloped prior information is a meaningful way to improve NLOS imaging quality.

Scene priors is an important kind of unexploited priors, which includes both low-level features (e.g., smoothness) and high-level features (e.g., semantic information) of hidden scenes. Scene priors are particularly important for specific reconstruction tasks, such as NLOS imaging in autonomous vehicles or medical imaging. Compared with other priors, most scene priors are implicit and difficult to extract manually. Therefore, conventional physics-based methods are challenging to formulate and efficiently utilize these scene priors.

The data-driven methods based on deep learning can learn the input data distribution, thereby efficiently completing the mapping from the input space to the output space, which has been utilized in a wide range of applications in recent years. For NLOS imaging, if there is enough reliable data, the data-driven methods can learn the scene prior effectively and improve reconstruction quality. Moreover, although the data-driven methods take a long time in the training phase and require high-performance computing hardware, it can almost achieve real-time processing in the test phase, which helps improve the real-time performance of NLOS imaging.

Limited by datasets and generalization, data-driven NLOS imaging research is still in the early stage. In recent years, several studies utilized deep learning to propose innovative reconstruction algorithms. The existing works can be roughly divided into two categories. One is to exploit the powerful representation ability of deep learning to build an end-to-end neural network directly (Fig. 6-(a)). The other is to combine the advantages of deep learning with physical models' constraints to improve imaging performance (Fig. 6-(b)).

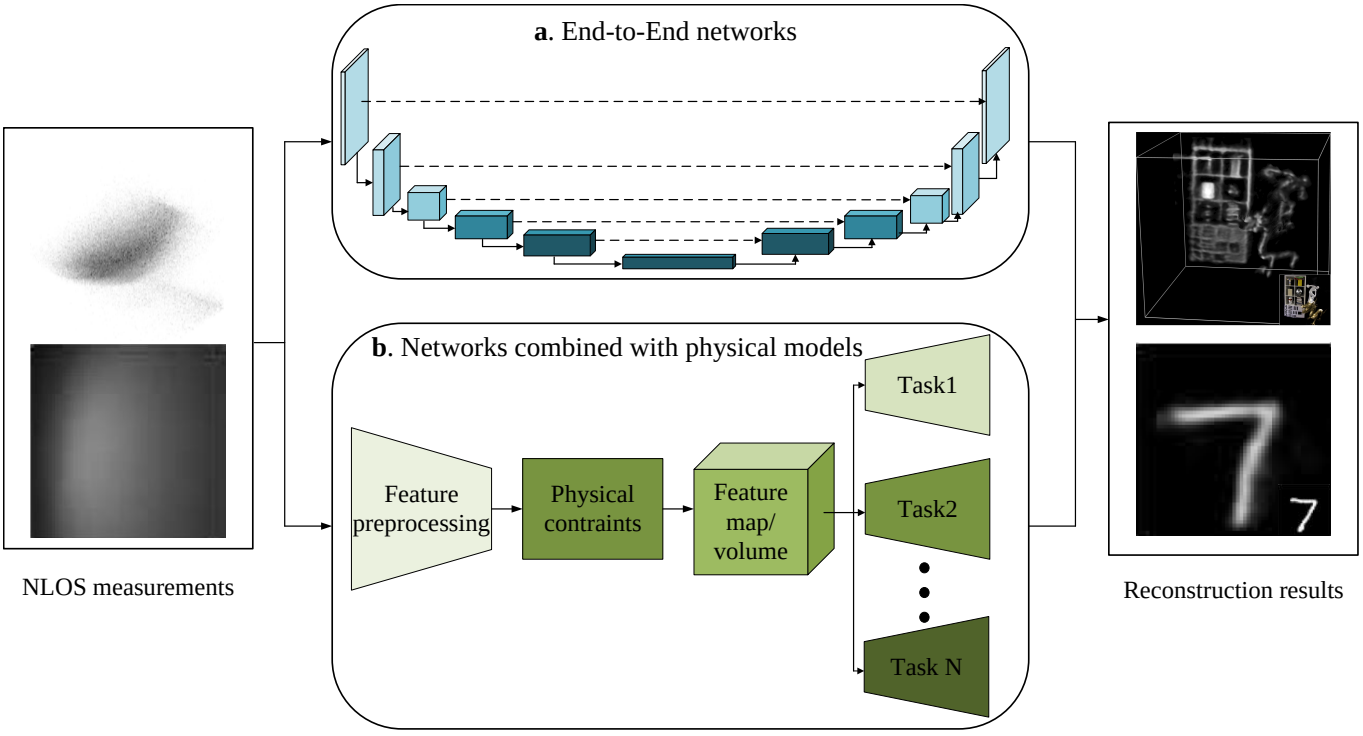


Fig. 6. Two types of models of NLOS imaging based on deep learning: (a) End-to-End Networks. (b) Physics-based Models. The reconstruction results in [18] and [314] are used as a demonstration.

B. End-to-End algorithms

NLOS imaging can be expressed as an inverse problem, as shown in Eq. (7). Therefore, an end-to-end deep learning model can be built, where the input is the measurement (e.g., τ), and the output is the reconstruction of the hidden scene (e.g., $\hat{\rho}$). Then, a loss function is used to measure the difference between the output $\hat{\rho}$ and ground truth ρ, followed by back-propagation used to optimize the network weights, and finally, a good reconstruction result can be obtained.

Most of the existed NLOS imaging works based on deep learning used the end-to-end network structure, as reviewed in Tab. III. Chen et al. used conventional cameras and continuous laser illumination to collect steady-state NLOS imaging data and corresponding hidden scenes [127]. After that, a U-Net [335] based network was trained to complete the mapping from the measurement to the corresponding hidden scenes. In [127], the loss function was a multi-scale (for different resolutions) L_2 loss function.

Diffusion priors for indoor steady-state NLOS. Gao et al. introduced NLOSdiffuser [336], a two-stage learned reconstruction pipeline for indoor steady-state NLOS. A coarse pixel-level inverse first extracts hidden-scene structure from relay-wall intensities; a pretrained diffusion model then refines semantic detail. This extends earlier steady-state U-Net inverses toward heterogeneous indoor scenes without requiring transient photon timing.

Cross-domain transfer for active steady-state NLOS. Liang et al. extended learned steady-state reconstruction beyond a direct wall-image-to-hidden-image mapping with HP-CDT [337]. Its hybrid perceptron uses hierarchical pooling and feature fusion to preserve primitive cues, combines convolutional local perception with Transformer-style global correlation modeling, and promotes interactions across feature levels. The cross-domain transfer stage then treats line-of-sight latent representations as structural priors that steer the optimization of NLOS features. Validation on rendered data and a measured active steady-state setup shows that priors learned from the easier LOS domain can reduce the ill-posedness of continuous multi-bounce inversion without requiring a heavy deployment model. This result connects the original U-Net steady-state baseline to later physics–data hybrids by making representation transfer, rather than only network depth, the source of additional hidden-scene information. Similarly, [314] and [315] used the U-Net based network structure, with cross-entropy and ordinary L_2 norm as loss function to complete the passive NLOS imaging in Fig. 1-(c) and (d). They achieved better imaging performance than conventional methods [8], [291], as shown in Fig. 5-(b). For ToF-based transient NLOS imaging scenes, Chopite et al. synthesized a large number of training images through rendering and noise models, and modified part of the input and output in U-Net from 2D tensor to 3D tensor using L_2 loss [17]. Finally, based on the end-to-end deep neural network, [17] completed the mapping from transient measurement to depth map.

Direct NLOS photography. Most active transient methods optimize an intermediate hidden volume, depth map, or surface before producing a human-interpretable image. Peng et al. instead formulated NLOS photography as direct synthesis of a two-dimensional view from the relay-wall viewpoint [338]. Their data-driven model maps measured transient transport

directly to a high-resolution photograph, bypassing explicit three-dimensional geometry and allowing a relatively small training set. This change of target complements LCT, f–k, phasor-field, and neural-field reconstruction: it treats hidden texture and appearance as first-class outputs rather than secondary attributes of a recovered volume, anticipating later learned systems that optimize perceptual fidelity in addition to geometric accuracy.

Compressive simulation-to-real transient learning. Zhu et al. directly targeted acquisition reduction with a convolutional network that maps sparse photon time-of-flight histograms to hidden two-dimensional images [97]. Training uses only simulated data, whereas evaluation includes physical captures; 8×8 relay samples reconstruct a 32×32 output. Relative to earlier dense end-to-end mappings, this study makes scan compression the central learning objective and foreshadows later signal-superresolution, under-scanning, virtual-scanning, and masked-transient-pretraining methods.

In addition to reconstruction tasks, recognition is also an important goal in NLOS scenes. Due to the limited capability of conventional algorithms in high-level semantics, NLOS recognition mainly uses data-driven end-to-end algorithms based on deep learning. Lei et al. completed the recognition of MNIST [339] and human posture under different NLOS settings [142]. [104] showed that the deep neural network can classify and recognize hidden scenes with only one SPAD scanning point. In addition, a recent study [108] has combined LSTM [340] and physical models to complete 3D human pose estimation. Besides, Satat et al. used deep neural networks to complete object classification through scattering media [341]. For passive NLOS scenes, Tancik et al. utilized CNN for positioning and recognition, achieving a high recognition accuracy [145]. Besides, [145] also explored the use of variational autoencoder (VAE) [342] to complete NLOS reconstruction. Du et al. designed a parallel encoder architecture for passive NLOS reconstruction that simultaneously processes multi-scale spatial features of the passive scattering measurement, enabling improved reconstruction fidelity with lower computational overhead compared to single-path encoder designs [321]. For joint NLOS scene understanding, Wang et al. proposed NLOS-R 2, which alternates between reconstruction and recognition in a mutual-refinement loop: semantic predictions from the recognition branch feed back as structural priors to sharpen reconstruction, while the improved reconstruction in turn enhances recognition accuracy [322]. This alternate training strategy achieves superior performance on both reconstruction and recognition tasks simultaneously, demonstrated on IEEE ICME 2025.

Bounded passive hidden-screen reconstruction. Zhou et al. studied single-frame passive reconstruction of a hidden screen from an ordinary RGB photograph of a diffuse relay wall [343]. Their survivability-guided representation learning (SGRL) approach constructs complementary feature paths with Gaussian-initialized depthwise operators and combines them using content-conditioned cross-band fusion. The learned wall-to-screen inverse recovers broad brightness and layout structure under controlled conditions without active illumination, transient sensing, or geometric calibration, but does not establish reliable fine-text recovery or cross-room generalization.

Learned illumination control and multi-person pose sensing. Semantic NLOS systems increasingly optimize not only the decoder but also how indirect evidence is acquired. Chandran et al. use an off-the-shelf projector and camera and train a message-passing network to infer scene structure and select the spotlight position that maximizes downstream NLOS localization and posture-classification performance [144]. The illumination position is therefore learned end-to-end rather than supplied as a separately supervised target, and synthetic plus real experiments extend learned steady-state sensing beyond a fixed planar-relay configuration. Hou et al. address a complementary limitation of transient semantic sensing: previous pose methods largely assume one hidden person. Their AMPE-NLOS pipeline first uses LCT to form coarse three-dimensional features, refines them with a 3D U-Net, and then predicts body-center heatmaps and SMPL parameter maps so a variable number of people can be separated by body-center-guided sampling [109]. Experiments on simulated multi-person transients and a self-built confocal laser/SPAD system move active NLOS from single-person pose inference toward multi-person three-dimensional mesh sensing.

Robust transient human-pose estimation. Xiao et al. further target semantic recovery when the transient signal itself becomes severely degraded [111]. Their pipeline reconstructs a hidden three-dimensional volume, derives complementary depth and intensity representations, and fuses them in a multi-stage network for three-dimensional joint estimation. A physics-based NLOS simulator converts ordinary smartphone human videos into large-scale pose training data, reducing dependence on costly transient capture. Experiments with a self-built laser/SPAD system retain useful pose estimates at relay depths up to 1.75 m and SNR as low as 0.13, shifting transient human-pose sensing from proof-of-concept inference toward robustness under weak-photon and longer-range conditions.

From reconstruction to recognition and clustering. Recent work increasingly treats reconstructed NLOS observations as inputs to semantic inference rather than as final outputs. Sun et al. introduced AME-Net for passive hidden-action recognition from subtle relay-wall video and released the NLOS-Action synthetic/real benchmark [262]. Lin et al. then considered multiple photon-corrupted NLOS reconstructions as complementary views: NCR-MVC repairs inconsistent neighborhood graphs with shared-neighbor reweighting, Huber-smoothed dual consistency gates, and adaptive inverse-loss view weights, while retaining closed-form alternating updates and a convergence guarantee [344]. The same group addressed temporally truncated photon sequences with CA-SlotNet [345]: CLAHE- and Sobel-guided local-contrast modulation adjusts slot-attention logits, while a physically guided slot regularizer stabilizes object routing when only part of the transient is observed. QSS-Net further makes efficient NLOS classification itself the target, representing the field's shift from complete geometric recovery toward time-sensitive semantic

decisions [346]. Together with learned feature embeddings and NLOS-R^2, these studies establish truncation-robust classification, recognition, action understanding, and clustering as a parallel trajectory to hidden-scene reconstruction.

Calibrated wall-mediated semantic inference. Dai et al. use only a calibrated wall-mediated observation of a display outside the camera field of view for 31-way category inference [347]. Scene-constrained regularized inversion reorganizes weak indirect evidence, while multi-stage Haar-subband attention preserves discriminative mid- and high-frequency cues under attenuation, ambient background, and matched operator re-parameterization. The method performs semantic NLOS sensing rather than full hidden-image, depth, or geometry reconstruction.

Physics-informed semantic learning from multipath radar. The semantic NLOS trajectory is not limited to optical relay-wall images. Zeng et al. fuse local scattering and global multipath structure with adaptive convolution scales for measured hidden-target classification [348], whereas Zhong et al. use separated radar paths as physically meaningful contrastive views and impose consistency between observed and regressed motion quantities [349]. The latter achieves strong low-label human-activity recognition, illustrating how propagation-path structure can replace generic image augmentations in self-supervised NLOS representation learning.

C. Network combined with physical models

Dual-model guidance for under-scanned transients. Yan et al. first recover sufficient-density measurements from sparse transients with three-dimensional pyramid pooling and window attention, then combine LCT and f–k branches to retain complementary global structure and fine texture [350]. Multi-scale refinement and adaptive fusion place classical geometric- and wave-optics inverses inside one learned under-scanning pipeline, extending measurement completion beyond a single fixed reconstruction backend.

The end-to-end method is convenient for training, but it is highly dependent on the dataset. When the dataset scale is large enough, end-to-end learning can learn the mapping well to achieve good results. However, end-to-end learning may have problems such as over-fitting and poor generalization ability when the data size is insufficient. Due to the cumbersome data acquisition process, NLOS imaging lacks real large-scale datasets (most of the existing large-scale datasets are synthetic). Hence, some state-of-the-art studies in recent years proposed a reconstruction algorithm combining deep learning and physical models instead of the end-to-end network only, as shown in Tab. III.

Metzler et al. used the optical memory effect (see Sec. III-III-C) to complete NLOS imaging [19]. Instead of using conventional phase recovery algorithms (e.g., hybrid input-output (HIO) [310] and alternating minimization (Alt-Min) [351]), [19] adopted U-Net to complete phase recovery, with the L_1 norm after autocorrelation as loss function. **Shared learned representations for reconstruction and recognition.** Chen et al. introduced a learned feature-embedding framework that maps transient measurements into a common hidden-scene representation and then specializes it for high-resolution image reconstruction, classification, and 2.5D object detection [18]. Rather than relying on a monolithic U-Net, the architecture incorporates differentiable modules with explicit physical roles, including transient propagation, visibility reasoning, image rendering, and depth estimation. Training on synthetically rendered diffuse and specular scenes while evaluating on measured transients demonstrated useful synthetic-to-real generalization. This work marks an early shift from single-output NLOS inversion toward reusable, task-aware representations, anticipating later physics-guided multi-task networks, transient pretraining, and transformer/operator models.

Masked transient pretraining. MARMOT moves reusable representation learning from multi-task supervision to self-supervised transient completion [352]. A Transformer encoder–decoder receives measurements hidden by a scanning-pattern mask; the retained subset is functionally equivalent to an arbitrary relay sampling pattern, and the network predicts the complete transient volume. In the final Visual Intelligence version, TransVerse contains one million confocal NLOS transients rendered from 500,000 Objaverse objects, enabling pretraining at a scale well beyond earlier task-specific transient datasets. The recovered dense transients support sparse-scan NLOS reconstruction, while the pretrained encoder transfers to classification, albedo estimation, and depth estimation. This development changes the role of a large transient dataset: instead of training one reconstruction mapping, it supplies a modality-level prior that can be adapted across sampling regimes and task heads.

Learnable inverse kernels and attention. Yu et al. subsequently made the inverse operator itself trainable by introducing a learnable inverse kernel tailored to the transient NLOS point-spread function, together with attention mechanisms for recovering hidden intensity and depth [52]. The method remains explicitly tied to the active transient forward model while allowing data-driven adaptation beyond a fixed analytic inverse. In the learned-reconstruction trajectory, it provides an intermediate step between the shared, task-aware feature embeddings of Chen et al. and later spatial–temporal transformers or adaptive physical-prior networks: physics determines the operator structure, whereas learned kernels and attention determine how measurement evidence is inverted and emphasized.

Learning the light-cone inverse. The original LCT obtains computational efficiency from a closed-form resampling and Wiener-deconvolution pipeline, but fixed low-pass filtering can suppress fine geometry. Gao et al. proposed a learned light-cone transform (L2CT) that retains the analytic LCT structure while replacing its frequency response with high-frequency-enhanced learnable Wiener filtering and adding spatiotemporal feature-extraction and volume-projection modules [64]. Its improvement on measurements from different acquisition systems illustrates a useful middle ground between fixed analytic inverses and unconstrained end-to-end networks: the milestone operator remains recognizable, while the poorly conditioned spectral components are learned from data.

Frequency-domain expert regularization. Zhang et al. introduce multi-regularization-experts fusion for robust transient NLOS reconstruction [353]. Complementary spectral regularizers replace reliance on a single fixed prior, continuing the transition from analytic inverse filtering to adaptive, physics-structured reconstruction.

Physics-initialized triplane reconstruction. Do and Nam proposed TriNLOS [354], which initializes a hidden-scene volume using a learnable enhanced light-cone transform and replaces cubic-cost dense 3D processing with quadratic-cost triplane features. ConvNeXt-style convolutions, Restormer attention, and axis-aware cross-attention refine hidden geometry. This ECCV work connects the milestone LCT inverse to compact neural representations and complements transient transformers by reducing volumetric representation cost.

Fermat-point representations for cross-system generalization. Shen et al. introduced FermatFormer, a physics-guided NLOS architecture that represents transient measurements through stable Fermat points [355]. By retaining light-path structure, this representation aims to improve transfer from synthetic training data to measured scenes and complements learned LCT and phasor-field inversion.

Graph and temporal-attention reconstruction. DG-NLOS converts dense transient grids into sparse graph features and uses separate albedo and depth branches to recover complementary hidden-scene attributes [356]. Time-Resolved Transformer instead models spatiotemporal dependencies across transient measurements for three-dimensional reconstruction [357]. These approaches complement fixed-grid convolutional inversion by emphasizing either sparse geometry or global temporal context. An earlier COSI 2024 conference-digest contribution by Yue et al. explored permutation-based transient attention with Permute-MLP mixing for spatiotemporal feature extraction [358]; as a short conference report, it is best viewed as a precursor rather than a fully validated reconstruction benchmark.

Diffusion-based transient video interpolation. Dynamic NLOS systems face coupled spatial and temporal undersampling before reconstruction begins. Sun et al. introduced TransVID, which uses a spatial–temporal attention encoder and latent conditional diffusion to interpolate transient videos jointly across scan position and time [66]. The method maps 16×16 measurements at 4 fps to 128×128 transient videos at 16 fps, providing a measurement-domain complement to ST-Mamba and TransiT: instead of reconstructing hidden video directly from sparse captures, it first synthesizes a dense, high-frame-rate transient sequence that can feed existing physical or learned inverses.

Noise-conditioned neural operators for transient inversion. Wang et al. recast confocal transient NLOS reconstruction as learning an inverse operator between continuous function spaces rather than a mapping between fixed-resolution tensors [65]. Their Noise-Adapted Neural Operator (NANO) first estimates the noise level of the acquired transient and uses it as a conditioning variable for an operator unfolded from a regularized gradient-flow formulation. Global–local transient lifting then couples large-scale structural information with local ballistic details before volumetric albedo projection. This operator viewpoint targets discretization invariance across different scan densities and irregular or severely down-sampled illumination patterns, while the explicit noise conditioning adapts the inverse kernel to photon-starved and non-stationary measurements. Tests on simulated and measured data therefore extend the trajectory from learnable inverse kernels and LCT-style physical priors toward mesh-independent operator learning for acquisition- and noise-robust NLOS reconstruction. At the time of this survey update, the work is available as arXiv:2508.09655 and no final accepted or published venue could be verified.

Attention-based Poisson denoising for single-photon NLOS. Tu et al. trained an attention-based feature-enhancement denoising network (AEF-Net) on Monte Carlo-simulated NLOS photon measurements [359]. Their method suppresses Poisson noise in reconstructed single-photon NLOS images and was also evaluated with a physical imaging system. This earlier, data-driven noise-removal strategy complements subsequent SPAD-aware, physics-informed cascades by isolating photon-noise suppression from the downstream hidden-scene inverse problem.

SPAD-aware physics-informed cascade learning. Zhao et al. addressed the coupling between detector noise and ill-posed transient inversion with physics-informed cascade learning (PICL) [360]. A lightweight first network separates SPAD-specific dark-count, timing-jitter, and mixed interference components, after which a second network reconstructs the hidden scene while embedding a differentiable forward model. Because physical measurement consistency supplies self-supervision, PICL avoids dependence on a large paired NLOS dataset and provides a direct bridge from trainable inverse kernels toward hardware-aware, noise-adaptive reconstruction.

Physics-enhanced white-light reconstruction. For inexpensive broadband acquisition, Fu et al. embedded a speckle-correlation image-formation model and denoising prior into a trainable reconstruction pipeline [242]. Unlike a generic wall-image U-Net, the network exposes the autocorrelation-to-object relationship as an internal physical constraint. This provides a passive/steady-state counterpart to differentiable transient physics: the measurement modality changes, but the learned solver remains anchored to an explicit transport statistic.

Hybrid physics–data passive enhancement. Purely data-driven passive reconstruction can be difficult to interpret and generalize, whereas a fixed light-transport inverse may preserve physics but lose perceptual detail. Liang et al. proposed HPDI, a dual-path architecture that combines a physics-informed coarse-to-fine pathway with a data-driven implicit reconstruction pathway and learns an adaptive fusion of their outputs [320]. Evaluation across passive datasets with and without explicit occluders shows how a recognizable transport model and scene-specific statistical priors can be complementary rather than competing constraints. This hybrid design extends computational periscopy and blind transport factorization toward trainable passive solvers with improved

fidelity, generalization, and data efficiency.

Another example of exploiting deep learning to promote the development of NLOS imaging is LiDAR-based imaging [126]. For traditional physical models, using commercial LiDAR as the equipment is still a very challenging problem. However, the recent work of Zhu et al. showed that by performing deep learning to complete feature extraction, LiDAR can be used to complete high-quality and robust 3D NLOS reconstruction [126].

Aittala et al. regarded the measurement data as the product of the hidden object and the light transport matrix [228] (as shown in Eq. (12)). Exploiting deep neural models to generate matrices satisfied matrix decomposition completes passive NLOS imaging based on unsupervised learning. [361] used unsupervised learning to complete active NLOS imaging. Specifically, [361] exploited multilayer perceptron layers (MLP) to construct a neural transient field that maps measurement data into a hidden scene, and utilized a physical model to constrain the results, thereby obtaining superior reconstruction results. Technically, such unsupervised learning methods [228], [361] are not data-driven methods, but the idea of using deep neural networks to simulate matrix factorization is instructive.

From hidden reconstruction to holistic 3D completion. HOLI-1-to-3 uses NLOS transients not only to reconstruct an isolated hidden scene, but to resolve the invisible side of an object observed from one conventional viewpoint [362]. Its neural plenoptic representation unifies an LOS radiance field with an NLOS transient field, and a two-stage optimization combines diffusion and transient priors to recover continuous visible and occluded geometry. Relative to NeTF, the trajectory expands from representing spherical hidden transport to fusing straight-ray and multi-bounce evidence for holistic generation; relative to purely RGB diffusion, measured transients provide physical constraints on otherwise ambiguous unseen shape.

Multi-bounce and outdoor transient scene reconstruction. Shoot-Bounce-3D treats normally discarded higher-order single-photon LiDAR returns as recoverable geometric evidence [363]. A multiplexed source illuminates several visible points simultaneously, and a learned transport prior trained on roughly 100,000 simulated transients separates the measured two-bounce contribution of each source point. The demultiplexed transport supports dense metric depth together with geometry hidden by occlusion or mirrors from one view and one capture. Although this is not the canonical relay-wall NLOS geometry, it directly extends the field's multi-bounce inverse problem from sequential scanning to practical single-shot acquisition. Transient LASSO broadens the same transient-learning trajectory from controlled indoor transport to outdoor, in-the-wild reconstruction [364]. Its neural scene representation operates on posed raw transient LiDAR, explicitly accounts for back-reflected and ambient components and high-photon-flux sensor behavior, and disentangles geometry and normals from reflectivity, retroreflectivity, and ambient illumination. Together, these studies move transient inverse rendering beyond isolated calibrated hidden volumes toward occlusion-aware single-shot capture and large-scale scene-and-sensor modeling.

Recent works have significantly extended this paradigm of combining physical constraints with deep learning.

Learning from under-scanned transients. Dense relay-wall scanning remained a practical acquisition bottleneck even after analytic reconstruction became fast. Li et al. introduced the first deep pipeline designed specifically for under-scanning measurements, coupling a transient recovery network that maps sparse scans to a dense measurement grid with a volume reconstruction network that embeds the linear light-path transport operator [60]. Multi-kernel feature extraction and fusion recover measurement structure, while regularized constraints preserve local detail and suppress oversmoothing. The method remains effective for 8×8 confocal scans and reports approximately $50\times$ faster inference than an iterative under-scanning baseline. This work shifts learned active NLOS from post-reconstruction refinement toward joint measurement completion and physics-aware inversion, preceding later signal-superresolution, virtual-scanning, diffusion, and transient-transformer approaches.

Neural implicit surfaces. Extending the neural transient field (NeTF) [361] to surface representations, Fujimura et al. proposed NLOS-NeuS [199], which represents the hidden scene as a neural implicit surface via a signed distance function (SDF). By introducing SDF constraints based on first-returning photon geometry and volume rendering, NLOS-NeuS reconstructs smooth and detailed 3D surfaces without voxel resolution limitations, representing a major step towards high-fidelity 3D NLOS scene understanding.

Learnable physical priors. Sun et al. introduced a framework with learnable physical priors [365], replacing manually designed fixed priors (such as single fixed path compensation) with adaptive priors learned from data. This enables significantly better generalization to diverse scene conditions and varying SNR levels, addressing the limited generalization capability that has been a persistent challenge for physics-based deep NLOS methods.

Gradient-coordinated physics-guided training. Ling et al. identify a complementary failure mode of physics-guided learning: heterogeneous reconstruction, measurement-consistency, and sensor-calibration objectives can produce conflicting gradients even when every loss term is physically meaningful [366]. Rather than tuning another scalar loss weighting, their framework coordinates the gradient families explicitly through soft conflict projection, protected physical routing, learnable instrument calibration, and staged unfreezing. Evaluated with NLOST-style transient reconstruction under multiple low-SNR conditions, the study shifts physics–data fusion from adding priors to the objective toward governing how competing physical constraints are allowed to update shared parameters; the coordination is used during training and does not require a heavier inference-time reconstruction architecture.

Diffusion model priors. Su et al. proposed a model-guided iterative diffusion sampling approach for NLOS reconstruction [32], where the physical forward model of NLOS imaging guides the diffusion sampling process to generate reconstructions that satisfy measurement constraints while leveraging

the strong generative prior of a pre-trained diffusion model. This approach achieves state-of-the-art NLOS reconstruction quality, demonstrating that diffusion models are a powerful new tool for solving the NLOS inverse problem.

Diffusion-prior Fourier single-pixel imaging. Fu et al. combined Fourier single-pixel acquisition with a generative diffusion prior for hidden imaging under severe scattering and extreme undersampling [367]. Measured low-frequency Fourier coefficients remain enforced as a data-consistency constraint during iterative denoising, while the learned prior supplies the missing high-frequency structure. Experiments through multiple paper layers at sampling rates down to 3% illustrate a distinct diffusion trajectory from transient-volume reconstruction: the generative prior operates directly on compressed single-pixel spectral measurements.

Plug-and-Play for dynamic NLOS. Ye et al. introduced a plug-and-play (PnP) framework for dynamic NLOS video reconstruction [368], incorporating a frequency-filtering step based on wave physics and a pre-trained deep neural network denoiser as an implicit prior. The flexible PnP structure allows easy integration of different denoisers and adapts to different NLOS imaging configurations, achieving superior temporal consistency over existing frame-by-frame reconstruction methods.

Unsupervised reconstruction from irregular measurements. Cui et al. proposed Virtual Scanning [213], an unsupervised framework for NLOS imaging from irregularly undersampled transients (IUT) that does not require paired training data. A virtual scanning process enables the network to learn within both the range and null spaces of the measurement operator, combined with a physics-guided SURE-based denoiser for robustness to noise. This method achieves higher fidelity and orders-of-magnitude faster inference than existing iterative solutions.

Aperture-limited diffusion and physics-guided high-speed reconstruction. TransDiff [369] extends unsupervised transient completion to aperture-limited relay surfaces using a latent diffusion prior constrained by the measured transients. In parallel, Physics to the Rescue [61] embeds complementary wave-propagation and volume-rendering priors into a feed-forward reconstruction model, improving robustness when a 5 Hz non-confocal acquisition system requires approximate light transport. Its flexible intensity- or raw-transient supervision and synthetic-to-real intensity/depth inference connect high-speed capture to later physics-constrained neural fields and differentiable transient rendering.

Untrained network priors. Wu et al. demonstrated that an untrained deep decoder network can serve as an effective image prior for NLOS reconstruction without any training data [370], leveraging the deep image prior inherent in the network architecture to regularize the ill-posed NLOS inverse problem, offering flexibility for novel experimental configurations where training data is unavailable.

Memory-efficient MetaFormer reconstruction. Transient networks improve quality by modeling the joint spatial–temporal volume, but their memory cost can prevent training and deployment on commodity hardware. Zhang et al. introduced CMFormer, whose multidimensional global and dilated convolutions act as a purely convolutional MetaFormer token mixer [371]. Aggregate feature transmission replaces width-expanding skip connections, cross-layer attention restores detail lost during downsampling, and gradient checkpointing further limits training memory. The reported 8-fps inference and synthetic-to-real evaluation make efficiency itself a learned-reconstruction objective, complementing NLOST-style global modeling rather than merely shrinking input resolution.

KAN-enhanced transient Transformers with semantic guidance. Liu et al. combined a Kolmogorov–Arnold Network-enhanced Transformer with convolutional attention for active transient NLOS reconstruction [372]. Learnable KAN activations model nonlinear global transport features, the convolutional branch preserves local structure, and EfficientSAM-derived contours enter through a semantic loss. The measured-data evaluation places this method in the trajectory from NLOST and TransiT toward hybrid architectures that jointly exploit global transient dependencies, local spatial evidence, and pretrained semantic priors.

Latent graph and zero-shot mixture regularization. Xu et al. compressed the hidden volume along its relatively redundant depth profile and built a group graph-Laplacian feature extractor in the latent space [373]. Used as a regularization-by-denoising structural term, it combines learned feature statistics with iterative measurement inversion while lowering the cost of full three-dimensional nonlocal graphs. Zhang et al. take a complementary training-free route: a mixture distribution separates target and background residual behavior, and dual-space adaptive weights become a zero-shot attention mechanism inside alternating minimization [374]. Together these methods show that attention-like adaptivity need not be confined to supervised neural networks; it can also emerge from probabilistic residual models and optimization dual variables.

Ground-truth-independent artifact cancellation. Zhou et al. introduced time-of-flight structural similarity (TOF-SSIM), which forward-projects a candidate hidden reconstruction and evaluates its consistency with the measured transient without requiring hidden-scene ground truth [375]. The resulting adaptive artifact-cancellation procedure modifies and backprojects the transient contribution associated with streaking artifacts and applies to both confocal and non-confocal data. This closes a practical gap between fast backprojection and heavier iterative solvers by making reconstruction-parameter selection and artifact suppression measurement-driven.

Reprojection-guided recovery from irregular relay samples. Cui et al. extended transient completion beyond fixed sparse grids to noisy, irregularly undersampled, and spatially fragmented relay measurements [376]. Their transient-recovery network is guided by a range-space reprojection module that extracts measurement-intrinsic structure; spatio-temporal modulation blocks then control where and how strongly this physical guidance affects denoising and interpolation. Evaluation on simulated and measured data shows improved transfer across relay layouts. In the development from under-scanning recovery and Virtual Scanning to TransDiff, this work makes reprojection consistency an explicit

conditioning signal rather than relying only on an implicit learned prior.

Physics-embedded thermal transport learning. NLOS-Former provides a modality-specific example of a network learning an effective transport operator rather than treating the relay observation as an unconstrained image-to-image mapping [276]. Its kernel-estimation branch encodes rough-surface thermal propagation and supports joint appearance and relative-depth inference; the primary modality discussion appears in Sec. V.

D. Challenges and Prospects

NLOS imaging based on deep learning still faces many challenges, including lack of datasets, lockstep network structure, and limited generalization capability. Here, we discuss these three challenges and the corresponding prospects separately.

1) Dataset: Data-driven algorithms' performance largely depends on the dataset's quality (such as scale and reliability). However, for transient NLOS imaging, acquiring a sample of data requires a raster scan of the imaging area, which takes about 1 to 5 minutes [1], [210]. Therefore, constantly changing hidden objects and collecting tens of thousands of measurements to form a large-scale dataset is extremely difficult. For NLOS imaging scenes based on traditional cameras, although data collection is less time-consuming, it is still complicated to continuously change scenes and construct large-scale dataset because data augmentation operations such as crop and flip are not suitable for NLOS imaging.

Most existing works construct a rendering model to synthesize datasets for training, i.e., they first use public datasets in other fields as the hidden object (ground truth) and then synthesize measurement data through a rendering model. For transient NLOS imaging, the rendering model includes two parts: transient rendering and sensor model [17], [18]. The SPAD-based photon calculation model has been studied in detail, and they take into account factors such as noise, crosstalk and afterpulsing [377]. For other types of NLOS imaging, Metzler et al. used Berkeley Segmentation Dataset 500 [378] as a hidden scene to calculate autocorrelation, which is then used to train the phase recovery network [19]. For passive NLOS imaging using traditional cameras and ambient light, diffuse surface illumination models, such as the Phong Model [379], can be used for rendering to obtain synthesized passive NLOS data [314].

Due to the sensitivity of deep learning algorithms to abnormal data, improving the accuracy of rendering models is extremely important to data-driven NLOS imaging methods. Another alternative is designing an automated/real-time data collection system to collect data in the experimental setup.

Since 2022, dedicated NLOS rendering tools have emerged to generate higher-fidelity synthetic training data. Royo et al. presented the first NLOS transient renderer designed to accurately simulate light transport in NLOS scenarios, enabling generation of realistic training data and testing of reconstruction algorithms [380]. Plack et al. subsequently developed a fast differentiable transient renderer for NLOS scenes that supports automatic differentiation, enabling gradient-based optimization of scene parameters and end-to-end training of NLOS reconstruction networks [381]. For passive NLOS tracking, Wang et al. released the NLOS-Track dataset, the first large-scale dynamic passive NLOS tracking dataset with thousands of real and synthetic video clips paired with trajectory labels [312], addressing the severe data scarcity problem in passive NLOS research.

Complementing physically complete transient renderers with a benchmark-oriented alternative, Shi et al. introduced a configurable simulator grounded in the light-cone transform and accelerated by three-dimensional FFT convolution [382]. The framework accepts depth and optional albedo maps, exposes relay-wall extent, stand-off distance, temporal binning, detector jitter, and Poisson noise as explicit parameters, and releases transient data for seven ShapeNet object categories together with common LCT, phasor-field, f–k, and back-projection baselines. This development lowers the cost of standardized algorithm comparison and large-scale learning, while remaining complementary to Monte Carlo transient rendering when higher-order transport, complex materials, or full sensor realism are required.

Differentiable transient simulation infrastructure. Royo et al. released *mitransient*, an extension of Mitsuba 3 for CPU/GPU time-resolved light transport [383]. The toolkit supports realistic materials and participating media, transient polarization tracking, differentiable rendering, NLOS scene construction, and capture-noise simulation. It complements task-specific synthetic datasets by providing a reusable forward-model layer for testing acquisition designs, generating training data, and optimizing transient inverse problems under a consistent physically based renderer.

2) Network Architecture: According to Sec. IV IV-B and IV-C, most NLOS imaging networks currently use the classic network structure (or with tiny changes), such as U-Net [335] and ResNet [384]. However, these well-known network structures are not necessarily suitable for NLOS imaging. Compared with recent works in other inverse problems, such as low-light image recovery [385] and image deblurring [386], networks designed for specific tasks achieve better results. Therefore, it can be expected that in the future, more novel network architectures based on the characteristics of NLOS imaging(e.g., [18]) will be proposed to improve the performance of NLOS imaging.

Since 2022, the NLOS community has indeed witnessed a wave of new architectures designed specifically for the challenges of 3D transient imaging.

Transformer-based networks. Li et al. proposed NLOST [28], the first transformer specifically designed for NLOS 3D reconstruction. NLOST uses spatial-temporal self-attention at multiple scales to capture both local and global correlations within the high-dimensional transient measurement volume, achieving superior performance on real-world confocal and non-confocal data. For NLOS videography, Li et al. developed TransiT [29], a transient transformer architecture that directly compresses the temporal dimension of input transients to extract features efficiently at high frame rates, combining feature fusion and spatial-

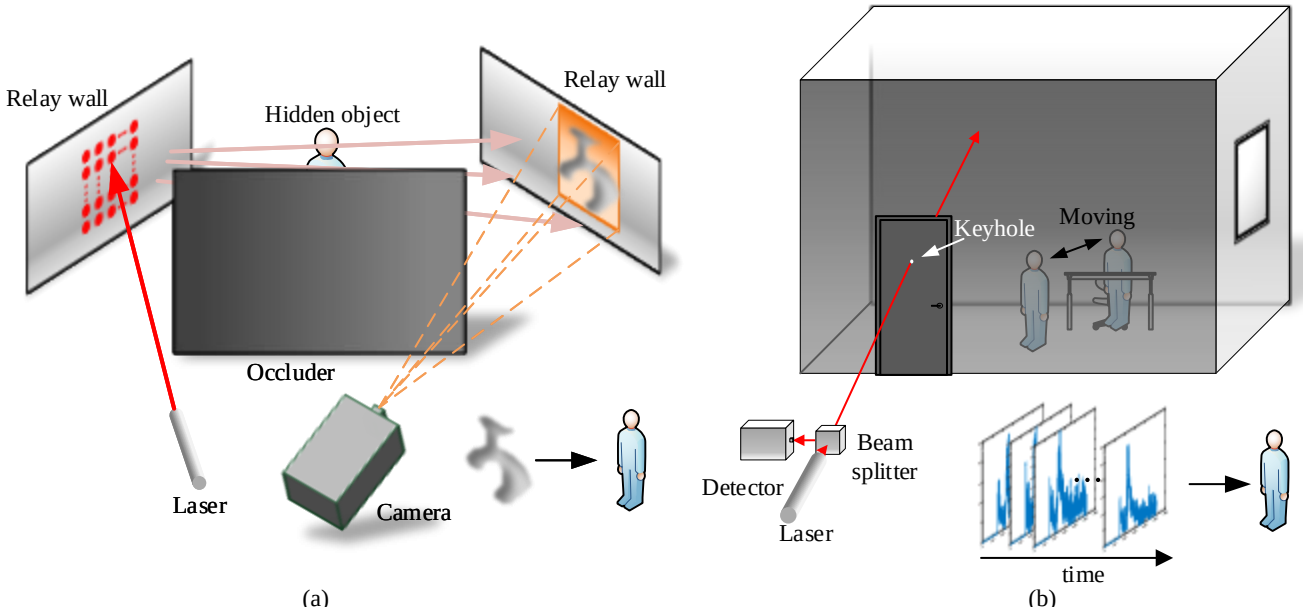


Fig. 7. New NLOS scenes. (a) Two bounce NLOS imaging. (b) Keyhole NLOS imaging.

temporal attention with transfer learning to bridge the synthetic-to-real gap.

State-space model (Mamba) for video NLOS. Li et al. proposed ST-Mamba [30], the first spatial-temporal Mamba model for dynamic NLOS reconstruction. ST-Mamba exploits the time-resolving causality in transient measurements through interleaved Mamba blocks, incorporating a wave-based loss function in the phasor field domain to handle measurement noise in NLOS video. This approach achieves superior temporal consistency across frames compared to per-frame reconstruction methods.

Hybrid state-space and windowed-attention reconstruction. Building on the Transformer and state-space branches, Jin et al. introduced NLOS-MT at ICPR 2026 [387]. The network first forms an initial hidden intensity representation and then applies a DeformMamba block, where deformable convolution adapts to object shape while Mamba's selective state-space mechanism captures long-range dependencies with linear-complexity sequence modeling. A U-Net equipped with windowed attention subsequently restores local contextual detail. This global–local division complements NLOST's spatio-temporal self-attention, TransiT's transient Transformer, and ST-Mamba's temporal state-space modeling: rather than choosing attention or state-space layers as the dominant backbone, NLOS-MT explicitly combines both for denoising and structural reconstruction, with experiments on synthetic and publicly available real NLOS datasets.

Graph neural networks. Su et al. proposed DG-NLOS [31], which employs a graph neural network (GNN) as its fundamental component to transform dense NLOS grid data into sparse structural features. DG-NLOS uses a dual-branch architecture with one branch dedicated to albedo recovery and another for depth/geometry, enabling simultaneous reconstruction of appearance and structure with reduced computational and memory cost. To the authors' knowledge, DG-NLOS is the first work to employ GNNs as a core module for NLOS reconstruction.

3) Generalization: Limited by the dataset and network structure discussed above, current data-driven NLOS imaging algorithms have limited generalization capabilities. [17] and [18] can get good test results on the public NLOS data [1], [388] through training on the synthetic training set, which means that they have a certain generalization ability. However, this does not mean that the generalization capabilities of current algorithms are sufficient. Faced with new parameters (such as BRDF, time jitter, aperture size) and new hidden scenes that do not exist in the training dataset, the trained network may not necessarily achieve acceptable results.

The most two direct ways to increase the generalization ability are improving the dataset's quality and the network structure, as discussed above. Another method is to use transfer learning so that a network trained on one kind of data can be quickly transferred to another kind of data. Besides, unsupervised learning is also a promising method. It does not rely on large amounts of data, but can use deep neural networks' powerful representation capabilities to train a model that meets the physical constraints (e.g., the optical transport matrix in [228]), which has more substantial generalization capabilities.

Recent advances have made meaningful progress on the generalization problem. Sun et al. proposed a framework with learnable physical priors [365] that adapts to diverse imaging conditions automatically, demonstrating substantially improved generalization across different SNR levels and scene configurations. Domain reduction strategies [197] that focus the reconstruction search space on the most likely hidden object locations also improve both speed and generalization. Transfer learning has been successfully applied in TransiT [29] to bridge the synthetic-to-real domain gap for NLOS videography. Looking forward, the combination of

physics-constrained learning, untrained priors [370], and self-supervised approaches [213] holds promise for enabling robust NLOS reconstruction even without large-scale real training data.

E. Dynamic NLOS Imaging

Real-time reconstruction of dynamic hidden scenes is one of the most practically significant challenges in NLOS imaging, and it has seen remarkable progress since 2022. The main bottleneck is that dense raster scanning limits the practical frame rate to well below real-time, creating a fundamental tension between acquisition density and temporal resolution.

Towards resolving this tension, several strategies have emerged. Ye et al. demonstrated real-time NLOS video at 4 fps for room-sized scenes by combining spectrum filtering (denoising/deblurring in the frequency domain) with an interleaved scanning scheme that computes high-resolution live video from low-quality sequences acquired during scanning [33]. Li et al. proposed ST-Mamba [30], which exploits temporal consistency across frames using spatial-temporal Mamba blocks and a wave-based phasor field loss to reconstruct high-resolution NLOS volumes from under-scanning transient videos. For videography with a transient transformer, TransiT [29] achieves 10 fps reconstruction at 64×64 resolution from 16×16 sparse transients captured at 0.4 ms exposure per point, demonstrating practical video NLOS reconstruction on real hardware. Ye et al. further developed a plug-and-play algorithm for dynamic NLOS video [368] that combines frequency-domain physics filtering with a deep denoiser prior, enabling flexible and high-quality reconstruction of moving hidden scenes.

V. New NLOS Scenes

In Sec. II-IV, most existing NLOS imaging studies have been introduced. These studies are accomplished in active and passive NLOS scenarios, as shown in Fig. 1. However, NLOS imaging is not limited to these scenes, and more new types of scenes are waiting to be developed. In this section, we take the recent two bounce NLOS imaging [20] and keyhole NLOS imaging [21] as examples to introduce potential new NLOS scenes and their prospects. We further summarize the 2022–2026 expansion of NLOS imaging into sparse and irregular active acquisition, active corner cameras, ptychographic depth-resolved NLOS, transient light-transport-matrix iteration, phasor-field enhancement under partial sampling, scattering-media robustness, few-prior passive transport estimation, low-cost LiDAR spatial reasoning, robotic navigation and exploration, radar/RF/mmWave sensing, passive electromagnetic-skin-aided radar imaging, acoustic and ultrasound sensing, terahertz waves, and semantic hidden-scene understanding.

A. Two bounce NLOS imaging

Henley et al. introduced the novel "two-bounce NLOS imaging", as shown in Fig. 7-(a) [20]. In this new scene, the hidden object is blocked by an occluder, but there are two diffuse reflection surfaces W_1 and W_2 on both sides of the hidden object. The observer can reconstruct the hidden scene by scanning the laser on one of the diffuse reflection surfaces (e.g., W_1) and using a traditional camera to capture the two-bounce light on the other reflection surface. In other NLOS imaging, hidden objects play a role in reflecting light. However, in two bounce NLOS imaging, hidden objects play a role in blocking light. By capturing the projection of light on W_2 after being blocked by a hidden object, whether there is a hidden object point in each voxel can be judged.

As discussed in [20], two bounce NLOS imaging can be applied to seeing behind trucks for autonomous vehicles and imaging between windows for search and rescue. These applications are extensions to traditional NLOS imaging (i.e., three-bounce NLOS imaging or passive NLOS imaging) applications and illustrate the broad application prospects of "turning walls into mirrors".

B. Two-Bounce NLOS: Role of Transients

Building on the two-bounce NLOS setup, Somasundaram et al. provided the first systematic analysis of the role of time-of-flight (transient) information in 2B-NLOS imaging under multiplexed illumination [389]. Their study quantifies the tradeoffs between temporal resolution, spatial resolution, and number of image captures, deriving formal mathematical constraints for 2B lidar and showing that transient information can significantly reduce the measurement requirements for shape reconstruction. This analytical framework provides design guidelines for future NLOS systems that incorporate time-gating in two-bounce configurations.

Implicit reconstruction and real-time two-bounce tracking. Between analytical two-bounce measurement design and the newer neural illumination/shadow fields, Chen et al. introduced a hierarchical-NeRF formulation for implicit shadow-based reconstruction [390]. Instead of explicitly carving every candidate voxel from all multi-view rays, the method decomposes shadow observations spatially and temporally, selects rays that carry the strongest geometric constraints, and optimizes a cumulative-transmittance neural field. This reduces redundant ray computation while supporting both static and changing hidden scenes. Zhang et al. then shifted the two-bounce branch from geometry reconstruction toward task-oriented dynamic perception with DCPNet [391]. Their time-multiplexed acquisition alternates two illumination views and fuses cross-view spatial consistency with single-view temporal coherence to recover hidden-object trajectories in corridor/tunnel-style scenes, using physics-rendered sequences for pretraining and real measurements for validation. Together these works fill the 2025 transition from explicit shadow carving toward implicit neural reconstruction and real-time two-bounce tracking, immediately preceding the NIF and D-NeSF neural-field trajectory discussed below.

A complementary steady-state trajectory replaces binary shadow carving with a continuous learned scene model. Zhang et al. introduced Neural Illumination Fields (NIF) [392], which jointly represents hidden-space density and illumination-dependent image formation with multilayer

perceptrons and optimizes the representation through differentiable intensity rendering. Because NIF synthesizes measured shadow images directly, it avoids error-prone binary shadow segmentation and remains effective under low shadow contrast and ambient-light interference. This work shifts two-bounce NLOS from discrete occupancy carving toward self-supervised neural inverse rendering for large hidden volumes.

Dynamic neural shadow fields for moving two-bounce scenes. The static-scene assumption remains restrictive when hidden people or objects move during a cyclic multi-position scan. Zhang et al. addressed this regime with D-NeSF [393], decomposing the four-dimensional hidden volume into six spatiotemporal feature planes whose interpolated features are mapped by an MLP to time-varying voxel density. A cyclic acquisition strategy aligns repeated illumination positions with periodic shadow observations, while a cumulative-transmittance constraint enables self-supervised fitting to captured binary shadow sequences. This work extends the two-bounce neural-field trajectory from high-fidelity static geometry to temporally consistent reconstruction of moving hidden targets, and exposes motion modeling and scan-time synchronization as central design variables for dynamic shadow-based NLOS.

Self-calibrating two-bounce imaging on non-planar relay walls. Moving two-bounce NLOS toward less controlled deployment, Chen et al. remove the assumption that the illumination wall is planar and pre-calibrated [394]. Their method automatically estimates the relative coordinate mapping between the two relay surfaces and then jointly refines ray vectors and shadow formation with a full-link MLP optimization, compensating for spot-position errors and wall unevenness. Real experiments cover a $2.37\,\mathrm{m} \times 4.6\,\mathrm{m} \times 3.1\,\mathrm{m}$ scene and report $2\,\mathrm{cm}$ lateral resolution. This shifts the two-bounce trajectory from increasingly expressive neural scene representations toward self-calibrating acquisition on irregular real relay geometry.

C. Keyhole Imaging

For all the NLOS imaging systems introduced above, there is a limitation that a large illumination/imaging area is required. However, in some NLOS scenes (e.g., hidden objects are blocked in a room with curtains), large imaging areas are not allowed, and only a few small holes can be used. Metzler et al. used a beam splitter to propose an imaging system similar to confocal NLOS but does not require scanning, called keyhole imaging, as shown in Fig. 7-(b) [21]. In this system, the pulsed laser and the time-resolved detector only use a small hole to illuminate and detect the hidden space. When the object in the hidden space moves with time, expectation-maximization can be used to complete the task of NLOS shape reconstruction and localization.

D. Active Corner Cameras

A newer line of work relaxes the need for dense multi-position illumination by turning a visible corner into a structured sensing device. Seidel et al. demonstrated an active corner camera that models the complete optical response of a visible corner and reconstructs both moving foreground objects and stationary hidden background geometry from NLOS snapshots. This direction is important because it moves NLOS imaging closer to deployed corner-mounted sensing: the system no longer only recovers a single isolated object, but can also maintain a background map of the hidden space while monitoring dynamic foreground changes.

E. Neuromorphic NLOS Tracking

Dynamic hidden-scene sensing does not always require reconstructing a complete albedo or geometry volume. Zhu et al. introduced computational neuromorphic NLOS tracking, using an event camera to respond selectively to motion-induced changes in the dynamic speckle field [395]. Because static relay-surface and background contributions generate few events, the event stream provides a compact, non-redundant representation for direct motion estimation. This work complements frame-based passive event-camera reconstruction and transient NLOS videography by establishing an event-driven task branch focused on efficient hidden-target tracking rather than full-scene recovery.

F. Ptychographic NLOS Imaging

Most transient NLOS methods treat the relay wall as an approximately diffuse surface and rely on time-of-flight inversion. Song et al. introduced ptychographic NLOS imaging [396], which instead exploits coherent diffraction and coded ptychography for depth-resolved visualization of hidden objects. In this formulation, a laser spot is scanned on the relay wall, the returning object wavefields are modulated by the complex-valued wall profile, and the resulting diffraction patterns form a ptychogram. Recovering both the hidden objects at different depths and the wall modulation profile connects NLOS imaging to computational wavefront retrieval, suggesting a complementary route for high-resolution depth-aware hidden-scene imaging when coherent measurements and wall-profile modulation are available.

G. Transient Light-Transport-Matrix Iteration

Beyond reconstructing a hidden volume from a single measured transient light transport matrix (TLTM), Sultan et al. proposed iterating the TLTM itself [90]. By computationally focusing the measured first-order TLTM onto surfaces in the hidden scene, the relay surface can be treated as a second-order active imaging system. This viewpoint enables hidden-scene relighting, separation of direct and indirect transport inside the hidden space, and dual-photography-style operations. It is therefore not only another reconstruction method, but also a step toward programmable transient probing of hidden scenes when dense SPAD-array measurements make fuller TLTM acquisition feasible.

Time-resolved plenoptic transport as an adjacent representation. Wang et al. introduced GenPIE, which recovers a high-dimensional time-resolved plenoptic light-transport representation from sparse transient observations using generative

priors and physically grounded refinement [397]. Although not restricted to around-corner NLOS, its separation of higher-order light paths and support for transient relighting provide a complementary representation for future hidden-scene transport analysis.

H. Sparse and Irregular Active NLOS Acquisition

Another important practical trajectory is the relaxation of dense rectangular relay scanning. Liu et al. introduced a Bayesian framework for NLOS imaging with arbitrary illumination and detection patterns [398]. By constructing virtual confocal signals and jointly regularizing the measured signal, virtual confocal signal, and hidden object, their CC-SOCR formulation reconstructs albedo and surface normals even when relay samples are sparse, fragmented, or geometrically irregular. A closely related few-shot extension [190] further emphasizes that the hidden target, the measured transient signal, and surface-based structure priors can be regularized in mixed dimensions, enabling high-quality reconstruction from very small confocal grids such as 5×5 measurements. Cho et al. take a complementary learning-based route with LEAP [399]: instead of directly reconstructing the hidden volume from noisy partial transients, the network predicts clean full-aperture phasor fields from partial measurements and constrains the learned representation to the informative frequency band of phasor-field propagation. This connects sparse acquisition, denoising, and learned measurement completion, showing that practical NLOS can reduce both sampling count and scan area without changing the downstream physical propagator. Together, these works bridge classical dense ToF NLOS and the newer arbitrary-relay / mobile-capture setting: they show that practical NLOS deployment depends not only on faster solvers, but also on inverse models that remain well posed when the relay surface is incomplete, irregular, or sparsely sampled.

Masked transient pretraining. Shen et al. introduced MARMOT, a self-supervised masked autoencoder that treats transient measurements as a reusable pretraining modality rather than training a separate reconstruction network for every NLOS task [352]. Its scanning-pattern mask removes structured subsets of relay measurements so that the visible tokens are functionally equivalent to arbitrary sparse sampling, while a Transformer encoder–decoder predicts the missing transient measurements. Pretraining on the TransVerse corpus of 500,000 synthetic 3D models allows the learned transient representation to transfer to downstream reconstruction through direct feature reuse or decoder fine-tuning. This work marks a shift from task-specific transient inversion toward foundation-style measurement priors that can support sparse acquisition, transient completion, and multiple NLOS objectives.

I. Arbitrary Relay Surfaces with Gaussian Transient Rendering

Wang et al. introduced a LOS-guided pipeline that estimates visible relay geometry and represents the hidden scene with 3D Gaussian primitives optimized by a differentiable transient renderer [400]. The method supports confocal and non-confocal measurements on spatially limited, sparse, planar, and strongly curved relay regions, then synthesizes transients on a virtual relay surface for conventional volumetric reconstruction. This work connects differentiable transient rendering with deployable arbitrary-relay capture and removes the large planar wall from the core geometric assumptions of learned NLOS inversion.

J. Scattering-Media NLOS Imaging

Most active optical NLOS models assume that light propagates through free space and scatters only at surfaces. In their final Optics Letters study, Luesia et al. consider a harder regime in which the hidden scene is submerged in scattering media such as fog- or smoke-like volumetric transport [401]. By testing phasor-field reconstructions under controlled simulated scattering parameters and real diffuser measurements, the work clarifies how virtual-wave NLOS behaves when the relay path itself contains strong volumetric scattering. This direction is important for applications such as rescue, surveillance, and biomedical imaging, where NLOS systems may need to operate in optically degraded media rather than clean laboratory air.

K. Passive NLOS with Limited Prior Data

Passive NLOS reconstruction is especially sensitive to changes in relay surfaces, illumination, and detection geometry because the transport matrix is usually unknown and strongly scene-dependent. Gui et al. proposed a High-Dimensional Projection Selection (HDPS) framework for passive NLOS imaging with arbitrary scene conditions and detection patterns using only a small amount of prior data [402]. Instead of assuming that a full transport matrix is available or training a separate network for every transport condition, HDPS models the hidden target, projection, and transport relation through a high-dimensional manifold whose low-dimensional projections can be adapted to different passive NLOS settings. This work is relevant to the broader trend from calibrated laboratory passive NLOS toward deployable ordinary-camera systems that must adapt to new walls, illuminations, and detection layouts with limited calibration.

L. Consumer LiDAR and Low-Cost Spatial Reasoning

Research-grade transient systems have traditionally required high-power lasers, dense calibrated scanning, and expensive timing electronics. Somasundaram et al. introduced a motion-induced aperture-sampling model that unifies object shape, object motion, and camera motion, enabling multi-frame NLOS reconstruction, tracking, and camera localization with smartphone-grade consumer LiDAR [403]. This Nature result turns previously harmful motion and low spatial resolution into a synthetic aperture and marks a shift from laboratory NLOS hardware toward plug-and-play sensing.

Low-cost sensing also changes the role of datasets and cross-modal priors. DENALI provides space–time histograms

and spatial-reasoning annotations for commodity LiDAR, establishing a CVPR benchmark for learned NLOS reasoning [404]. In RF imaging, GeRaF 2.0 combines visible line-of-sight geometry outside an enclosure with radar propagation inside it, using the exterior geometry to regularize neural signed-distance reconstruction of hidden surfaces [405]. Together these works show that deployable NLOS increasingly couples inexpensive measurements with motion, visible-scene geometry, and learned priors rather than relying only on denser transient acquisition.

Commodity continuous-wave ToF scattering maps. Fang et al. used a low-cost continuous-wave ToF camera to acquire relay-surface intensity and depth images and proposed scalable scattering mapping (SSM) for learned decoupling of object and relay-surface scattering signatures [406]. The study collected 210,000 depth images across polypropylene, acrylic, Lambertian, painted-door, and plaster-wall relays, and demonstrated reconstruction on laboratory objects and clothing in a corridor. Unlike picosecond transient inversion, SSM calibrates a scene-dependent degradation-to-clear mapping from commodity correlation-depth measurements, broadening deployable NLOS toward inexpensive depth cameras while making the need for coverage of new object–relay combinations explicit.

M. Thermal NLOS through rough relay surfaces

Ye et al. extend passive long-wave-infrared NLOS beyond conveniently smooth relay materials with NLOSFormer [276]. A physics-embedded network estimates a scene-dependent thermal convolution kernel and jointly recovers hidden appearance and relative depth, while the ThermalNLOS dataset covers rough-wall conditions and dynamic sequences. Reported operation at approximately 4 fps shows that learned thermal transport can relax idealized relay-surface assumptions while retaining physical structure, expanding NLOS toward darkness, adverse visible illumination, and practical building materials.

N. Photonic hidden vital-sign sensing

Beyond recovering hidden geometry or object identity, indirect optical measurements can support physiological sensing. Chen et al. demonstrated photonic sensing of hidden vital signs [407], extending optical NLOS toward contactless hidden-human monitoring. This application should be distinguished from full hidden-scene imaging: the recovered quantities describe physiological activity rather than dense three-dimensional shape.

O. Radar-Based NLOS Imaging

Millimeter-wave (mmWave) radar offers a compelling alternative modality for NLOS sensing due to its ability to penetrate non-metallic obstacles, operate in adverse lighting conditions, and provide Doppler velocity information. Early radar NLOS works [408] demonstrated detection and tracking of hidden vehicles using Doppler radars in automotive scenarios. More recently, Lai et al. proposed HoloRadar [34], a practical system that achieves full 3D scene reconstruction (rather than mere detection and tracking) using a single mmWave radar mounted on a mobile robot. HoloRadar employs a two-stage pipeline: the first stage generates high-resolution multi-return range images capturing both line-of-sight (LOS) and NLOS reflections, and the second stage uses a physics-guided architecture that models mirror-image ray interactions to map NLOS reflections to their true locations. HoloRadar has been validated across diverse real-world indoor environments and represents a major step toward full NLOS 3D reconstruction in uncontrolled settings. The radar approach is complementary to optical NLOS: it operates through walls and in total darkness, but at lower spatial resolution than optical methods.

Semantic-guided radar NLOS tracking. Li et al. [409] use camera semantics and 4D radar multipath to localize hidden targets (0.419 m MAE; 80% tracking success), rather than reconstructing full 3D scenes.

Cellular ISAC hardware for NLOS sensing. A complementary 5G/6G ISAC trajectory asks whether communication hardware can use multipath as an around-corner sensor rather than treating it only as a channel impairment. Tosi et al. first demonstrated the feasibility of NLOS target detection with a 27.4 GHz commercial mmWave ISAC proof-of-concept, including channel-state-information processing that suppresses spectral replicas caused by time-division-duplex gaps [410]. The later ICT study moved from feasibility to reliable intrusion monitoring of fully occluded moving targets, adding range–Doppler detection and probability-hypothesis-density filtering for tracking and false-alarm rejection in an industrial testbed [411]. Together, these works connect radar NLOS with standards-compatible cellular infrastructure and show a deployment path in which communication radios become opportunistic hidden-region sensors.

A parallel RF/mmWave lineage has progressively broadened the task from around-corner localization to practical hidden-shape and full-scene reconstruction. CornerRadar learned a propagation hint map for robust indoor localization across diverse layouts [412], while Mosaic exploited curved and planar urban reflectors to widen automotive-radar blind-spot coverage [413]. RFlect then used practical environmental reflectors, including poles and curved/composite surfaces, for around-corner shape imaging [414]. The geometry branch advanced with mmNorm, which estimates mmWave surface normals and integrates them into hidden 3D shape [415], and with Jeon et al. , who infer T-junction layout from static radar points and ray-trace dynamic multipath to localize multiple hidden pedestrians in measured outdoor scenes [416]. Attention-based BiScalar-AA and its related two-stage attention network convert sparse radar points to pseudo-images for learned NLOS object detection and tracking [417], [418]. Most recently, Wave-Former introduced physics-aware wireless shape completion for complete 3D geometry of fully occluded objects [419], whereas RISE combines AoA/AoD multipath enhancement with hierarchical diffusion for layout reconstruction and object detection from a single static radar [420]. Together, these works show a clear RF trajectory from point localization, through reflector-aware imaging and surface-normal recovery, to learned full-object and scene-level

geometry.

Passive urban RF emitter localization via multipath geometry. Gajewski and Łopatka [421] studied single-site localization of non-cooperative radio emitters in an urban NLOS setting using a five-element uniform circular SDR array. Their correlation-robust CM-MUSIC direction estimator separates coherent multipath components; backward ray tracing through a known city map produces a spatial likelihood field, and a learned sector mask suppresses implausible candidate locations. Measured FM-signal experiments report a median localization error of 9.22 m, but a 55.52 m mean and outliers reaching 410.68 m reveal sensitivity to incorrect sector predictions. This branch is complementary to reflector-aware radar imaging: it infers hidden emitter position rather than reconstructing object reflectivity or three-dimensional shape, and requires prior urban geometry.

Higher-order RF paths and joint LOS/NLOS inference. Chen et al. jointly localize visible and hidden radar targets by combining low-rank clutter removal with sparsity and group-sparsity constraints, recovering target-position images in mixed LOS/NLOS corner scenes [422]. Su et al. extend mixed LOS/NLOS inference with geometry-partitioned range–angle–Doppler sparse inversion and deep-unfolded IAA-Net/RIRM-Net solvers [423]. Their simulated and measured T-junction tests improve weak hidden-target localization and false-response suppression despite stronger direct returns, but do not reconstruct dense hidden-object geometry. Extending multipath from a nuisance to a geometric cue, Zhang et al. derive radio-SLAM estimation bounds for double-bounce paths and use measured 5G mmWave signals to recover landmarks unobservable from single-bounce paths [424]. Tong et al. further couple NLOS-assisted over-the-air phase synchronization with off-grid sparse imaging in distributed MIMO, using alternating refinement to improve both tasks [425]. These studies broaden radar NLOS toward joint environment mapping and calibrated multipath sensing; unlike HoloRadar-style methods, they do not primarily target dense hidden-object 3D shape reconstruction.

Rough, uncertain, and unknown relay geometry in mmWave NLOS. A complementary radar trajectory removes the ideal planar/specular relay assumption itself. Xu et al. model rough relay surfaces as stochastic microfacet scatterers and deliberately exploit the resulting multipath diversity with a stochastic-geometry-aided recovery procedure [426]. Their later multi-angle formulation uses an automotive-squint SAR model and a double-sparse image prior to separate structure shared across heterogeneous relay orientations [427]; when those orientations are only approximately known, a Bayesian extension treats the relay angles as latent dictionary parameters and alternates expectation-propagation inference with parameter updates [428]. Lv et al. push calibration further by reconstructing the relay reflector directly from mmWave measurements before around-corner human sensing, removing the dependence on LiDAR-provided reflector geometry [429]. In parallel, Hydra explicitly models multi-bounce scattering from uncontrolled intermediate objects and localizes targets beyond the transmit field of view without prior environment knowledge [430]. A 2026 Aerospace and Electronic Systems Magazine overview consolidates these rough-relay challenges and solutions [431]. Taken together, this branch changes relay geometry from a fixed prerequisite into an estimated, uncertain, or even beneficial part of the inverse problem, complementing HoloRadar/RFlect-style reflector-aware reconstruction and the newer learned full-scene systems.

Multipath-exploitation inverse scattering for RF NLOS. Suenobu et al. first applied a physical-optics linearization to a known indoor T-junction, using a numerically evaluated Green tensor to encode the wall-mediated paths between the radar aperture and hidden imaging region [432]. Their subsequent journal study extended this formulation to multipath-exploitation imaging of PEC objects [433]. Rather than suppressing indirect returns as ghosts, the method incorporates them into the forward operator and recovers hidden target position and planar extent. Three-dimensional full-wave simulations and measured 2 GHz anechoic-chamber experiments establish a complementary RF trajectory to mirror-symmetry backprojection and learned mmWave reconstruction: environment geometry is assumed known, while target scattering is inferred through a linearized inverse problem.

Du et al. extended radar NLOS from room-scale mmWave perception toward long-range X-band imaging [434]. Their system exploits the longer wavelength to make relay interactions predominantly specular and combines learned dense prediction with geometry-aware reconstruction to compensate for the low native spatial resolution of radar. Prototype experiments reconstruct hidden objects at distances up to 40 m, demonstrating that modality selection can change not only robustness but also the practical range envelope of NLOS imaging.

Measured mmWave reconstruction from physical models to fast regularized migration. Wei et al. established a measured 77 GHz MIMO foundation for around-corner 3D radar imaging, deriving mirror-symmetry backprojection, resolution behavior, and relay-surface phase-error compensation [435]. Cai et al. then introduced NSIR, which separates LOS and multipath NLOS echoes before jointly estimating hidden reflectivity, total-variation structure, and reflective-surface phase error [436]. RM-CSTV [437] replaced a large matrix operator with range migration while retaining complex sparsity and total-variation constraints, correcting virtual-target positions and preserving contours under aperture loss and sparse sampling. These works form a model-driven radar lineage in which environmental reflections are calibrated as an imaging aperture rather than treated only as nuisance multipath.

Self-supervised and model-unfolded radar NLOS reconstruction. The radar branch subsequently adopted learned optimization without abandoning its propagation operators. ACTE-Net [438] combines a Gaussian-convolution artifact suppressor, an energy-enhancement operator, and holographic reconstruction in an unsupervised network validated on measured hidden targets. Cai et al. further exploit group invariance in an equivariant-imaging framework whose TV-constrained unfolding network and adaptive peak-convolution thresholds reconstruct 2D/3D mmWave SAR images from

partial noisy echoes [439]. Chen et al. embed a fast range-migration operator in a FISTA-derived network for measured near-field mmWave reconstruction under phase error, aperture shadowing, and multipath [440]; a parallel 121 GHz system replaces range migration with a compact holographic operator and demonstrates the same model-unfolding trajectory at sub-terahertz wavelengths [441]. Together with HoloRadar's mobile, ray-interaction-aware scene reconstruction [34], this progression shifts radar NLOS from isolated target focusing toward practical full-scene geometry and adaptive learned inversion while retaining explicit propagation structure.

Visually constrained penetrative RF neural reconstruction. Lu et al. introduced GeRaF 2.0 for joint line-of-sight and non-line-of-sight geometry reconstruction from penetrative radar measurements [405]. Rather than optimizing the hidden signed-distance field in isolation, the method uses visually observed exterior geometry to constrain RF propagation from the visible region into an enclosed hidden volume. These line-of-sight priors stabilize the zero-level set, reduce surface ambiguity, and permit physically consistent recovery of both visible and concealed geometry. This direction complements mirror-path mmWave systems such as HoloRadar: instead of relying only on environmental specular relays, it combines penetrative RF transport with cross-modal geometric evidence to regularize a neural implicit scene representation.

Recent RF/mmWave work further expands the meaning of NLOS perception beyond monostatic reconstruction. MITO [442] provides a multi-spectral mmWave dataset and simulation tool with both LOS and NLOS captures of everyday objects, lowering the barrier for learning-based mmWave NLOS perception. N^2LoS [443] studies tag-assisted mmWave NLOS localization, using a single backscatter tag to separate tag and environmental reflections in non-penetrable NLOS settings. Backscatter-assisted indoor NLOS positioning [444] extends this branch from dedicated tag modulation toward passive asynchronous backscatter devices that act as virtual anchors for corridor-constrained user tracking, showing that low-power environmental tags can turn otherwise difficult multipath into useful NLOS localization evidence. Yiğitler et al. then moved this passive-anchor branch from measured tracking to calibration-aware fundamental limits [445]. For an uplink NLOS model in which the direct and backscatter-assisted paths share an unknown scatterer, they treat the common gain, relative backscatter response, and residual device phases as nuisance variables and derive closed-form equivalent Fisher information matrices. The analysis isolates which carrier-phase and bandwidth-dependent delay information survives partial or absent calibration and shows that joint single-snapshot UE–scatterer identification requires at least two backscatter devices in two dimensions or three in three dimensions, together with sufficient angular diversity. This provides a theoretical deployment counterpart to measured backscatter-assisted NLOS localization rather than another hidden-shape reconstruction method. Jeon et al. first infer T-junction layout from static 2D mmWave points and ray-trace dynamic multipath returns to localize multiple hidden pedestrians in measured outdoor experiments [416]. Park et al. then use camera-derived road layout to interpret multipath-distorted mmWave point clouds for NLOS pedestrian localization at urban T-junctions [446]. Kim et al. address a less stationary darting-out regime in which parked vehicles form temporary occluders: monocular segmentation and depth estimate the current vehicle geometry, 2D radar points refine that geometry, and the resulting spatial interpretation supports early hidden-pedestrian localization in real urban driving experiments [447]. Park et al. subsequently move the same automotive NLOS branch into ego-dynamic driving with reflection-aware reasoning [448]. Their camera–radar BEV fusion predicts reflection-type labels and reflective-surface structure before physics-guided ray tracing unfolds distorted multipath; the project reports 120 real outdoor scenarios, 12,539 frames, and 1.23-m localization accuracy with a moving ego-vehicle. Together, these works trace a progression from map/layout-assisted single-target localization to multi-target ray tracing, temporary-obstacle reasoning, and learned reflection-aware multipath interpretation under vehicle motion. NLOS-aided distributed MIMO/ISAC treats reconstructed NLOS multipath components as useful information for joint over-the-air synchronization and sparse off-grid environment imaging. Zhang et al. show that even higher-order double-bounce NLOS paths, often discarded as nuisance multipath, can improve snapshot radio SLAM and reveal landmarks that are unobservable from single-bounce paths alone. Yasmeen et al. experimentally used a custom one-bit RIS with a 5.5 GHz monostatic radar to steer illumination around a corridor corner and recover human walking micro-Doppler signatures [449]. A follow-on dual-beam RIS radar study [450] examines a lower-complexity one-bit quantized RIS configuration that produces dual symmetric beams, benchmarking the resulting beam steering and radar cross-section against metal reflectors and ideal single-beam RIS baselines. Liaquat et al. extend reconfigurable propagation from a single engineered relay to multi-RIS-assisted monostatic radar, deriving received-power, path-loss, and SNR expressions that quantify when additional programmable surfaces recover blocked sensing links and improve NLOS target detectability [451]. Goïcoechea et al. move beyond beam redirection to matrix imaging: a single transmit–receive antenna interrogates programmable RIS masks to reconstruct the full reflection matrix, effectively synthesizing an array for hidden-scene imaging, selective focusing through clutter, and moving-target tracking [452].

Road-layout-aware IRS localization for automotive radar. Chakraborty et al. use road geometry to guide intelligent reflecting surface beam-pattern synthesis, while sparse Bayesian estimation identifies multiple hidden radar targets [453]. The simulation-based method optimizes spatial coverage and remains NLOS localization rather than volumetric imaging.

Frequency-shifting RIS for practical NLOS mmWave localization. Madnaik and Sundaresan introduce PRISM, which applies space–time modulation at a reconfigurable intelligent surface so that the angular direction of a hidden target is encoded as a frequency shift in a monostatic mmWave radar return [454]. The resulting angle-dependent frequency labels permit multiple-target detection and localization without tight real-time synchronization between a commodity radar and the

RIS. A narrowband approximation and genetic configuration search reduce offline design computation; reported experiments achieve 31 cm median localization error and detect up to six NLOS targets with approximately 80% accuracy. This advances the RIS branch from static beam redirection toward programmable propagation coding, but remains target sensing rather than full hidden-scene reconstruction.

From assumed relay geometry to joint environment and target inference. A parallel radar trajectory removes the common assumption that the reflective environment is known a priori. Luo et al. estimate the reflective surface directly from the tangency between the reflector and multipath ellipses, using dictionary matching and temporal smoothing to recover relay geometry from radar measurements [455]. Xue et al. jointly recover an NLOS target and part of a complex L-shaped building layout by tracking multipath time-of-arrival sequences and assigning diffraction, first-order reflection, and second-order reflection paths to complementary target- and wall-estimation roles [456]. Jia et al. similarly use two back-projection views and path-length consistency to estimate both hidden targets and the width of an unknown L-shaped corridor even when part of the multipath set is missing [457]. Once this geometry uncertainty is exposed, the problem also becomes one of robust dynamic inference: Pham et al. explicitly model the multimodal likelihoods created when several multipath hypotheses are difficult to distinguish, and use particle filtering to preserve competing hidden-target states rather than committing to an incorrect path association [458]. Wu et al. use CFAR-derived 0–1 non-coherent binary accumulation followed by two-stage positioning to mitigate false alarms and missed detections [459], while Xu et al. extend multipath exploitation to distributed-radar tracking with multipath-assisted JPDA [460]. Zhu et al. then close this loop by first performing 3-D environmental perception from weak reflector echoes and transferring the inferred reflector parameters into path-oriented NLOS moving-target reconstruction with mmWave MIMO imaging radar [461]. Together, these works shift RF NLOS from reconstruction under a calibrated relay map toward joint estimation of the environment, propagation paths, hidden-target state, and motion.

Diffraction, multipath-ghost correlation, and cross-regional inference. The same unknown-environment trajectory is also visible in recent conference work that makes more explicit use of diffraction and multipath hypotheses. Yu et al. use diffraction-signal information to estimate building-corner and NLOS-target parameters rather than treating the corner geometry only as fixed prior knowledge [462], extending the earlier diffraction-based around-corner localization lineage. Wei et al. jointly estimate hidden-target locations and building layouts by extracting multipath Range–Doppler features, estimating directions and ghost positions with the iterative adaptive approach, and spatially matching the resulting ghosts to candidate targets and walls [463]. This turns multipath ghosts from nuisance artifacts into geometric constraints that couple target localization and environment reconstruction. At IGARSS 2026, Yu et al. further reported a cross-regional NLOS target-localization method based on a joint multipath generalized likelihood ratio test (GLRT) [464]. The official conference program verifies the paper and its 11 August 2026 presentation; because a proceedings DOI was not yet indexed at the time of this update, we retain the final conference designation without inventing publication metadata. Together, these works sharpen the progression from known-corner diffraction models to joint target–layout inference and then to detection/localization rules designed to combine multipath evidence across hidden regions.

Reconfigurable propagation for physiological NLOS sensing. A complementary RF branch treats the relay path as a controllable component rather than a fixed environmental reflector. Li et al. combine visual target localization with programmable RIS beam control so that the sensing field is redirected toward a hidden chest region, and then estimate respiration and heartbeat from the returned signal using an improved variational-mode-decomposition pipeline [465]. Tripathy et al. integrate a liquid-crystal RIS with a self-injection-locked radar and experimentally demonstrate contactless vital-sign monitoring after electronically steering the sensing path into an NLOS region [466]. Mao et al. subsequently demonstrated broadband 1-bit RIS focusing over 6.88–13.62 GHz in an obstructed radar setting and used enhanced variational mode decomposition to recover respiration and heartbeat, reporting mean estimation errors of 2.82 RPM and 5.49 BPM, respectively [467]. These works extend NLOS RF from geometry, localization, and micro-Doppler toward physiological sensing; they should therefore be interpreted as hidden-region semantic/biomedical sensing rather than hidden-shape reconstruction. Passive or static metasurfaces provide another route: Tornielli Bellini et al. study multi-view near-field NLOS radar imaging with non-reconfigurable electromagnetic skins [468], where a moving radar synthesizes an aperture over phase-configured passive modules that focus energy into the hidden region. Chen et al. further analyze RIS-aided NLOS bistatic MIMO radar with arbitrary electromagnetic-vector-sensor (EMVS) arrays [469], showing that polarization-rich EMVS measurements and PARAFAC-style factorization can resolve phase ambiguities even when sensor spacings exceed the conventional $\lambda/2$ limit. Newly identified adjacent RF works further show how engineered or naturally occurring reflectors can make hidden-space RF sensing more deployable: mmMirror [470] uses a Van-Atta-array IRS with commodity FMCW radars for device-free around-corner localization, while See and Beam [471] uses LiDAR-visible specular surfaces to infer NLOS mmWave reflection paths and localize NLOS users for beam steering. Together, these works suggest that future NLOS systems will not be limited to single-sensor hidden-shape reconstruction, but will also include localization, mapping, synchronization, dataset-driven representation learning, reflector-aware communication/sensing, passive backscatter infrastructure, higher-order radio-SLAM multipath exploitation, engineered static and reconfigurable metasurfaces, practical quantized RIS beam control, array-geometry design, and environment-aware radio perception.

Engineered metasurface relays and static-target RIS

demonstrations. Woodford et al. introduced Metasight, which replaces the large ambient reflectors assumed by earlier automotive NLOS radars with compact passive printed mmWave metasurfaces and an angular encoding/decoding scheme [472]. This establishes an engineered-relay alternative to relying on opportunistic specular reflections, while retaining the task of hidden-target sensing rather than dense volumetric reconstruction. A later citing hardware demonstration by Angjo et al. uses a programmable 28-GHz RIS and a monostatic FMCW radar to illuminate a static passive target behind an obstacle [473]. By subtracting empty-scene from target-present range–RIS-angle power-delay maps, it reveals a target-associated angular peak and estimates its range from known geometry. The latter is a three-page accepted MobiCom demonstration, not a general 3D reconstruction method; it requires reference calibration and reports a controlled single-target setup. Together, the two works illustrate a shift from passive encoded metasurface relays to actively steered RIS-assisted NLOS radar detection.

RIS-assisted gridless and monostatic angular sensing. A complementary RIS trajectory targets hidden-object direction rather than reflectivity or surface reconstruction. Yuan et al. formulated RIS-enabled NLOS direction finding in the covariance domain [474]: after estimating the noise variance, atomic-norm minimization recovers a Hermitian Toeplitz representation and an ADMM solver provides gridless multi-source DoA estimates with limited receive hardware. Zhang et al. considered a monostatic radar–RIS–target–RIS–radar path [475]. A phase-codebook scans the hidden angular sector, steering-vector decoupling recovers the target outer-product matrix from the RIS-element superposition, and Root-MUSIC estimates the target angles. Both studies are numerical, so they should be read as theoretical RIS-assisted NLOS localization advances rather than measured hidden-scene imaging systems.

RIS-assisted detection, estimation bounds, and quantized-surface hardware. Moving beyond direction-only estimation, Yan et al. analyzed RIS-assisted detection of low-radar-cross-section targets by combining conventional monostatic echoes with bistatic returns redirected by a programmable surface [476]. Their detector architectures and performance analysis address weak-target detection rather than volumetric hidden-scene recovery. Sanjeeva Reddy and Veera Reddy studied a bistatic IRS-assisted geometry in which a spatially separated passive panel redirects target-scattered energy back to a monostatic receiver, deriving Fisher-information and Cramér–Rao bounds for target azimuth and elevation estimation [477]. At the hardware level, Yasmeen et al. characterized the radar cross section of phase-quantized RIS apertures using an analytical scattering model, full-wave simulations, and a fabricated 5.5 GHz one-bit surface; non-specular illumination enabled otherwise inaccessible micro-Doppler measurements [478]. These studies connect RIS-assisted NLOS inference to realistic surface quantization and detection limits, while remaining adjacent radar sensing rather than dense NLOS 3D imaging.

A closely related tensor-processing branch uses reconfigurable propagation together with multidimensional radar structure. Yu et al. use an IRS to establish a controllable NLoS path for bistatic FDA–MIMO radar, estimate target count by sequential MDL, and factorize a third-order echo tensor with PARAFAC to decouple DOD/range and recover 2-D DOA before geometric multi-target localization [479]. Xie et al. then construct a fourth-order covariance tensor for RIS-aided bistatic MIMO radar, extract its signal subspace with HOSVD, reconstruct the receive steering matrix through Khatri–Rao structure, and obtain automatically paired DOD/DOA estimates [480]. Yu et al. further address fast angle estimation for both coherent and noncoherent NLoS targets with tensor decomposition [481]. These works trace a progression from IRS-created hidden paths and full target localization to increasingly structured and efficient angle inference; the current evidence is primarily simulation-based, so this branch should remain distinct from measured around-corner imaging and reconstruction systems.

Urban-intersection FMCW radar perception. Lin and Chen studied an application-facing 77 GHz automotive-radar setting in which specular reflections from buildings provide around-corner observations at occluded urban intersections [482]. Their compact one-dimensional, AlexNet-derived regression network restores interference-corrupted chirps before standard range and angular estimation, reducing simulated severe-interference errors from 5.48 m/$18.95°$/$10.77°$ to 0.56 m/$0.46°$/$0.73°$ for range, angle, and azimuth deviation, respectively. Unlike measured radar geometry reconstruction systems, the study is evaluated entirely in MATLAB simulation; its value is therefore as a learned signal-restoration and autonomous-driving NLOS perception branch, with real-road multipath and material validation remaining open.

Cyclostationary micro-Doppler detection in fully NLOS scenes. Liyanage et al. built a 2.47 GHz continuous-wave MIMO radar with one transmitter and four receivers and used spectral correlation density images of propeller micro-Doppler as inputs to a compact EfficientNet-B0 detector [483]. Measurements collected in indoor and semi-urban settings used cabinets, doors, and walls to block the direct path; cyclic rotor features remained visible under multipath, and the four-channel model achieved 86.11% overall accuracy compared with 73.77% for a single-channel baseline. This result broadens NLOS radar from vehicles, pedestrians, and static hidden geometry toward low-altitude aerial-threat sensing. It should nevertheless be distinguished from imaging: the airframes were stationary while the propellers rotated, and the reported task was binary detection rather than localization, tracking, or hidden-shape reconstruction.

Radar–LiDAR blind-spot fusion in open-pit mines. Yang et al. studied a deployment-oriented NLOS perception problem in which haul trucks and mining structures fully occlude rear targets from roof-mounted LiDAR [484]. Their Blind-Spot Complementary Fusion framework calibrates 3D LiDAR and 4D mmWave radar, suppresses ground-coupled multipath artifacts, identifies physical LiDAR blind regions, and injects only radar points that pass local-density and spatial-consistency tests. Real measurements at 35 m and 70 m and a fully occluded test-field target show improved cross-modal proximity and nonzero envelope-level evidence where LiDAR

alone misses the target. The proposed Volume Recovery Rate is explicitly a proxy for hidden-target existence and coarse envelope coverage, not complete shape reconstruction or semantic detection. By citing HoloRadar as the reconstruction-oriented radar milestone while pursuing training-free safety cues, this work adds a complementary branch in which NLOS sensing is evaluated by downstream blind-zone risk coverage.

Measured multipath radar recognition and activity understanding. Radar NLOS can support semantic decisions even when the measurements are too sparse for hidden-shape reconstruction. Zeng et al. used a 15 GHz stepped-frequency system and an adaptive multi-scale residual network to combine fine high-amplitude single-path cues with global multipath structure and attention-weighted local scattering features [348]. Measured concrete- and wood-wall experiments distinguish pedestrians, weapon-carrying human models, quadrotor drones, and tank models with 99.6% accuracy. Zhong et al. then treated separated UWB-radar propagation paths as natural multiview positives for self-supervised contrastive learning [349]. MuPhyCoNet aligns learned features with observation- and prediction-level physical quantities, reaching 94.32% six-action accuracy with only 10% labels on 19,500 measured spectrograms. These studies extend around-corner RF from detection and localization toward target identity and human activity, but should remain categorized as semantic sensing rather than full 2D/3D imaging.

Tornielli Bellini et al. push reflector-assisted RF NLOS imaging toward communication infrastructure by embedding a monostatic hidden-scene imaging mode into the initial-access beam sweep of a next-generation base station [485]. A low-cost non-reconfigurable modular reflector and imaging-specific codebook extend the standard communication beams; coherent processing across the resulting viewpoints forms an effective aperture for near-field reflectivity imaging, while a maximum-likelihood velocity estimator compensates moving targets. By deriving resolution, aperture, latency, SNR, and communication–imaging trade-offs and validating the design experimentally, this work connects earlier electromagnetic-skin NLOS prototypes to standard-aware 6G ISAC deployment.

Tag-assisted multipath disambiguation. Shi et al. introduced N^{2}LoS, which combines a single 24 GHz radar with an active backscatter tag [443]. Hybrid frequency-hopping/direct-sequence signaling separates tag returns from environmental reflectors, while frequency–spatial MUSIC resolves additional multipath components. Experiments report roughly 10–12 cm median coordinate error at 5 m, illustrating how a lightweight tag can convert uncontrolled around-corner multipath into identifiable localization paths. The final journal record appears in IEEE Transactions on Mobile Computing, vol. 25, no. 5, pp. 6002–6016 (2026), superseding the earlier arXiv-only metadata.

Sub-terahertz polarimetric indoor mapping. Alburadi et al. demonstrated that a 222–228 GHz polarimetric FMCW radar with a mechanically scanned fan-beam antenna can combine rapid high-resolution indoor mapping with measured around-corner imaging [486]. The system interprets double-bounce specular paths as hidden-scene evidence, while co- and cross-polarized channels help distinguish useful NLOS returns from direct leakage and environmental clutter. This work extends the measured mmWave lineage toward sub-terahertz carrier frequencies and emphasizes polarization as an additional physical cue for practical RF NLOS imaging.

P. Acoustic NLOS Imaging

Acoustic waves provide yet another physically distinct modality for NLOS sensing. Similar to optical ToF systems, acoustic NLOS imaging exploits the time-of-flight of sound waves reflected off a relay surface. The Lindell et al. acoustic NLOS work [146] established that microphone arrays and audio sources can reconstruct hidden 3D scenes using the same algorithmic framework as optical NLOS. Li et al. translated the optical ToF and f–k migration lineage into a coherent synthetic-aperture ultrasound system [35]. A scanning emitter–receiver pair preserves acoustic phase and reconstructs multiple hidden targets and complex 3D scenes at distances up to 2 m from the scattering surface, with measured lateral and depth resolution of approximately 1 cm. The study also reports the NLOS modulation-transfer function and releases data and code, establishing ultrasound as a quantitatively comparable, low-power and eye-safe alternative to optical transient NLOS. Acoustic NLOS imaging is particularly suited to scenarios where electromagnetic NLOS is impractical, such as underwater or through highly opaque materials.

Acoustic NLOS localization taxonomy. Wang et al. surveyed NLOS sound-source localization through multi-sensor, geometrical-acoustics, and statistical-acoustics models [487]. Their taxonomy places reflected-path inference, edge diffraction, microphone-array processing, and learned acoustic localization within a broader sensing framework; localization of a hidden emitter should be distinguished from reconstruction of a hidden 3D scene.

Boger-Lombard, Slobodkin, and Katz first showed that acoustic interferometry can retrieve effective Green functions from cross-correlations of uncontrolled broadband noise, enabling passive localization and tracking of a human hidden around a corner without controlled active probing [488].

The scope of acoustic NLOS has also moved beyond relay-surface reflection. Sommer and Katz exploit knife-edge diffraction at doorways and convex corners to passively localize a hidden acoustic point source in three dimensions [489]. For a doorway, its two edges act as virtual detector arrays; for a corner, the method uses the frequency-dependent diffraction-loss signature in a manner analogous to head-related transfer functions. This relay-free setting shows that edge diffraction, rather than diffuse reflection alone, can be treated as an exploitable physical transport path for hidden-scene sensing.

Laser–acoustic fusion for hidden-human orientation sensing. Doğan combined laser and acoustic chirp measurements at the feature level for four-class hidden-person orientation recognition [490]. Each sensing channel contributes 21 engineered features, yielding a 42-dimensional fused representation evaluated with classical classifiers, the proposed LAO-Net, and explainable-AI analysis under controlled NLOS ex-

periments. This work extends acoustic NLOS from source localization and geometric reconstruction toward multimodal semantic inference; its output is an orientation class rather than a hidden image, position map, or 3D surface.

Recent acoustic work has also separated hidden-source localization from volumetric echo reconstruction. Jeon et al. track vehicles approaching from fully occluded road regions using an Acoustic–Spatial Pseudo-Likelihood inside a particle filter [491]. Their ARIL dataset adds bird's-eye-view position ground truth and, together with OVAD, turns passive acoustic NLOS sensing into an application-oriented trajectory estimate for collision avoidance. Zhai et al. instead model the finite obstacle edge itself: the Biot–Tolstoy–Medwin first-order diffraction response forms a non-free-field steering vector for irregular-grid frequency-domain beamforming [492]. The resulting method limits main-lobe deformation as the hidden source moves and is validated with both simulations and a 32-channel microphone-array experiment. Together, these systems extend the acoustic trajectory beyond relay-wall tomography toward passive tracking and diffraction-aware source localization in practical occluded environments.

Wang et al. take a complementary data-driven route with CorridorLocNet [493]. Instead of requiring an explicit room or edge model, the network concatenates Mel-spectrogram and GCC-PHAT observations, extracts local time–frequency structure with a residual convolutional branch and long-range temporal dependencies with a lightweight Conformer branch, and uses cross-attention to regress the hidden footstep position. A real dataset captured with footsteps behind a corridor corner yields 98.83% classification accuracy and a 2.56-m reduction in average localization error relative to the compared reflection-aware method. This result complements physics-explicit diffraction localization by showing that complex corridor multipath can also serve as a learned spatial fingerprint for passive acoustic NLOS sensing.

A complementary second-order regime arises when both the direct path and first-order diffraction are unavailable. Zhai et al. construct a Biot–Tolstoy–Medwin second-order edge-diffraction transfer model and use it as the sensing matrix of a block-sparse inverse problem [494]. Fast marginalized block sparse Bayesian learning then estimates hidden-source position and strength; simulations and a 32-channel array experiment show that higher-order diffraction, usually treated as a weak residual, can become the primary sensing path under stronger occlusion.

At the task level, Wang et al. combine bird's-eye-view scene geometry with time–frequency and spatiotemporal acoustic spectra for detecting vehicles that are fully occluded from the camera [495]. Their scene-aware network uses CNN, LSTM, and Conformer modules to model local spectral structure, temporal evolution, and long-range cross-modal context, reporting 94.1% and 97.0% accuracy on the OVAD and AOVD datasets, respectively. Unlike acoustic echo tomography or diffraction-based source localization, this work does not reconstruct hidden geometry; it establishes a complementary semantic branch in which visible environment layout conditions the interpretation of sound propagated from an unseen traffic participant.

Material recognition from wall-mediated acoustic echoes. Alakuş and Türkoğlu first introduced ANLOS-R as a dedicated multi-channel acoustic NLOS dataset and semantic-sensing benchmark [496]. The acquisition uses eight loudspeakers and eight microphones facing a relay wall while the direct path to the target is blocked, records three sensor positions and single-channel, combined/MIMO, and background measurements, and contains 1,440 echo samples. Their accompanying pipeline isolates target-related reflection regions with an attention-enhanced U-Net and fuses spectral and temporal representations for material classification. Building on the same acquisition, the subsequent Sensors study [497] combines classical acoustic descriptors with multi-scale wavelet energy and entropy into a 70-dimensional representation, uses recurrent models for hidden-material recognition, and adds SHAP-based interpretation. Together, the two works extend acoustic NLOS from localization and geometry toward dataset-driven material-aware environmental perception. More recently, Çalışan [498] studied the semantic task of classifying *pairs* of occluded objects from secondary wall-reflected acoustic chirps. In one controlled room, 120 recordings across four paired-object classes were summarized by short-time Fourier spectra; repeated recording-level cross-validation gave 98.5% mean accuracy for a random-forest classifier, while additional microphones did not consistently improve discrimination. The study does not establish cross-room or cross-session generalization, and its output is object-combination classification rather than hidden-scene geometry. It extends the ANLOS-R material-recognition lineage toward multi-object acoustic NLOS sensing while emphasizing the need for independent scene-level validation.

Foundations of acoustic hidden-source localization. Before modern acoustic NLOS reconstruction, Mak and Furukawa used map-corrected time-of-arrival measurements to localize controlled sound sources beyond direct visibility [499]. Singh et al. then demonstrated diffraction-aware matched-field localization using microphone arrays fully shadowed by a wedge [500]. Kitić et al. formulated through-wall source inference as a cosparse wave-equation inverse problem [501]. For mobile robots, Takami et al. fused microphone-array signals with RGB-D geometry to account for reflected and diffracted paths [502], while An et al. combined the uniform theory of diffraction, acoustic ray tracing, and particle filtering to estimate hidden source positions [503]. These studies establish an acoustic localization lineage that precedes later around-corner scene imaging and physics-informed hidden-source reconstruction; their primary output is a source position rather than volumetric geometry.

Audio–visual geometry for acoustic NLOS localization. King et al. combine four-channel microphone time-difference-of-arrival estimates with robot-acquired LiDAR maps and reverse acoustic ray tracing to infer a hidden sound-source location [504]. Feasibility experiments reported localization within approximately 0.5 m; the result is source localization rather than volumetric scene reconstruction.

Equivalent-source time reversal for hidden acoustic sources. Tian et al. combine equivalent-source modeling of

candidate-to-microphone transfer functions with time-reversal focusing to localize obstructed sound sources [505]. Validated in simulation and experiment, this physics-based precursor complements later diffraction-aware, passive-correlation, and boundary-aware inverse approaches while remaining a source-localization method rather than volumetric scene reconstruction.

Boundary-aware acoustic inversion for fully hidden sources. Sroka et al. extend adjoint-based acoustic localization by introducing volume penalization for rigid environmental boundaries, explicitly accounting for reflected and diffracted waves instead of treating multipath solely as an error source [506]. They recover a fully hidden sound source in space and time in a controlled test and examine experimental microphone measurements of a moving vehicle. This complements diffraction- and correlation-based around-corner acoustic sensing by showing that a boundary-aware wave model can support hidden-source inference; the output is source localization, not volumetric scene reconstruction.

Acoustic liveness detection in cluttered NLOS scenes. Cetin et al. extend acoustic NLOS semantic sensing from hidden-source localization and material recognition to live-human versus inanimate-object discrimination for search-and-rescue settings [507]. Their blocked-path multichannel setup measures wall-mediated echoes from hidden people and clutter, and compares raw eight-channel signals, arithmetic channel averaging, and a convolutional–transformer autoencoder latent representation across convolutional, recurrent, and classical ensemble classifiers. The results expose a representation–model interaction: fusion that suppresses apparently redundant signal structure can improve one classifier while removing local phase/temporal cues needed by another. In the acoustic NLOS trajectory, this work therefore follows relay-free/source-localization and ANLOS-R material recognition with a task-oriented liveness branch, shifting evaluation from recovering hidden geometry or material labels toward deciding whether an occluded return corresponds to a living person or debris.

Identity recognition from secondary acoustic echoes. Olgun further extends task-oriented acoustic NLOS from liveness to person identity by treating delayed multipath as a discriminative biometric signal rather than suppressing it as nuisance noise [508]. In an eight-speaker/eight-microphone blocked-path setup with ten subjects, the method selects late secondary-echo segments, combines time-averaged STFT spectra with Limited Penetrable Visibility Graph (LPVG) topology features, and evaluates identity recognition under leave-one-scene-out changes in object configuration. The fused representation reaches 96.17% mean accuracy, showing that person–environment multipath interactions can encode identity-level information. This result marks a further semantic expansion of acoustic NLOS: indirect echoes are used not only to infer hidden source position, material, or liveness, but also who is present behind the occluder.

Q. Robotic Exploration with NLOS Perception

Most classical NLOS experiments focus on reconstructing a pre-defined hidden volume. In robotics, however, hidden perception is valuable because it can actively guide where the robot should move next. Young et al. studied single-photon-LiDAR NLOS sensing for autonomous navigation [509], combining multi-bounce histogram capture, hidden-region occupancy prediction, and a control module that lets a robot choose safer paths in an L-shaped corridor. This work is important because it evaluates NLOS not only by reconstruction fidelity, but by downstream navigation utility. SuperEx [510] later integrates single-photon LiDAR NLOS sensing into indoor mapping and exploration. Instead of treating NLOS reconstruction as a standalone output, it uses timing histograms for NLOS empty-space carving and hidden-structure recovery, then feeds the recovered hidden-space information back into the exploration policy. Together, these works close the loop between NLOS perception and robotic decision making, making NLOS useful even when the reconstruction is partial or uncertain.

R. Low-Cost LiDAR NLOS Spatial Reasoning

A parallel 2026 trend is the use of consumer or low-cost LiDAR hardware for NLOS reasoning. Somasundaram et al. showed that hidden objects can be imaged with consumer LiDAR by using ego-motion to induce a synthetic sampling aperture [511]. DENALI [404] shifts the emphasis from hand-engineered inversion to large-scale data-driven spatial reasoning: it releases 72,000 real-world hidden-object scenes captured as space-time histograms from low-cost LiDARs, enabling models to learn NLOS cues directly from the multi-bounce returns that are usually discarded by consumer depth sensors. Together, these works indicate that mobile NLOS may develop along two complementary paths: physics-based reconstruction with motion-induced sampling, and dataset-driven perception from inexpensive histogram sensors.

S. Terahertz NLOS Imaging

Terahertz and sub-terahertz waves provide another route to around-obstacle perception. Cui and Trichopoulos [512] showed that building surfaces can act as lossy mirrors for sub-THz signals, enabling hidden-object reconstruction by mirror-folding the propagation geometry around the obstacle. This modality sits between optical and RF NLOS: it has shorter wavelength and potentially higher spatial resolution than many RF systems, while still interacting with common environmental surfaces in a way that can be exploited for around-corner imaging.

T. NLOS Human Pose Estimation

Estimating 3D human poses of persons hidden around a corner is an example of semantic NLOS understanding beyond shape reconstruction.

Semantic human-pose recovery from transients. Xiao et al. move beyond applying a conventional pose estimator to a reconstructed hidden image by directly fusing transient-derived intensity and depth features and regressing semantic body joints end to end [111]. A physics-based pipeline synthesizes training measurements from ordinary smartphone

videos, while real experiments demonstrate pose estimation at a 1.75 m relay distance and under an SNR of 0.13. This development advances the earlier physics-based HiddenPose concept toward detailed, low-SNR measured-scene semantics.

Adaptive multi-person pose and mesh sensing. Hou et al. extend active transient semantics from single-person pose recovery to scenes containing an unknown number of hidden people [109]. Their AMPE-NLOS pipeline propagates transient features with LCT to obtain a coarse physical 3D representation, refines it with a 3D U-Net, and predicts both a body-center heatmap and an SMPL parameter map. Body-center-guided sampling then associates mesh parameters with individual people, enabling adaptive multi-person 3D mesh regression. Simulated training data and measurements from a self-built confocal pulsed-laser/SPAD system demonstrate a trajectory in which NLOS reconstruction becomes a front end for structured, multi-instance human understanding rather than only hidden geometry or single-person joints. Isogawa et al. introduced an optical NLOS physics-based 3D human-pose-estimation pipeline that converts transient measurements into physically valid body-pose sequences. Although this direction remains less developed than geometric NLOS reconstruction, it illustrates an important path for future systems: hidden-scene perception should ultimately support semantic tasks such as human pose estimation, action recognition, and risk assessment, not only voxelized geometry recovery.

VI. Conclusion

In this paper, we have presented a comprehensive survey of non-line-of-sight (NLOS) imaging, covering the foundational physical models and algorithms established through 2021 and the significant advances made from 2022 to 2026. The field has matured substantially during this period, progressing from laboratory demonstrations with minute-long acquisition times to real-time video of dynamic room-sized scenes, and from a primarily optical inverse problem to a broader computational sensing paradigm spanning light, RF, sound, terahertz waves, and robotic perception.

Progress on fundamental challenges. The three core challenges identified in the 2021 survey—ill-posedness with low SNR, limited resolution, and long data acquisition time—have all seen meaningful progress. On SNR, photon-efficient imaging [216] and first-photon event stamping [215] have demonstrated high-quality reconstruction at sub-photon-per-pixel levels. Resolution has been pushed to ~ 0.6 mm via active wavefront shaping (UNCOVER [162]), while picosecond-resolution up-conversion detection has shown that sub-millimeter axial resolution can also be pursued through extreme timing precision. Most dramatically, real-time 4 fps NLOS video of room-sized dynamic scenes has been demonstrated [33], and scan-free systems operating at up to 19 fps have been reported [217], fundamentally changing the prospects for practical NLOS deployment.

Deep learning advances. The deep learning component of NLOS imaging has seen a transformation in architecture diversity and sophistication. Beyond the U-Net and ResNet baselines of the 2021 era, the field has adopted transformers (NLOST [28], TransiT [29]), Mamba/state-space models for video [30], graph neural networks [31], and diffusion model priors [32]. Neural implicit surface representations (NLOS-NeuS [199]) have extended beyond voxel volumes to enable smooth 3D surface reconstruction. Recent works further broaden this trajectory: Time-Resolved Transformer studies photon-efficient LOS/NLOS transient 3D reconstruction with spatio-temporal attention; MARMOT introduces masked autoencoder pretraining for transient/NLOS data; and NANO uses a noise-adapted neural operator to improve sparse and photon-starved reconstruction. Learnable physical priors [365], phasor-field enhancement under partial acquisition [399], and domain reduction strategies [197] have improved generalization, while unsupervised methods [213] and untrained network priors [370] have reduced dependence on large training datasets.

Forward models, rendering, and reconstruction infrastructure. A parallel trend is the improvement of the computational infrastructure around NLOS imaging. Dedicated NLOS rendering tools [380], [381] and real-world datasets such as NLOS-Track [312] have better supported training and evaluation. Newer tools such as mitransient extend this direction by bringing time-resolved, polarized, and differentiable transient light transport into Mitsuba 3, including NLOS scene setups and capture-noise modeling. On the reconstruction side, structure-sparsity regularization improves DLCT-style inversion under low-SNR conditions, while Quasi-Fresnel Transform and optimized FFT/NUFFT sampling strategies [59] explore lighter-weight and more flexible transient inversion. Sparse and irregular active acquisition has also become a central modeling issue: arbitrary illumination/detection reconstruction [398], few-shot signal-surface collaborative regularization [190], and LEAP [399] show how Bayesian virtual-confocal formulations, mixed-dimensional priors, and phasor-domain measurement completion can preserve reconstruction quality when the relay samples are incomplete, irregular, noisy, or very sparse. Ptychographic NLOS [396] adds a coherent wavefront-retrieval route by using relay-wall modulation and ptychographic diversity for depth-resolved hidden-object visualization. Iterating the transient light transport matrix [513] pushes this model-centric trajectory further by treating a measured relay-surface TLTM as a first-order system that can be computationally focused into the hidden scene to obtain second-order transport, hidden-scene relighting, and dual-photography-style measurements. Complementing these method papers, the recent comparative ToF study by Marco et al. [204] is valuable because it places representative NLOS inverse models under a common forward-model and hardware-aware evaluation framework, clarifying which limitations arise from physics, capture geometry, photon counts, and algorithm-specific assumptions.

Passive NLOS advances. Passive NLOS imaging has progressed from simple 2D reconstruction with occluders to real-time passive tracking of moving persons (PAC-Net [312]) and full-color 3D passive NLOS imaging with ordinary cameras [313]. Hyperspectral and multispectral fusion (Hyper-NLOS [286]) and the exploitation of rough surfaces as relay surfaces [325] have broadened the environmental condi-

tions and surface types compatible with passive NLOS sensing. Event-camera passive NLOS, represented by NLOS-ES and event-enhanced passive NLOS with physical embedding, shows that neuromorphic sensing can preserve fast motion cues that are difficult to capture with conventional frame cameras.

New modalities and scenes. Beyond optical NLOS, mmWave radar NLOS reconstruction (HoloRadar [34]) and ultrasound synthetic aperture NLOS imaging [35] have demonstrated full 3D scene reconstruction in non-optical modalities, enabling operation in darkness, through walls, and under conditions where optical methods fail. Optical NLOS itself is also moving beyond clean free-space propagation: scattering-media phasor-field analysis [401] studies how virtual-wave NLOS behaves when the hidden scene and relay path are affected by volumetric scattering. The latest frontier extends even further: GeRaF 2.0 studies RF/radar 3D reconstruction through boxes; RIS-assisted radar actively controls the RF propagation environment for around-corner sensing, while dual-beam RIS radar [450] studies the practical effect of one-bit quantized RIS beam control and reciprocal two-way scattering; passive non-reconfigurable electromagnetic skins [468] provide a lower-complexity metasurface route for multi-view near-field NLOS radar imaging; distributed MIMO/ISAC NLOS imaging uses NLOS multipath for joint synchronization and off-grid imaging; N^2LoS explores tag-assisted mmWave NLOS localization; and terahertz NLOS uses lossy-mirror environmental reflections for around-obstacle imaging. Acoustic NLOS has also diversified from relay-surface reconstruction to relay-free passive acoustic localization via edge diffraction. In robotics, single-photon-LiDAR navigation work [509] and SuperEx connect NLOS measurements with indoor exploration and mapping, demonstrating that partial hidden-space information can be valuable even when it is used primarily for decision making rather than final-quality reconstruction.

Geometry, relay surfaces, and representation. The classical assumption of a planar diffuse relay wall is increasingly being relaxed. 3D Gaussian Transient Rendering models arbitrary relay surface geometries with differentiable transient rendering, while ptychographic NLOS [396], arbitrary illumination and detection patterns [398], scattering-media NLOS [401], and active corner-camera systems expand the range of feasible acquisition layouts. In the RF regime, engineered reflectors such as RISs, Van-Atta-array IRSs, and static electromagnetic skins show that the propagation environment itself can become part of the sensing hardware. These developments suggest that future NLOS systems will be less tied to laboratory walls and more capable of exploiting real environmental geometry.

Remaining challenges and future directions. Despite remarkable progress, several important challenges remain open. (1) *Outdoor and strong ambient light operation:* most existing systems require controlled or low-ambient-light conditions. Coherent sensing approaches [225] and temporal gating designs [159] are promising, but robust outdoor operation at long range remains difficult. (2) *Large-scale real training data:* the gap between synthetic and real-world NLOS measurements continues to limit generalization. Automated data collection systems, more realistic rendering, and domain adaptation methods are needed. (3) *Non-Lambertian and complex relay surfaces:* most algorithms still assume diffuse relay walls; exploiting specular [325], non-planar surfaces [188], arbitrary relay geometries, and volumetric scattering media in a unified framework remains an open problem. (4) *End-to-end system integration:* bridging the gap between research demonstrations and practical deployed systems—handling calibration, robustness, and computational cost—is a critical next step. (5) *Semantic NLOS understanding:* going beyond shape reconstruction to semantic tasks such as human pose estimation, action recognition, and object identification in hidden scenes represents an important emerging frontier.

In summary, NLOS imaging has transitioned from a laboratory curiosity to a technology on the verge of practical deployment across multiple modalities and application domains. Continued progress on the challenges outlined above will be essential for realizing the transformative potential of seeing around corners.

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